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High versus low-added sugar consumption for the primary prevention of cardiovascular disease (Review)

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[Intervention Review]

High versus low-added sugar consumption for the primary prevention of cardiovascular disease

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ABSTRACT

Background

High intake of added sugar have been suggested to impact the risk for cardiovascular disease (CVD). Knowledge on the subject can contribute to preventing CVD.

Objectives

To assess the effects of a high versus low-added sugar consumption for primary prevention of CVD in the general population.

Search methods

We searched Cochrane Central Register of Controlled Trials (CENTRAL) in the Cochrane Library, MEDLINE, Embase, Conference Proceedings Citation Index-Science (CPCI-S) on 2 July 2021. We also conducted a search of ClinicalTrials.gov and the WHO International Clinical Trials Registry Platform (ICTRP) Search Portal for ongoing or unpublished trials. The search was performed together with reference checking, citation searching and contact with study authors to identify additional studies. We imposed no restriction on language of publication or publication status.

Selection criteria

We included randomised controlled trials (RCTs), including cross-over trials, that compared different levels of added sugar intake. Exclusion criteria were: participants aged below 18 years; diabetes mellitus (type 1 and 2); and previous CVD. Primary outcomes were incident cardiovascular events (coronary, carotid, cerebral and peripheral arterial disease) and all-cause mortality. Secondary outcomes were changes in systolic and diastolic blood pressure, total cholesterol, LDL-cholesterol, HDL-cholesterol, triglycerides, fasting plasma glucose and adverse events (gastrointestinal symptoms and impaired dental health).

Data collection and analysis

We used the standard methodological procedures expected by Cochrane.

Main results

We included 21 RCTs (1110 participants completing the interventions) examining the effects of different levels of added sugar intake with a mean duration of 14 weeks. The study participants were generally described as healthy and the mean age ranged from 22 to 57 years.

No studies reported on cardiovascular events or all-cause mortality. There was minimal effect of low intake of added sugar on total cholesterol levels (MD 0.11, 95% CI 0.01 to 0.21; $I^2 = 0\%$; 16 studies; 763 participants; low certainty of evidence) and triglycerides (MD 0.10, 95% CI 0.03 to 0.17; $I^2 = 3\%$; 14 studies; 725 participants) but no evidence of effect on LDL-cholesterol and HDL-cholesterol. There was minimal effect on diastolic blood pressure (MD 1.52, 95% CI 0.67 to 2.37; $I^2 = 0\%$; 13 studies; 873 participants) and on systolic blood pressure (MD 1.44, 95% CI 0.08 to 2.80; $I^2 = 27\%$; 14 studies; 873 participants; low certainty of evidence), but no evidence of effect on fasting plasma glucose.

Only one study reported on dental health, with no events. No other trials reported adverse events (impaired dental health or gastrointestinal symptoms).

All results were judged as low-quality evidence according to GRADE. The risk of bias was generally unclear, five studies were classified at an overall low risk of bias (low risk in at least four domains, not including other bias).

Authors' conclusions

No trials investigating the effect of added sugar on cardiovascular events or all-cause mortality were identified in our searches. Evidence is uncertain whether low intake of added sugar has an effect on risk factors for CVD; the effect was small and the clinical relevance is, therefore, uncertain. Practical ways to achieve reductions in dietary added sugar includes following current dietary recommendations.

Future trials should have longer follow-up time and report on all-cause mortality and cardiovascular events in order to clarify the effect of added sugar on these outcomes. Future trials should also aim for more direct interventions and preferably be more independent of industry funding.

PLAIN LANGUAGE SUMMARY

Low levels of sugar to prevent cardiovascular disease

Background

Cardiovascular disease (CVD) is a group of disorders affecting the heart and blood vessels and the number one cause of death worldwide. It is important to detect modifiable risk factors and find strategies to prevent CVD. There are several established modifiable risk factors for developing CVD, one of them being eating an unhealthy diet rich in sugar. Sugar can be divided in two categories; sugars naturally occurring in food and sugars that are *added* to food. A high level of added sugar intake is suggested to cause weight gain and affect blood lipids, increasing the risk of CVD. This review assessed different levels of added sugars in the diet and the effect on cardiovascular events (e.g. heart attack or stroke), death, and CVD risk factors in healthy adults.

Study characteristics

Databases for randomised controlled trials (clinical trials in which participants are randomly assigned to either an experimental or a control treatment) were searched. The trials that were included compared different levels of added sugar intake and its effect on risk factors for CVD in healthy adults. People with previous CVD or diabetes were not included in the review.

Key results

Twenty-one trials were found with 1110 participants. None of the trials looked at cardiovascular events or death. The trials reported on blood pressure, blood lipid levels and blood sugar levels. The review found that low levels of added sugar intake led to a small reduction in blood pressure and blood lipid levels, but no effect was seen on blood sugar. The evidence is current to July 2021.

Quality of the evidence

The studies included in the review provide low-quality evidence that low levels of added sugar in the diet indirectly reduces the risk of cardiovascular disease. More long-term studies of high quality assessing effects of different levels of sugar on CVD risk factors, cardiovascular events and death are needed.

SUMMARY OF FINDINGS

Summary of findings 1. Effects of high- compared to low-added sugar consumption for the primary prevention of cardiovascular disease

Effects of high- compared to low-added sugar consumption for the primary prevention of cardiovascular disease

Patient or population: Adults without diabetes or CVD

Setting: Community

Intervention: High-added sugar intake

Comparison: Low-added sugar intake

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with low-added sugar intake	Risk with Effects of high-added sugar intake				
Cardiovascular event	No trials reported cardiovascular events					
All-cause mortality	No trials reported all-cause mortality					
Systolic blood pressure (mmHg) Mean follow-up 12 weeks ¹	The mean systolic blood pressure (mmHg) was 117.3	MD 1.44 higher (0.08 lower to 2.80 higher)	—	873 (14 RCTs)	⊕⊕⊕⊕ LOW 2 3	Lowndes 2014b reported no differences in systolic blood pressure, but could not be included in the analysis as no numerical data were provided.
Total cholesterol (mmol/L) Mean follow-up 10 weeks ¹	The mean total cholesterol (mmol/L) was 4.7	MD 0.11 higher (0.01 lower to 0.21 higher)	—	763 (16 RCTs)	⊕⊕⊕⊕ LOW 2 3	
LDL-cholesterol (mmol/L) Mean follow-up 12 weeks ¹	The mean LDL-cholesterol (mmol/L) was 2.9	MD 0.01 higher (-0.08 lower to 0.10 higher)	—	712 (12 RCTs)	⊕⊕⊕⊕ LOW 2 3	
HDL-cholesterol (mmol/L) Mean follow-up 14 weeks ¹	The mean HDL- cholesterol (mmol/L) was 1.2	MD 0.01 higher (-0.02 lower to 0.04 higher)	—	817 (16 RCTs)	⊕⊕⊕⊕ LOW 2 3	
Fasting plasma glucose(mmol/L)	The mean fasting plasma glucose(mmol/L) was 5.1	MD 0.06 higher (-0.01 lower to 0.13 higher)	—	720 (14 RCTs)	⊕⊕⊕⊕ LOW 2 3	

Mean follow-up 12 weeks¹

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: Confidence interval; **RR:** Risk ratio; **OR:** Odds ratio;

GRADE Working Group grades of evidence

High certainty: We are very confident that the true effect lies close to that of the estimate of the effect

Moderate certainty: We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different

Low certainty: Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect

Very low certainty: We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect

¹ Minimum study duration was 4 weeks for the secondary outcomes.

²**Risk of bias.** Unclear risk of bias in most domains. Downgraded once

³**Indirectness.** Indirect interventions. Only surrogate outcomes reported. Downgraded twice

BACKGROUND

High intake of added sugars has been suggested as having an impact on risk for cardiovascular disease (CVD). Many studies support this claim, in particular, in regard to sugar-sweetened beverages (SSB). However, for total added sugar intake, the evidence is much weaker. Hence, there is a need to review the current evidence from randomised controlled trials (RCTs) in this area.

Description of the condition

Cardiovascular diseases (CVDs) are disorders of the heart and blood vessels and incorporate, for instance, acute coronary and cerebrovascular events. The origin of many CVDs is atherosclerosis, a condition that develops over many years and is often initially asymptomatic. Atherosclerosis is defined as the progressive accumulation of lipids and inflammatory cells in arteries of the body (Torres 2015). Atherosclerosis develops when arteries are narrowed or completely blocked, thus restricting the flow of blood or oxygen to organs or tissue. CVDs are a major cause of disability and are the number one cause of death globally. In 2015, it was estimated that 31% of all global deaths were due to CVDs (World Health Organization 2017). As the world's population increases and becomes older, the prevalence of CVDs is expected to increase further in the future (Roth 2015).

The most established modifiable risk factors for CVDs are physical inactivity, tobacco use, excessive alcohol consumption and maintaining an unhealthy diet. Non-modifiable factors that can affect CVD risk include: higher age, male sex and heredity (Mendis 2011). In order to decrease mortality and morbidity resulting from CVDs, early detection methods as well as prevention strategies focusing on the modifiable risk factors are crucial. Prominent markers of detection and treating targets for CVDs are overweight and obesity conditions, dyslipidaemia and hypertension (Mendis 2011), which are all becoming increasingly prevalent health issues worldwide (Global Burden of Disease 2016).

Description of the intervention

Sugars (all mono- and disaccharides) can be divided into two categories: those occurring naturally in foods (intrinsic), such as lactose in milk and fructose in fruit, and those that are added to foods (extrinsic). There is no globally unanimous way to define the term 'added sugar', but the definition by the American Heart Association is clear, simple and corresponds to the majority of other definitions: "Added sugars include *any* sugars or caloric sweeteners that are added to foods or beverages during processing or preparation", for instance, white sugar (sucrose), honey, or high-fructose corn syrups (HFCS) (American Heart Association 2017). The World Health Organisation (WHO) utilises the term 'free sugars' which includes all added sugars and the naturally occurring sugars in fruit juices (World Health Organization 2015). The WHO recommends that, for both children and adults, the total intake of free sugars should be less than 10% of the total energy intake and, ideally, not exceed 5%. It is a growing public health concern that high-added sugar intake is causing an increased overall intake of calories for many individuals. Also, of concern is that a high intake of sugar-rich foods can replace healthier and more nutritious foods, resulting in a less nutrient-dense diet. Impaired nutrient density and the effects on increased energy intake and body weight, as well

as the risk for dental caries, lies at the foundation of today's dietary guidelines regarding sugar (World Health Organization 2015).

The consumption of added sugar in high-income countries is, in general, exceeding this recommended upper level. However, the current overall trends in these countries are actually suggesting a reached plateau or even a small decrease (Wittekind 2014), while the global trends, driven by low- and middle-income countries, are suggesting overall increasing sugar consumption (World Cancer Research Fund International 2015).

How the intervention might work

One of the main driving forces behind why high sugar intake is hypothesised to increase risk of CVD is because of the potential weight gain it is causing. Sugar-rich products are generally very palatable, or "high rewarding" (Stice 2013) and, at the same time, less satiating than more fibrous foods (Rebello 2013). This can result in over-consumption and an energy intake higher than required, hence leading to weight gain. In a meta-analysis of RCTs and cohort studies on sugar intake and body weight, of a total of 38 cohort studies and 30 trials, the authors concluded that sugar intake is a determinant of body weight, which likely is due to skewed energy balance rather than any biological effects of the sugar molecules. However, the effects of a reduced sugar intake have modest effects on obesity (Te Morenga 2012).

Higher sugar intakes are also associated with increased triglycerides, total cholesterol and LDL-cholesterol in a large meta-analysis of RCTs (Te Morenga 2014), which is considered to be a serum lipid profile that induces atherosclerosis (NECP 2002). In addition, it is suggested that sugar intake can be related to CVD risk through decreasing insulin sensitivity (Aeberli 2011) and can possibly also promote low-grade inflammation (Aeberli 2011; Liu 2002; Sorensen 2005). However, several studies do not support these findings (Black 2006; Lewis 2013; Maersk 2012) and it is difficult to say with certainty whether this is a cause of the actual sugar itself or from altered adipose tissue (body fat) and blood lipid profile. In addition, blood pressure is one of the strongest risk factors and first treatment targets for CVD and is, therefore, also considered an important risk marker to evaluate (Karmali 2018).

Fructose has been in particular focus for a lot of research, possibly because of the large increase in usage of HFCS for sweetening, which contains 55% fructose and 45% glucose. Fructose has a lower glycaemic index (GI) compared to both sucrose and especially glucose, and does not yield as high insulin responses (Bantle 1986; Teff 2009), but there are findings suggesting that fructose, in particular, contributes to the increased accumulation of lipids in both vessels and liver and to visceral accumulation of adipose tissue surrounding the organs (Silbernagel 2011; Stanhope 2009). However, fructose and glucose are seldom consumed separately, but rather joined as sucrose or HFCS, and findings are not unanimous. Thus, it remains unclear how much the different metabolic pathways of glucose and fructose actually matter for the adipose tissue, blood lipid profile and cardiovascular risk (Silbernagel 2011). Therefore, this review focusses on different intake levels of all added sugars rather than comparing different sugars against each other.

Why it is important to do this review

Regarding diet and CVD prevention, meta-analyses of either RCTs, cohort studies, or both, have previously shown that low-sodium intake (Aburto 2013), low intakes of saturated fatty acids (Hooper 2020), high intake of dietary fibre (Threapleton 2013) and whole grains (Ye 2012), and a diet rich in fruits and vegetables (Hartley 2013) can decrease CVD risk. As for sugars, evidence from a meta-analysis of prospective cohort studies on SSB and its effect on increased cardiovascular risk is quite conclusive (Xi 2015). Regarding total added sugar intake, two cohort studies have observed a positive relationship between added sugar intake and CVD mortality (Ramne 2018; Yang 2014). However, the study by Ramne and colleagues could also see tendencies for increased risk with the lowest intakes, i.e. a U-shaped association (Ramne 2018). A third cohort study, investigating sugar and mortality, could not see an association (Tasevska 2014). Additionally, both the study by Ramne and colleagues and Tasevska and colleagues could see tendencies for opposite associations when separately examining added sugar intake from beverages (positive) and solid foods (negative). This inconclusive epidemiology goes in line with the existing evidence for total added sugar intake and CVDs, which currently is not very persuasive or conclusive (EFSA 2010; Hauner 2012).

As mentioned, the previous focus for research has primarily been on SSB, where the majority of data comes from observational studies, or comparing different types of sugar against each other, i.e. glucose versus fructose, rather than investigating an altered dose of total added sugars. (Chiavaroli 2015; Evans 2017; Ha 2012; Zhang 2013). Several Cochrane reviews of RCTs investigating different dietary components for prevention of CVD have been conducted (Clar 2017; Hartley 2016; Hooper 2018; Kelly 2017), however, none regarding added sugar intake. A few other systematic reviews have investigated added sugar intake and cardiovascular risk markers, but none have aimed to include CVD incidence as the primary outcome (Fattore 2017; Te Morenga 2014). Hence, it is of great importance to elucidate the evidence from RCTs that attend to the whole spectra of added sugar consumption from all sources and its potential role in primary prevention of CVD.

OBJECTIVES

To assess the effects of a high versus low-added sugar consumption for primary prevention of CVD in the general population.

METHODS

Criteria for considering studies for this review

Types of studies

We included randomised controlled trials (RCTs), including cross-over trials. We included studies reported as full text, those published as abstract only, and unpublished data.

Types of participants

The study participants were required to have participated in an intervention comparing different levels of added sugar intake. Studies were not included if they fulfilled one or more of the following criteria:

- Study population aged below 18 years;

- Study population with diabetes mellitus (type 1 and 2);
- Study population with previous CVD.

Types of interventions

We included trials aimed to manipulate sugar intake, hence, we compared a high sugar intake with a low sugar intake. The studies included any type of added sugar (e.g. sucrose, fructose or glucose) or sugar-rich foods and beverages consumed orally. Limiting to added sugars meant that interventions of, for instance, high or low-fruit intake (high in intrinsic sugars) were not included. The included interventions consisted of dietary advice where added sugar intake was either increased or decreased, provision of sugar-rich products or encompassing a diet alteration where added sugar intake is either at a high or low level (as defined in the individual studies). If the intervention aimed towards increasing sugar intake, then the comparison group (or no treatment group) represented the low sugar group. And if the intervention group aimed to decrease sugar intake, then the comparison group (or no treatment group) represented the high sugar group.

The comparison group could be given no advice/supplementation (continue with their usual diet), a placebo product or any control diet. Studies that compared corresponding amounts of different types of sugars, for example, glucose versus fructose, were excluded from the review. We did not include studies where an altered sugar consumption was part of an overall diet or lifestyle intervention which, for instance, additionally modified exercise routines in order to avoid potential confounding.

In order to include as many studies as possible, while still ensuring enough time for stabilisation of, for instance, blood lipid concentrations, we included interventions that lasted a minimum of four weeks between baseline and follow-up. Four weeks was not expected to be enough time to have an effect on our primary outcomes, therefore, a minimum of six months follow-up time was required for these outcomes. Cross-over trials were included, but only data from the first period was extracted for evaluation of the primary outcomes, and from both periods for the secondary outcomes. A minimum wash-out period of two weeks was necessary to minimise carry-over effects for the secondary outcomes. Trials not adhering to these study duration requirements were excluded from the analyses.

Types of outcome measures

Primary outcomes

- Incident cardiovascular event (coronary, carotid, cerebral and peripheral arterial disease)
- All-cause mortality

Secondary outcomes

- Changes in systolic blood pressure
- Changes in diastolic blood pressure
- Changes in total cholesterol
- Changes in LDL-cholesterol
- Changes in HDL-cholesterol
- Changes in triglycerides
- Changes in fasting plasma glucose
- Adverse events: gastrointestinal symptoms (such as nausea, abdominal pain, constipation and diarrhoea)

- Adverse events: impaired dental health (such as dental caries)

Where a published report did not appear to report one of these outcomes, we attempted to access the trial protocol and contacted the trial authors to ascertain whether the outcomes were measured but not reported. Relevant trials which measured these outcomes but did not report the data at all, or not in a usable format, were included in the review as part of the narrative. We did not report on other measured outcomes not relevant for the present review.

Search methods for identification of studies

Electronic searches

The electronic search was conducted on 2 July 2021, and we identified trials through systematic searches of the following bibliographic databases:

- Cochrane Central Register of Controlled Trials (CENTRAL) in the Cochrane Library (Issue 7 of 12, July 2021);
- MEDLINE (Ovid, from 1946 to 1 July 2021);
- Embase (Ovid, from 1980 to 2021 week 25);
- Conference Proceedings Citation Index — Science (CPCI-S; Web of Science, from 1990 to 2 July 2021).

The search strategies for the databases ([Appendix 1](#)) were adapted from the draft search strategy for MEDLINE (Ovid). The Cochrane sensitivity-precision maximising RCT filter ([Lefebvre 2011](#)) was applied to MEDLINE (Ovid) and adaptations of it to the other databases, except CENTRAL. We also conducted a search of ClinicalTrials.gov (www.ClinicalTrials.gov) and the WHO International Clinical Trials Registry Platform (ICTRP) Search Portal (<http://apps.who.int/trialsearch/>) for ongoing or unpublished trials on 2 July 2021.

We imposed no restriction on language of publication or publication status. We did not perform a separate search for adverse effects of interventions of altered sugar intake. We considered adverse effects described in included studies only.

Searching other resources

We checked reference lists of all included studies and any relevant systematic reviews identified for additional references to RCTs. We

also examined any relevant retraction statements and errata for included studies. We contacted authors for missing data, where necessary.

Data collection and analysis

Selection of studies

We used Cochrane's Screen4Me workflow to help assess some search results. Screen4Me comprises three components: known assessments — a service that matches records in the search results to records that have already been screened in Cochrane Crowd and been labelled as an 'RCT' or as 'Not an RCT'; the RCT classifier — a machine learning model that distinguishes RCTs from non-RCTs and, if appropriate, Cochrane Crowd — Cochrane's citizen science platform where the Crowd help to identify and describe health evidence. For more information about Screen4Me and the evaluations that have been done, please go to the Screen4Me webpage on the Cochrane Information Specialist's portal: <https://community.cochrane.org/organizational-info/resources/resourcesgroups/information-specialists-portal>. In addition, more detailed information regarding evaluations of the Screen4Me components can be found in the following publications: [Marshall 2018](#); [McDonald 2017](#); [Noel-Storr 2018](#); [Thomas 2017](#).

Two review authors (SB and AJ) independently screened titles and abstracts for inclusion and coded them as 'retrieve' (eligible or potentially eligible/unclear) or 'do not retrieve'. If there were any disagreements, a third and fourth author were asked to arbitrate (SA and ES). We retrieved the full-text study reports/publications and two review authors (SB and AJ) independently screened the full text, identified studies for inclusion, and identified and recorded reasons for exclusion of the ineligible studies. We resolved any disagreement through discussion or, where required, we consulted a third and fourth person (SA and ES). We identified and excluded duplicates and collated multiple articles of the same study so that each study, rather than each article, was the unit of interest in the review. We recorded the selection process in sufficient detail to complete a PRISMA flow diagram ([Figure 1](#)) and [Characteristics of excluded studies](#) table.

Figure 1. Figure 1. Study flow diagram

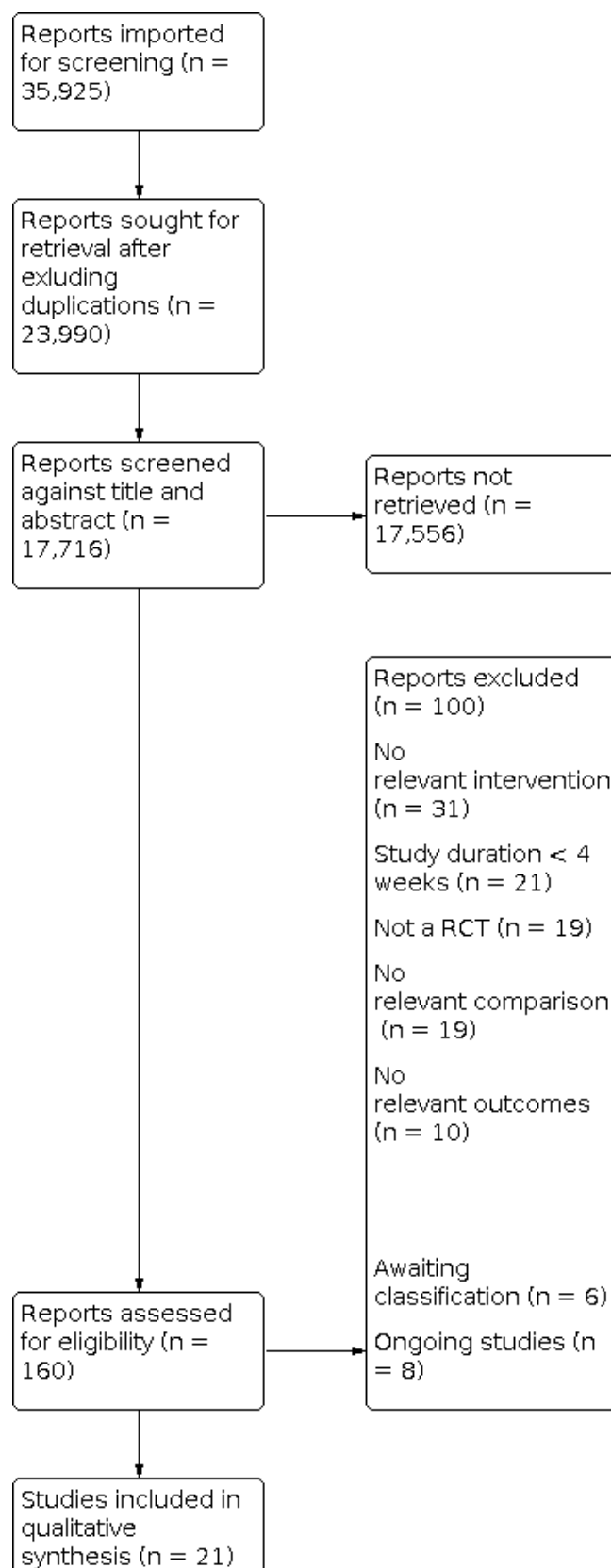
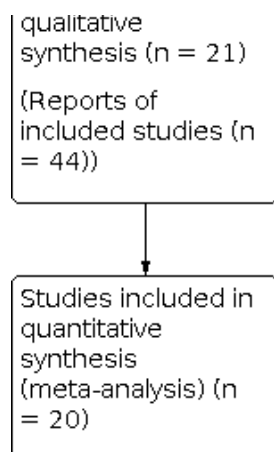


Figure 1. (Continued)



Data extraction and management

We used a data collection form for study characteristics and outcome data.

We extracted the following study characteristics, if available:

- **Methods:** study design, total duration of study, number of study centres and location, study setting, date of study and wash-out period for cross-over trials.
- **Participants:** N randomised, N lost to follow-up/withdrawn, N analysed, mean age, age range, gender, mean BMI, parameters of metabolic syndrome, other diseases, eventual weight change, compliance with intervention, inclusion criteria, and exclusion criteria. Baseline data on lipids and other characteristics were collected and assessed for imbalance between groups.
- **Interventions:** intervention details, and comparison details in sufficient detail to allow subgroups to be determined.
- **Outcomes:** primary and secondary outcomes (including adverse events) specified and collected, and time points reported.
- **Notes:** funding source of trial and notable conflicts of interest of trial authors.

Two review authors (SB and AJ) independently extracted outcome data from included studies. We resolved disagreements by consensus or by involving a third and fourth person (SA and ES). One review author (AJ) transferred data into the Review Manager (RevMan 2014) file. We double-checked that data were entered correctly by comparing the data presented in the systematic review with the data extraction form. Two review authors (SB and AJ) spot-checked study characteristics for accuracy against the trial report.

Assessment of risk of bias in included studies

Two review authors (SB and AJ) independently assessed risk of bias for each study using the criteria outlined in the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins 2011). We resolved any disagreements by discussion or by involving two other authors (SA and ES). We assessed the risk of bias according to the following domains.

- Random sequence generation
- Allocation concealment
- Blinding of participants and personnel

- Blinding of outcome assessment
- Incomplete outcome data
- Selective outcome reporting
- Other bias

The primary aspect of 'other biases' to consider was compliance with the intervention. Each potential source of bias was graded as high, low or unclear and provided a quote from the study report together with a justification for our judgement in the Risk of bias table. A study needed to be rated as low risk in at least four domains (not including 'other bias') to be considered at low summary risk of bias. The risk of bias judgements were summarised across different studies for each of the domains listed.

For cross-over trials, the risk of biases that we considered were:

- Whether the cross-over design was suitable
- Whether there was a carry-over effect
- Whether only first-period data were available
- Incorrect analysis
- Comparability of results with those from parallel-group trials

Measures of treatment effect

Mean difference and standard error were used for continuous data. If the same study presented both change scores and end of follow-up data, we used the *Cochrane Handbook* (Deeks 2018) guidelines to identify the most appropriate form to include. Differences in units across studies were converted to the unit most used to avoid expressions in standardised mean difference. We entered data presented as a scale with a consistent direction of effect. We narratively described skewed data reported as medians and interquartile ranges.

Unit of analysis issues

If studies reported results at several time points, we analysed the time point with the longest follow-up.

Studies with multiple intervention groups: Where we identified relevant studies with multiple intervention groups, we only included the highest intake group and the lowest intake group, to avoid splitting the control group. Moreover, since it might be different between studies if the intervention arm aimed to

increase or decrease sugar intake, these were combined in the main analyses. This allowed us to always compare low sugar intake with high sugar intake so that trials aiming to increase sugar intake appeared with the control arm in the low sugar intake side of the equation.

Cross-over trials: We evaluated the evidence based on data from the first phase of the intervention for the primary outcomes since these are conditions evolving over time, have long-term effects, and it cannot be ascertained that any length of wash-out period is adequate for eliminating the risk of carry-over effects. Both periods were considered for the continuous secondary outcomes, so long as there was a minimum wash-out period of two weeks between phases. We adhered to the recommended methods outlined in the *Cochrane Handbook* regarding cross-over trials (Higgins 2011), and we used the generic inverse variance (GIV) method and reported the mean difference and standard error for each trial. For the trials that did not report this and instead reported the results as for instance standard deviation, we used the Revman calculator to compute the standard error.

Dealing with missing data

We contacted study authors in order to obtain missing numerical outcome data where possible (e.g. when a study was identified as abstract only). If this was not possible, the risk of bias assessment reflected this potential shortcoming. If relevant trials did not report the SD, we used the Revman calculator to attain the standard error (SE), from other available data, such as standard deviation (SD), 95% confidence interval (CI) and P values. For missing participant data, we based the evidence on an intention-to-treat analysis.

Assessment of heterogeneity

We used the I^2 statistic to measure heterogeneity among the trials in each analysis. If we identified substantial heterogeneity, we reported it and explored possible causes by prespecified subgroup analysis. We also visually assessed forest plots in order to evaluate the direction and magnitude of effects and potential overlap of CIs. Statistical heterogeneity was assessed using the I^2 test. We also, in addition to using the I^2 test, considered the P value from the χ^2 tests. The significance level was set at a P value < 0.05. I^2 was calculated using a random-effects model and interpreted according to the *Cochrane Handbook* (Deeks 2018) guidelines:

- 0% to 40%: might not be important;
- 30% to 60%: may represent moderate heterogeneity;
- 50% to 90%: may represent substantial heterogeneity;
- 75% to 100%: considerable heterogeneity.

Assessment of reporting biases

We were not able to pool more than 10 trials in order to create and examine a funnel plot to explore possible small study biases for the primary outcomes as planned; instead a funnel plot was created for the outcome total cholesterol, as described in [Effects of interventions](#).

Data synthesis

We undertook meta-analyses only where this was meaningful i.e. if the treatments, participants and the underlying clinical question were similar enough for pooling to make sense.

We used a random-effects model, as we were expecting some degree of heterogeneity in the results. Clinical diversity among the included studies was expected, since they could have varying interventions, doses and comparisons. In addition, previous reviews that had a similar approach and focus, had chosen the random-effects model.

If some studies reported continuous outcomes using change from baseline and some reported end of follow-up values, we combined these in the same meta-analysis and reported the mean difference and standard error.

We converted units, when applicable, so that all results were expressed in the same units. Total cholesterol, HDL-cholesterol and LDL-cholesterol were converted from mg/dL to mmol/L by dividing mg/dL by 38.67. Triglycerides were converted from mg/dL to mmol/L by dividing mg/dL by 88.57. Fasting plasma glucose was converted from mg/dL to mmol/L by dividing mg/dL by 18.

Subgroup analysis and investigation of heterogeneity

We carried out the following subgroup analyses:

- Increased or decreased added sugar intake;
- Isocaloric exchange or ad libitum intake;
- Liquid or solid state of added sugar source;
- Control in form of starch/refined grains;
- Weight change (> 0.5 kg difference in weight change between the two study arms) or no weight change;
- Healthy population or high-risk population (metabolic syndrome or obesity, hypertension, dyslipidaemia and elevated fasting blood glucose levels/pre-diabetes);
- Duration of intervention (more or less than six months);
- Industry-funded studies or no involvement with industry.

We investigated all outcomes in all subgroup analyses, and we used the formal test for subgroup interactions in Review Manager (RevMan 2014).

Sensitivity analysis

We carried out the following sensitivity analyses:

- Only including studies with a low risk of bias. A study was judged to be at low risk of bias when there was low risk in at least four domains (not including 'other bias').
- Excluding cross-over trials.

Summary of findings and assessment of the certainty of the evidence

We created a [Summary of findings 1](#) for high versus low-added sugar intake, presenting the following outcomes: cardiovascular events, all-cause mortality, systolic blood pressure, total cholesterol, LDL-cholesterol, HDL-cholesterol, and fasting plasma glucose. We used the five GRADE considerations (study limitations, consistency of effect, imprecision, indirectness and publication bias) to assess the quality of a body of evidence as it related to the studies which contributed data to the meta-analyses for the prespecified outcomes. We used methods and recommendations described in Section 8.5 (Higgins 2011) and Chapter 15 (Schünemann 2018) of the *Cochrane Handbook for*

Systematic Reviews of Interventions using GRADEpro software (<https://grade.pro.org/>).

We justified all decisions to downgrade the quality of studies using footnotes, and we made comments to aid the reader's understanding of the review, where necessary.

Judgements about evidence quality were made by two review authors (SB and AJ) working independently, with disagreements resolved by discussion or involving a third and fourth author (SA and ES). Judgements were justified, documented and incorporated into reporting of results for each outcome.

RESULTS

Description of studies

Descriptions of the included and excluded studies can be found in the [Characteristics of included studies](#) and in the [Characteristics of excluded studies](#) tables.

Results of the search

The search generated 17,723 hits, and the subsequent screening of titles and abstract resulted in 145 studies to be assessed as full text. Twenty-one studies adhered to the inclusion criteria and were included in the review. The study flow diagram can be seen in [Figure 1](#). In addition, eight studies were categorised as ongoing and are outlined in the [Characteristics of ongoing studies](#) table.

Included studies

Types of studies

Twenty-one studies were included in total and are outlined in the [Characteristics of included studies](#) table. Seven were cross-over trials ([Ahmad 2020](#); [Black 2006](#); [Lewis 2013](#); [Maki 2015](#); [Markey 2016](#); [Njike 2011](#); [Umpleby 2017](#)) and 14 were parallel-group RCTs. The studies were published between 1976 and 2021 with a clear majority published in the 2000s. Six of the included studies were conducted in the USA, five in the UK, two in Denmark, Norway, Switzerland and Mexico and one in Finland and New Zealand. Twelve of the included studies were two-arm trials, five trials had three arms ([Ebbeling 2020](#); [Mann 1972](#); [Njike 2011](#); [Sanchez-Delgado 2021](#); [Vazquez-Duran 2016](#)), but only two arms were used for this review for all analyses. Three trials had four arms ([Geidl-Flueck 2021](#); [Lowndes 2014b](#); [Maersk 2012](#); for [Geidl-Flueck 2021](#) and [Maersk 2012](#), only two arms were relevant for this review) and one study had six arms ([Lowndes 2014a](#)), four that were relevant for the purposes of this review.

Overview of study populations

In total, there were 1110 study participants across the included trials included in the meta-analysis. The study populations ranged from 13 ([Black 2006](#); [Lewis 2013](#)) to 231 ([Lowndes 2014a](#)). The mean study population was 53 participants. The age of the study participants ranged from a mean of 22 years ([Vazquez-Duran 2016](#)) to a mean of 57 years ([Ahmad 2020](#); [Umpleby 2017](#)). Two studies did not report mean age ([Mann 1972](#); [Madero 2015](#)). The mean of age for all the remaining study participants was 39.2 years. Four studies only included men ([Ahmad 2020](#); [Black 2006](#); [Geidl-Flueck 2021](#); [Umpleby 2017](#)) and two studies only included women ([Johnson 2015](#); [Surwit 1997](#)). Two studies did not report on the sex distribution of the study participants ([Campos 2015](#); [Smith](#)

[1996](#)). Excluding the trials not reporting on sex distribution, the total study population consisted of 47% men and 53% women. The study participants were generally described as 'healthy' (as defined in the individual studies) ([Black 2006](#); [Geidl-Flueck 2021](#); [Mann 1972](#); [Markey 2016](#); [Sanchez-Delgado 2021](#); [Smith 1996](#)) or overweight or obese ([Campos 2015](#); [Lewis 2013](#); [Lowndes 2014a](#); [Lowndes 2014b](#); [Madero 2015](#); [Maersk 2012](#); [Njike 2011](#); [Raben 2002](#); [Surwit 1997](#)). One study described the participants as being of increased CVD risk ([Ahmad 2020](#)) and some trials requested that the participants had a habitually high SSB consumption ([Campos 2015](#); [Ebbeling 2020](#); [Maki 2015](#); [Vazquez-Duran 2016](#)). Two trials included individuals with a specific diagnosis: polycystic ovary syndrome (PCOS) ([Johnson 2015](#)) and non-alcoholic fatty liver disease (NAFLD) ([Umpleby 2017](#)).

Description of interventions

The most common intervention was comparing different levels of sucrose (expressed as the percentage of the daily total energy intake), and was determined by adjusting the participants' intake of foods and/or beverages ([Black 2006](#); [Lewis 2013](#); [Mann 1972](#); [Markey 2016](#); [Surwit 1997](#); [Umpleby 2017](#)). Two studies had a similar design, but focused on fructose instead ([Johnson 2015](#); [Madero 2015](#)). [Ahmad 2020](#) compared high versus low sugar intake, but did not specify what type of sugar used. Studies also compared foods and beverages sweetened with sucrose with artificial sweeteners ([Raben 2002](#)), steviol glycosides ([Sanchez-Delgado 2021](#)) or HFCS ([Lowndes 2014a](#); [Lowndes 2014b](#)). Five trials compared SSBs with unsweetened or artificially sweetened beverages ([Campos 2015](#); [Ebbeling 2020](#); [Maersk 2012](#); [Njike 2011](#); [Vazquez-Duran 2016](#)) and one trial compared SSBs with SSB abstinence ([Geidl-Flueck 2021](#)). One study compared sugar-sweetened products with dairy products ([Maki 2015](#)) and one study compared a sugar-free diet with participants' usual diet ([Smith 1996](#)). Eight trials ([Campos 2015](#); [Ebbeling 2020](#); [Geidl-Flueck 2021](#); [Maersk 2012](#); [Njike 2011](#); [Raben 2002](#); [Sanchez-Delgado 2021](#); [Smith 1996](#)) used an ad libitum diet. The remaining trials described their dietary exchange as isocaloric.

Outcomes included

None of the included trials reported on the primary outcomes; cardiovascular events and all-cause mortality. For the seven secondary outcomes, all were reported in at least 10 different trials. [Mann 1972](#) did not provide any standard deviations for reported outcomes and could not be included in the meta-analysis. For adverse events, one study reported on dental health ([Maersk 2012](#)).

Funding sources and conflicts of interests

Eleven of the included studies reported funding sources from the sugar industry with potential vested interests and other conflicts of interests ([Black 2006](#); [Lewis 2013](#); [Lowndes 2014a](#); [Lowndes 2014b](#); [Maersk 2012](#); [Maki 2015](#); [Mann 1972](#); [Markey 2016](#); [Njike 2011](#); [Raben 2002](#); [Surwit 1997](#)). Four studies stated that there were no conflicts of interests ([Ahmad 2020](#); [Geidl-Flueck 2021](#); [Johnson 2015](#); [Sanchez-Delgado 2021](#)). The remaining studies did not report funding source or potential conflicts of interests ([Campos 2015](#); [Ebbeling 2020](#); [Madero 2015](#); [Smith 1996](#); [Umpleby 2017](#); [Vazquez-Duran 2016](#)).

Excluded studies

Reasons for exclusion for studies that did not fulfil the inclusion criteria are outlined in the [Characteristics of excluded studies](#) table. The most common reasons for exclusion were: not relevant intervention, no relevant comparison and study duration less than four weeks, as can be seen in [Figure 1](#).

Studies awaiting classification and ongoing studies

Six studies are awaiting classification (see [Characteristics of studies awaiting classification](#)). Eight studies are ongoing studies (see [Characteristics of ongoing studies](#)).

Risk of bias in included studies

The risk of bias assessments for the included studies can be seen in [Figure 2](#) and [Figure 3](#).

Figure 2. Figure 2. Risk of bias summary: review authors' judgment about each risk of bias domain for each included study.

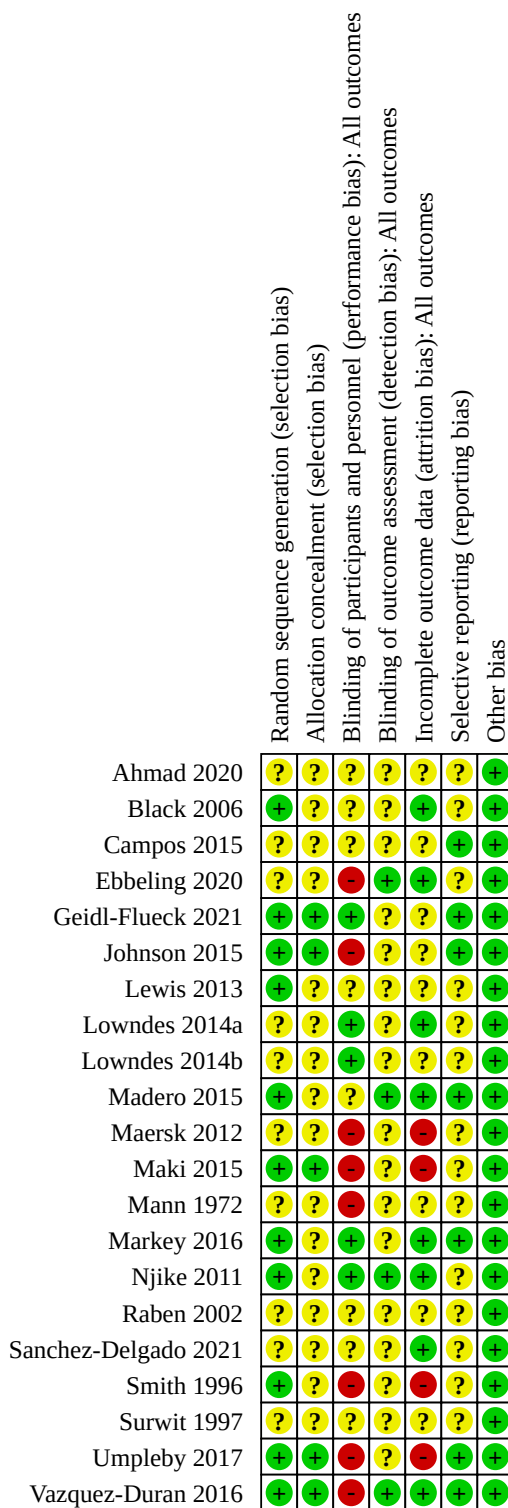
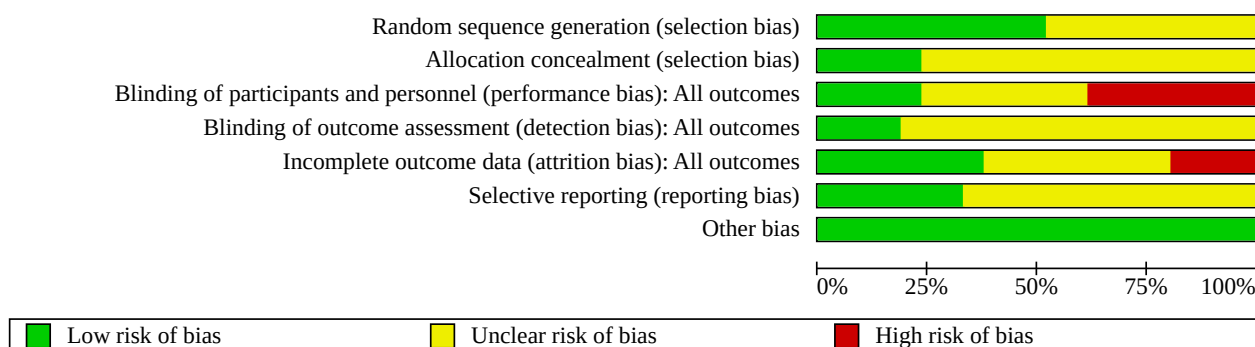


Figure 3. Figure 3. Risk of bias graph: review authors' judgements about each risk of bias domain presented as percentages across all included studies.



Allocation

Random sequence generation was adequately described in 11 studies which were hence rated as low risk of bias in this domain (Black 2006; Geidl-Flueck 2021; Johnson 2015; Lewis 2013; Madero 2015; Maki 2015; Markey 2016; Njike 2011; Smith 1996; Umpleby 2017; Vazquez-Duran 2016). Ten studies were reported as randomised but no information on how the randomisation process was conducted was included; they were therefore rated as having 'unclear' risk of bias (Ahmad 2020; Campos 2015; Ebbeling 2020; Lowndes 2014a; Lowndes 2014b; Maersk 2012; Mann 1972; Raben 2002; Sanchez-Delgado 2021; Surwit 1997).

Allocation concealment was deemed unclear in all but five studies. These studies adequately described allocation concealment and were rated as having low risk of bias (Geidl-Flueck 2021; Johnson 2015; Maki 2015; Umpleby 2017; Vazquez-Duran 2016).

Blinding

Adequate blinding procedures for participants and personnel were reported in five studies (Geidl-Flueck 2021; Lowndes 2014a; Lowndes 2014b; Markey 2016; Njike 2011). Eight trials were not blinded, the primary reason being that blinding was not possible due to the nature of the intervention; they were therefore rated as having high risk of bias (Ebbeling 2020; Johnson 2015; Maersk 2012; Maki 2015; Mann 1972; Smith 1996; Umpleby 2017; Vazquez-Duran 2016). The remaining trials did not mention blinding of participants and personnel and were rated as unclear (Ahmad 2020; Black 2006; Campos 2015; Lewis 2013; Madero 2015; Raben 2002; Sanchez-Delgado 2021; Surwit 1997).

Outcome assessors were adequately blinded in four studies (Ebbeling 2020; Madero 2015; Njike 2011; Vazquez-Duran 2016) and not mentioned in the remaining studies.

Incomplete outcome data

Eight trials included information on number of individuals lost to follow-up, provided reasons for attrition and had a similar rate of dropouts in the different intervention groups (Black 2006; Ebbeling 2020; Lowndes 2014a; Madero 2015; Markey 2016; Njike 2011; Sanchez-Delgado 2021; Vazquez-Duran 2016). Studies rated at a high risk of bias had high dropout rates, did not specify dropout proportions between study groups or had low compliance in the

intervention group (Maersk 2012; Maki 2015; Smith 1996; Umpleby 2017). Nine studies did not provide enough information to judge this domain and were rated as 'unclear' (Ahmad 2020; Campos 2015; Geidl-Flueck 2021; Johnson 2015; Lewis 2013; Lowndes 2014b; Mann 1972; Raben 2002; Surwit 1997).

Selective reporting

Studies that prospectively registered the trial on, for instance, clinicaltrials.gov and had no major deviations between the protocol and the published article, were rated as having low risk of bias in this domain (Campos 2015; Geidl-Flueck 2021; Johnson 2015; Madero 2015; Markey 2016; Umpleby 2017; Vazquez-Duran 2016). For the remaining trials, it was not possible to fully judge this domain based on the information provided (Ahmad 2020; Black 2006; Ebbeling 2020; Lewis 2013; Lowndes 2014a; Lowndes 2014b; Maersk 2012; Maki 2015; Mann 1972; Njike 2011; Raben 2002; Sanchez-Delgado 2021; Smith 1996; Surwit 1997).

Other potential sources of bias

Compliance with intervention

All included studies assessed compliance with the intervention. Some studies (Black 2006; Johnson 2015; Lewis 2013; Surwit 1997) contacted participants daily or several times per week to ensure dietary adherence. Other studies assessed compliance through interviews weekly (Sanchez-Delgado 2021), every two weeks (Ebbeling 2020) or every month (Smith 1996). Several studies monitored self-reported dietary intake weekly (Lowndes 2014a; Lowndes 2014b; Madero 2015; Vazquez-Duran 2016), biweekly (Ahmad 2020; Mann 1972) or every four to six weeks (Njike 2011; Raben 2002; Umpleby 2017). Campos 2015 estimated more than 90% compliance from returned packages of beverages; a method also used by Geidl-Flueck 2021, Maersk 2012, Maki 2015 and Markey 2016. Lack of dietary compliance was one of the main reasons for participant attrition in many of the included studies.

Risk of bias for cross-over trials

The seven included studies (Ahmad 2020; Black 2006; Lewis 2013; Maki 2015; Markey 2016; Njike 2011; Umpleby 2017) had a suitable cross-over design. All studies met the inclusion criteria of a minimum wash-out period of two weeks between interventions and the number of participants allocated to each of the two sequences were nearly equal, minimising risk for carry-over effects.

Data from both periods were available and analyses appeared to be correctly executed across studies. The results were comparable with those from parallel-group trials.

Summary risk of bias

All studies were assessed in relation to secondary outcomes due to lack of reported primary outcomes. Five studies were rated at an overall low risk of bias (Geidl-Flueck 2021; Madero 2015; Markey 2016; Njike 2011; Vazquez-Duran 2016), as these were rated as low risk in at least four domains (not including 'other bias'). The risk of bias assessment for three trials resulted in an unclear assessment, but no domains were rated at a high risk of bias (Black 2006; Lowndes 2014a; Sanchez-Delgado 2021). The remaining studies had at least one domain assessed as high risk.

Effects of interventions

See: [Summary of findings 1 Effects of high- compared to low-added sugar consumption for the primary prevention of cardiovascular disease](#)

Primary outcomes

No trials reported on the primary outcomes, cardiovascular events and all-cause mortality.

Secondary outcomes

Blood pressure

Systolic blood pressure

Fourteen studies reported systolic blood pressure and were summarised in a meta-analysis of 15 comparisons. A difference in effect was found between the comparison groups, favouring a low-added sugar intake (MD 1.44, 95% 0.08 to 2.80; $I^2 = 27\%$; 14 studies; 873 participants; low certainty of evidence; [Analysis 1.1](#)). The results were pooled using a random-effects model.

Lowndes 2014b reported no statistically significant difference in systolic blood pressure between the two intervention groups but provided no numerical data and could therefore not be included in the meta-analysis.

Five studies were judged to be at low risk of bias (Geidl-Flueck 2021; Madero 2015; Markey 2016; Njike 2011; Vazquez-Duran 2016) and were included in a sensitivity analysis (MD -0.07, 95% -0.69 to 0.56; $I^2 = 0\%$; 5 studies; 303 participants; low certainty of evidence; [Analysis 8.1](#)). Owing to the low number of studies included in the analysis, these results should be interpreted with caution.

After excluding the seven cross-over studies (no cluster-randomised trials were found) and Mann 1972 (as no standard deviation for reported outcomes was provided), another sensitivity analysis was carried out on the 13 remaining parallel trials (MD 1.77; 95% CI 0.05 to 3.50; $I^2 = 42\%$; 9 studies; 735 participants; low certainty of evidence; [Analysis 9.1](#)). Excluding cross-over studies did not lead to a substantial change in the effect estimate.

Diastolic blood pressure

Fourteen studies reported diastolic blood pressure and were summarised in a meta-analysis comprising 15 comparisons. A difference in effect was noted between the comparison groups, favouring a low-added sugar intake (MD 1.52, 95% CI 0.67 to 2.37; $I^2 = 0\%$; 13 studies; 873 participants; [Analysis 1.2](#)). The results were pooled using a random-effects model.

Lowndes 2014b reported no statistically significant difference in diastolic blood pressure between the two intervention groups but provided no numerical data and could therefore not be included in the meta-analysis.

The five studies judged at low risk of bias were included in a sensitivity analysis (MD 0.19, 95% -1.93 to 2.31; $I^2 = 64\%$; 303 participants; [Analysis 8.2](#)). Owing to the low number of studies included in the analysis, these results should be interpreted with caution.

A sensitivity analysis including all parallel trials was performed (MD 1.98, 95% CI 1.23 to 2.72; $I^2 = 7\%$; 9 studies; 735 participants; [Analysis 9.2](#)). Excluding cross-over studies did not lead to a substantial change in the effect estimate.

Blood lipids

Total cholesterol

Sixteen studies reported total cholesterol and were summarised in a meta-analysis of 19 comparisons. The pooled analysis showed a difference in effect between the comparison groups and favoured a low-added sugar intake (MD 0.11, 95% CI 0.01 to 0.21; $I^2 = 0\%$; 16 studies; 763 participants; low certainty of evidence; [Analysis 1.3](#)). The results were pooled with a random-effects model.

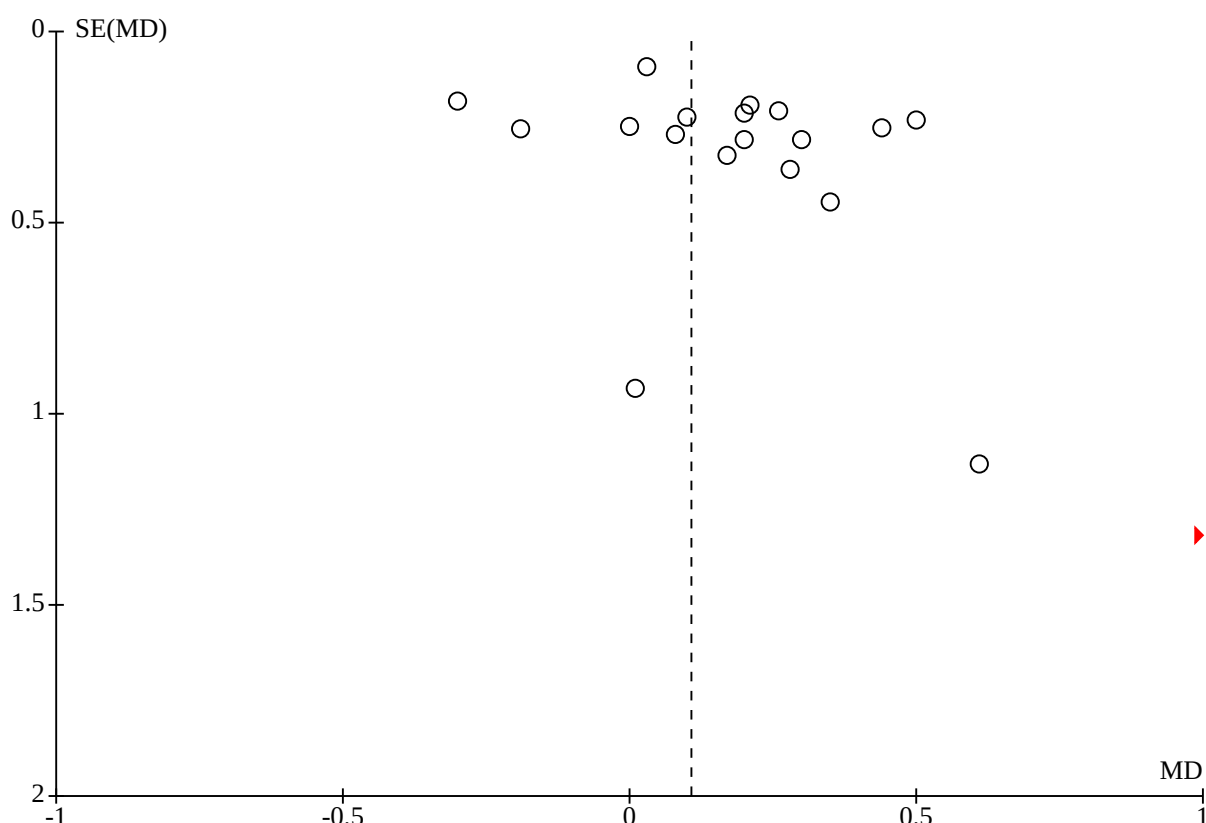
Sanchez-Delgado 2021 reported total cholesterol as median and IQR and could not be included in the pooled analysis. Total cholesterol concentrations increased in the high-added sugar intake group; no change was observed in the low-added sugar intake group.

The five studies judged at low risk of bias were included in a sensitivity analysis (MD 0.19, 95% -1.93 to 2.31; $I^2 = 15\%$; 3 studies; 159 participants; low certainty of evidence; [Analysis 8.3](#)). Owing to the low number of studies included in the analysis, these results should be interpreted with caution.

A sensitivity analysis including all parallel trials was performed (MD 0.13, 95% CI -0.02 to 0.27; $I^2 = 14\%$; 10 studies; 600 participants; low certainty of evidence; [Analysis 9.3](#)). Excluding cross-over studies did not lead to a substantial change in the effect estimate.

As no studies reported on primary outcomes, a funnel plot for total cholesterol was generated, since this analysis included the highest number of studies; [Figure 4](#). By visual inspection, the funnel plot appeared asymmetrical, suggesting the possibility of publication bias.

Figure 4. Figure 4: Funnel plot for studies that reported total cholesterol.



LDL-cholesterol

Twelve studies reported LDL-cholesterol and were summarised in a meta-analysis of 15 comparisons. The pooled analysis showed no difference in effect between comparison groups (MD 0.01, 95% CI -0.08 to 0.10; $I^2 = 20\%$; 12 studies; 712 participants; low certainty of evidence; [Analysis 1.4](#)). The results were pooled with a random-effects model.

The five studies judged at low risk of bias were included in a sensitivity analysis (MD -0.04, 95% -0.11 to 0.03; $I^2 = 0\%$, 2 studies; 87 participants; low certainty of evidence; [Analysis 8.4](#)). Owing to the low number of studies included in the analysis, these results should be interpreted with caution.

A sensitivity analysis including all parallel trials was performed (MD 0.02, 95% -0.18 to 0.22; $I^2 = 55\%$; 5 studies; 515 participants; low certainty of evidence; [Analysis 9.4](#)). Excluding cross-over studies did not lead to a substantial change in the effect estimate.

HDL-cholesterol

Sixteen studies reported HDL-cholesterol and were summarised in a meta-analysis of 19 comparisons. The pooled analysis showed no difference in effect between comparison groups (MD 0.01, 95% CI -0.02 to 0.04; $I^2 = 0\%$; 16 studies; 817 participants; low certainty of evidence; [Analysis 1.5](#)). The results were pooled with a random-effects model.

The five studies judged at low risk of bias were included in a sensitivity analysis (MD 0.01, 95% -0.07 to 0.09; $I^2 = 0\%$; 2 studies; 87

participants; low certainty of evidence; [Analysis 8.5](#)). Owing to the low number of studies included in the analysis, these results should be interpreted with caution.

A sensitivity analysis including all parallel trials was performed (MD 0.01, 95% -0.04 to 0.06; $I^2 = 22\%$; 9 studies; 620 participants; low certainty of evidence; [Analysis 9.5](#)). Excluding cross-over studies did not lead to a substantial change in the effect estimate.

Triglycerides

Sixteen studies reported on triglycerides and 14 of these were summarised in a meta-analysis of 16 comparisons. The pooled analysis showed a difference in effect between comparison groups, favouring a low-added sugar intake (MD 0.10, 95% CI 0.03 to 0.17; $I^2 = 3\%$; 14 studies; 725 participants; [Analysis 1.6](#)). The results were pooled with a random-effects model.

[Black 2006](#) reported triglycerides as a median and IQR and could not be included in the pooled analysis. Triglyceride levels were similar in both diet groups. [Sanchez-Delgado 2021](#) also reported triglycerides as a median and IQR and could not be included in the pooled analysis. A significant increase in triglyceride concentration was observed in the high-added sugar group and a decrease was observed in the low-added sugar group. [Mann 1972](#) did not report any standard deviations for triglyceride levels and was therefore not included in the meta-analysis. A significant decrease in triglyceride levels was seen in the group with low-level added sugar intake.

The five studies judged at low risk of bias were included in a sensitivity analysis (MD 0.07, 95% -0.01 to 0.16; $I^2 = 0\%$; 3 studies; 159 participants; [Analysis 8.6](#)). Owing to the low number of studies included in the analysis, these results should be interpreted with caution.

A sensitivity analysis including all parallel trials was performed (MD 0.14, 95% CI 0.01 to 0.28; $I^2 = 28\%$; 9 studies; 566 participants; [Analysis 9.6](#)). Excluding cross-over studies did not lead to a substantial change in the effect estimate.

Fasting plasma glucose

Fourteen studies reported fasting plasma glucose and were summarised in a meta-analysis consisting of 16 comparisons. No difference in effect was found between the comparison groups (MD 0.06, 95% CI -0.01 to 0.13; $I^2 = 26\%$; 14 studies; 720 participants; low certainty of evidence; [Analysis 1.7](#)). The results were pooled using a random-effects model.

[Sanchez-Delgado 2021](#) reported fasting plasma glucose as a median and IQR and could not be included in the pooled analysis. No significant differences between the two diet groups were observed.

The five studies judged at low risk of bias were included in a sensitivity analysis (MD -0.06, 95% -0.19 to 0.08; $I^2 = 0\%$; 2 studies; 87 participants; low certainty of evidence; [Analysis 8.7](#)). Owing to the low number of studies included in the analysis, these results should be interpreted with caution.

A sensitivity analysis including the parallel trials was performed (MD 0.11, 95% CI 0.00 to 0.21; $I^2 = 43\%$; 7 studies; 472 participants; low certainty of evidence; [Analysis 9.7](#)). Excluding cross-over studies did not lead to a substantial change in the effect estimate.

Adverse events

One study reported on dental health ([Maersk 2012](#)), with no events. No other trials reported on the adverse events dental health or gastrointestinal symptoms.

Subgroup analyses

No subgroup analysis was carried out on high or low-added sugar intake, since this was covered in the main analyses. No control arms used starch/refined grains and therefore no subgroup analysis was conducted on this.

Isocaloric intake or ad libitum

Testing for subgroup differences between studies with isocaloric exchange and ad libitum intake did not lead to any subgroup differences ([Analysis 2.1](#); [Analysis 2.2](#); [Analysis 2.3](#); [Analysis 2.4](#); [Analysis 2.5](#); [Analysis 2.6](#); [Analysis 2.7](#)).

Solid and liquid or liquid state

No studies investigated altered sugar intake in solid foods only and therefore the comparison was changed to solid and liquid versus liquid state.

Testing for subgroup differences between solid and liquid and liquid state did not lead to any subgroup differences ([Analysis 3.1](#); [Analysis 3.2](#); [Analysis 3.3](#); [Analysis 3.4](#); [Analysis 3.5](#); [Analysis 3.6](#); [Analysis 3.7](#)).

Difference in weight change

One study did not report the participants weight at follow-up and was therefore excluded from this subgroup analysis ([Vazquez-Duran 2016](#)).

Trials that reported a greater than 0.5 kg difference in weight change between the two study arms and the studies that reported no weight change were separated, and the test for subgroup difference suggested that high-added sugar compared with low added sugar may have a different effect on diastolic blood pressure depending on weight change ($P = 0.0004$, $I^2 = 86.6\%$; [Analysis 4.2](#)). The remainder of the subgroup analyses for difference in weight change did not lead to any subgroup differences ([Analysis 4.1](#); [Analysis 4.3](#); [Analysis 4.4](#); [Analysis 4.5](#); [Analysis 4.6](#); [Analysis 4.7](#)).

Healthy or high-risk population

For the subgroup analysis of healthy or high-risk populations, the study was categorised as high-risk if more than half of the participants met these requirements, and vice versa.

Testing for subgroup differences between healthy and high-risk populations did not lead to any subgroup differences ([Analysis 5.1](#); [Analysis 5.2](#); [Analysis 5.3](#); [Analysis 5.4](#); [Analysis 5.5](#); [Analysis 5.6](#); [Analysis 5.7](#)).

Intervention < 6 months or ≥ 6 months

Comparing the length of the intervention by dividing the trials into less than six months or six months or more for diastolic blood pressure suggested that a high or low-added sugar intake may have a different effect depending on the study duration ($P = 0.03$, $I^2 = 79.7\%$; [Analysis 6.2](#)). No differences between the subgroups were observed for the remainder of the analyses ([Analysis 6.1](#); [Analysis 6.3](#); [Analysis 6.4](#); [Analysis 6.5](#); [Analysis 6.6](#); [Analysis 6.7](#)).

Industry funding

The trials were divided according to the presence of industry funding/involvement and did not result in any subgroup differences ([Analysis 7.1](#); [Analysis 7.2](#); [Analysis 7.3](#); [Analysis 7.4](#); [Analysis 7.5](#); [Analysis 7.6](#); [Analysis 7.7](#)).

DISCUSSION

Summary of main results

This systematic review summarised 21 RCTS examining the effect of added sugar on risk factors for cardiovascular disease. None of the studies reported on cardiovascular event or all-cause mortality, the review's primary outcomes. The risk of bias was generally unclear, and only five studies were classified at an overall low risk of bias (low risk in at least four domains, not including other bias). The interventions generally compared different levels of fructose or sucrose in foods and/or beverages, with some trials using artificial sweeteners as the comparator.

The study participants who were in the high-added sugar intake group had 1.44 mmHg higher systolic blood pressure mean (MD 1.44, 95% CI 0.08 to 2.80; $I^2 = 27\%$; 14 studies; 873 participants; low certainty of evidence; [Analysis 1.1](#)), 1.52 mmHg higher diastolic blood pressure mean (MD 1.52, 95% CI 0.67 to 2.37; $I^2 = 0\%$; 13 studies; 873 participants; [Analysis 1.2](#)), 0.11 mmol/L higher total cholesterol mean (MD 0.11, 95% CI 0.01 to 0.21; $I^2 = 0\%$; 16 studies; 763 participants; low certainty of evidence; [Analysis 1.3](#)) and a

0.10 mmol/L higher triglyceride level mean (MD 0.10, 95% CI 0.03 to 0.17; $I^2 = 3\%$; 14 studies; 725 participants; [Analysis 1.6](#)) when comparing with the participants in the low-added sugar intake group. There did not appear to be an association between a high intake of added sugar and increased levels of LDL-cholesterol, HDL-cholesterol and fasting plasma glucose. Adverse events were only reported in [Maersk 2012](#), (reporting no events of impaired dental health). The evidence in the present review is uncertain and the effect was small and, therefore, the clinical relevance is uncertain.

The subgroup analyses, comparing isocaloric exchange with ad libitum intake, solid with liquid state, healthy with high-risk population and industry-funding with no involvement, did not generate any subgroup differences. Separating the trials according to weight change (less or greater than 0.5 kg change between study arms) resulted in a difference in effect for diastolic blood pressure ($P = 0.0004$, $I^2 = 86.6\%$; [Analysis 4.2](#)). The considerable heterogeneity in this comparison indicates that these results should be interpreted with caution.

A sensitivity analysis was conducted where only studies with an overall low risk of bias were included ($n = 5$). No statistically significant results appeared in the sensitivity analysis. The results should be viewed with caution due to the low number of studies included. A sensitivity analysis, where the cross-over trials ($n = 7$) and [Mann 1972](#) (as no standard deviation for reported outcomes was provided) were excluded, was also conducted. A difference in effect between high and low-added sugar intake groups was observed for systolic blood pressure (MD 1.77; 95% CI 0.05 to 3.50; $I^2 = 42\%$; 9 studies; 735 participants; low certainty of evidence; [Analysis 9.1](#)), diastolic blood pressure (MD 1.98; 95% CI 1.23 to 2.72; $I^2 = 7\%$; 9 studies; 735 participants; [Analysis 9.2](#)), triglycerides MD 0.14; 95% CI 0.01 to 0.28; $I^2 = 28\%$; 9 studies; 566 participants; [Analysis 9.6](#)) and fasting plasma glucose (MD 0.11, 95% CI 0.00 to 0.21; $I^2 = 43\%$; 7 studies; 472 participants; low certainty of evidence; [Analysis 9.7](#)), all favouring a low-added sugar intake. The statistical heterogeneity observed for diastolic blood pressure means that this result needs to be viewed with caution. The results of the sensitivity analyses did not lead to a substantial change in effect estimate for any of the outcomes.

Overall completeness and applicability of evidence

The number of trials meeting the inclusion criteria was relatively large, and allowed for analyses of our secondary outcomes. No studies reported on our primary outcomes and therefore no analyses could be conducted on these. Hence, no conclusions can be drawn regarding high versus low-added sugar intake for the primary prevention of CVDs. The conclusions drawn in this review rely entirely on results attained from analyses of the secondary outcomes. However, the secondary outcomes were chosen primarily on the basis of the role these have in the development of cardiovascular disease and even though no conclusions can be drawn regarding the primary prevention of cardiovascular disease, a decreased sugar intake seems to decrease blood pressure and lipid levels and potentially decreases the risk of future cardiovascular events.

The included trials compared different levels of added sugar or compared with a placebo on cardio-metabolic outcomes. The results showed a higher mean value on blood pressure and total cholesterol and triglycerides for the individuals having high-added

sugar intake compared with individuals who had a low-added sugar intake, but the effect was relatively small and the clinical relevance limited. Heterogeneity between the trials was a potential concern and was explored in subgroup and sensitivity analyses. No differences were observed in these analyses, apart from the subgroup analysis on weight change that resulted in a difference between the two groups for diastolic blood pressure, but with considerable heterogeneity. A concern in both the subgroup and sensitivity analyses was the small number of studies in certain groups, meaning that the results should be interpreted cautiously.

Quality of the evidence

As no trials were identified that focused on our primary outcomes, these could not be assessed in terms of quality of the evidence. For all reported outcomes in this review, the certainty of evidence was graded as low. Uncertain risk of bias for many trials and domains and indirectness of trials motivated the choice to downgrade.

The main concerns regarding risk of bias were blinding of participants (performance bias). Lack of blinding of the participants might affect the performance of the individuals. However, since none of the included outcomes in the review was based on self-administered data, a lack of blinding will likely not be as a significant issue. The study authors generally justified the lack of blinding by stating that blinding was impossible due to the nature of the intervention. Nonetheless, it is not possible to ascertain the impact on the results by not blinding the participants.

There was a lack of information provided from many studies regarding allocation concealment (selection bias) and selective reporting (reporting bias), and hence the majority of trials were rated as unclear in these domains. Allocation concealment was adequately reported in four trials. The seven trials that were registered on clinicaltrials.gov (or equivalent) were rated at a low risk of bias. The remainder of the trials did not report any prespecified study protocol, and it was therefore not possible to judge the risk of bias in this domain for those trials. Five studies were rated as at an overall low risk of bias, as these were rated as low risk in at least four domains ([Geidl-Flueck 2021](#); [Madero 2015](#); [Markey 2016](#); [Njike 2011](#); [Vazquez-Duran 2016](#)).

In addition to the risk of bias assessment, indirectness was also considered a concern when assessing the quality of the evidence. No trials reported on our primary outcomes, hence, the results of this review depended solely upon surrogate outcomes, causing indirectness. Another concern was that the trials might not be comparable due to varying interventions, populations and/or duration. Attempts were made to investigate this further in the subgroup analyses, and no major differences between subgroups were disclosed. Lastly, many studies stated industry involvement with potential vested interests. However, in a subgroup analysis comparing industry involvement with no involvement, no significant subgroup differences were noted.

Potential biases in the review process

In addition to the comprehensive search across databases which was conducted, we also went through the reference list of relevant trials to ensure that no trials were missed. We attempted to contact study authors when necessary. In order to evaluate publication bias, we created a funnel plot of the outcome with the highest

number of trials (total cholesterol). The plot indicated no presence of publication bias by visual inspection.

Four cross-over studies (Reiser 1981; Reiser 1989; Hallfrisch 1983; Gostner 2005) matched our inclusion criteria but failed to mention any wash-out period and were therefore excluded after unsuccessful attempts to contact the study authors. A summary of the results of these four studies now follows. In 1981, Reiser and colleagues studied 24 hyperinsulinaemic individuals and found a positive association between a high sucrose intake and total cholesterol, triglycerides and LDL-cholesterol. A high sugar intake was associated with lower HDL-cholesterol. Hallfrisch and colleagues compared different levels of fructose intake among 12 hyperinsulinaemic men and 12 men with normal insulin response. Total cholesterol and LDL-cholesterol increased with fructose intake, in both groups, and triglycerides increased for the hyperinsulinaemic men only. In the Reiser trial (1989), ten hyperinsulinaemic and 11 non-hyperinsulinaemic men consumed either fructose or high-amylose cornstarch following a cross-over design. Fructose intake was associated with an increase in total cholesterol, VLDL-cholesterol and triglycerides among the hyperinsulinaemics, when compared to cornstarch, and similar results were reported for the non-hyperinsulinaemics. In the Gostner trial, nineteen volunteers received 30 g of isomalt or 30 g of sucrose daily as part of a controlled diet during two four-week study periods. No effect on blood lipids or other measured outcomes was seen.

Comprehensive lifestyle interventions were excluded from the present review, in order to avoid potential confounding. This meant that trials aiming to change, for instance, diet and physical activity were not included, since it would be difficult to disentangle the effects of sugar intake from the effects of physical activity. However, this led to fewer studies being included in the review, since a multifactorial approach is fairly common in RCTs on the current topic. It was also challenging to assess the appropriateness of older studies, as these were not as comprehensive as current ones. For this reason, some studies were excluded, or only reported narratively and excluded from the meta-analysis. Other confounders that might have affected the results, that were not accounted for in this review, included family history of coronary artery disease, smoking and the presence of regular exercise (outside the scope of the intervention).

The shortest study duration that was included was four weeks, in order to include as many studies as possible. The majority of the studies were relatively short-term; only six had a duration of six months or longer. Substantial effects of an altered added sugar intake might not appear during a shorter trial, potentially contributing to lack of evidence in this review.

Agreements and disagreements with other studies or reviews

There is currently no evidence from RCTs on the effect of added sugar intake and cardiovascular disease, as is shown in the present systematic review, and the existing research on the topic is observational in nature (Tasevska 2014; Xi 2015).

A previous systematic review of RCTs found an association between high sugar intake and increased total cholesterol, triglycerides and LDL-cholesterol (Te Morenga 2014). The results of the present systematic review are in line with the conclusions presented

by Te Morenga and colleagues regarding triglycerides and total cholesterol, but no association between high-added sugar intake and LDL-cholesterol was found in this review. The results also deviated for blood pressure, as we found an association between added sugar intake and diastolic blood pressure, and Te Morenga and colleagues did not. However, the evidence in the present review is uncertain and the effect was small and, therefore, the clinical relevance is uncertain. In contrast to the present review, the review by Te Morenga and colleagues included diabetics. Te Morenga used the wider definition of free sugar and not added sugar, and the endpoint for the Te Morenga review was 2013, whereas for this review the endpoint was 2020. Nonetheless, our review is in broad agreement with the results presented by Te Morenga and colleagues as both concluded that sugar intake impacts upon blood lipids and blood pressure levels.

Another systematic review of RCTs was not able to establish an association between sugar intake and cardiovascular risk factors, when comparing free sugars with complex carbohydrates (Fattore 2017). Heterogeneity, as well as publication bias, were major concerns in the review conducted by Fattore and colleagues and the variability between the interventions made it difficult to draw conclusions. The present review, although not faced with the challenges associated with heterogeneity, struggled with indirectness that resulted in the overall quality of the evidence being rated as low.

AUTHORS' CONCLUSIONS

Implications for practice

Currently, there is no evidence regarding the effect of added sugar consumption on CV events and all-cause mortality. This review provides low-quality evidence that a low intake of added sugar intake reduces total cholesterol, triglycerides and blood pressure levels, potentially decreasing the risk of future CV events. The observed effect was small and the clinical relevance is unclear. Potential bias related to confounding factors, such as regular exercise, was not taken into consideration in this review and, given the limited data, the results should be interpreted carefully.

Implications for research

Since we were unable to detect trials conducted on the primary outcomes, more RCTs on CV events and all-cause mortality are needed. This requires trials with a longer follow-up, as most of the included trials in the present review were relatively short-term. There is also a need for well-conducted RCTs where the published papers are written in sufficient detail enabling a complete risk of bias assessment. Finally, more trials conducted without the involvement of sugar organisations or companies with potential vested interests are needed.

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* Indicates the major publication for the study

CHARACTERISTICS OF STUDIES

Characteristics of included studies [ordered by study ID]

Ahmad 2020

<i>Study characteristics</i>	
Methods	Study design: Randomised controlled trial Study grouping: Cross-over Duration of study: Two periods of 12 weeks each Wash-out period: 4 weeks
Participants	Baseline Characteristics • <i>Number of participants at baseline:</i> Not reported • <i>Number lost to follow-up:</i> Not reported • <i>Number analysed:</i> 25 • <i>Age, mean:</i> 56.8 • <i>Women, %:</i> 0 • <i>BMI, mean:</i> 28.6 Included criteria: Male sex; increased cardio-metabolic risk Excluded criteria: Not reported Pretreatment: Not reported
Interventions	Intervention Characteristics High-added sugar intake • <i>Dose:</i> 28% of total energy intake provided as sugar • <i>Type of sugar:</i> Not specified • <i>Type:</i> Participants supplied with all food items required for the intervention Low-added sugar intake • <i>Dose:</i> 9% of total energy intake provided as sugar • <i>Type of sugar:</i> Not specified • <i>Type:</i> Same
Outcomes	Secondary outcomes: systolic blood pressure, diastolic blood pressure, triglycerides, total cholesterol, HDL-cholesterol, LDL-cholesterol, fasting plasma glucose
Identification	Sponsorship source: Funded by BBSRC Country: UK Setting: Not specified Authors name: Aryati Ahmad Institution: Faculty of Health Sciences, Univeriti Sultan Zainal Abidin; Faculty of Health and Medical Sciences, University of Surrey Email: aryatiahmad@unisza.edu.my

Ahmad 2020 (Continued)

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Notes

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, but no further details provided
Allocation concealment (selection bias)	Unclear risk	No details provided
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	No details provided
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	No details provided
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	No details provided
Selective reporting (reporting bias)	Unclear risk	All stated outcomes were included in the analyses but no prespecified analysis plan provided
Other bias	Low risk	Participants were instructed to finish food portion sheets daily; monitored every two weeks

Black 2006

Study characteristics

Methods	Study design: Randomised controlled trial Study grouping: Cross-over Duration of study: Two periods of 6 weeks each Wash-out period: 4 weeks
Participants	Baseline Characteristics • Number of participants at baseline: 14 • Number lost to follow-up: 1 • Number analysed: 13 • Age, mean: 33.3 • Women, %: 0 • BMI, mean: 26.6 Included criteria: Male sex

Black 2006 (Continued)

Excluded criteria: BMI > 35; cardiac, hepatic or renal disease; a history of diabetes; blood pressure >140/80 mmHg; hyperlipidaemia (LDL > 3.0 mmol/L and triglycerides > 2.0 mmol/L)

Pretreatment: not reported

Interventions	Intervention Characteristics High-added sugar intake • <i>Dose:</i> 25% of total energy intake provided as sucrose • <i>Type of sugar:</i> Sucrose • <i>Type:</i> Participants supplied with all food items that were pre-weighed into daily portions Low-added sugar intake • <i>Dose:</i> 10% of total energy intake provided as sucrose • <i>Type of sugar:</i> Sucrose • <i>Type:</i> Same	
Outcomes	Secondary outcomes: total cholesterol, LDL-cholesterol, HDL-cholesterol, fasting plasma glucose	
Identification	Sponsorship source: Unrestricted research grant from The Sugar Bureau and Suikerstichting, the Netherlands Country: UK Setting: Not specified Authors name: R. Neil, A. Black Institution: Regional Centre for Endocrinology and Diabetes, Royal Victoria Hospital, Belfast Email: steven.hunter@royalhospitals.nhs.n-i.uk Address: Dr. Steven J. Hunter, Regional Centre for Endocrinology and Diabetes, Royal Victoria Hospital, Grosvenor Road, Belfast, U.K., BT12 6BA	
Notes		
Risk of bias		
Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Judgement Comment: Subjects were randomised in blocks of four using a random-number generator to ensure that equal numbers of volunteers received high or low-sucrose diets during the first phase.
Allocation concealment (selection bias)	Unclear risk	Judgement Comment: No information provided
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Judgement Comment: Participants were not aware of the assigned intervention. Unclear if carers/people delivering were aware
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	Judgement Comment: No information provided
Incomplete outcome data (attrition bias)	Low risk	Judgement Comment: One person left for reasons unrelated to the study. Otherwise complete follow-up

Black 2006 (Continued)

All outcomes

Selective reporting (reporting bias)	Unclear risk	Judgement Comment: All stated outcomes are included in the analyses but no prespecified analysis plan provided
Other bias	Low risk	Daily visits by study personnel to ensure dietary compliance

Campos 2015

Study characteristics

Methods	Study design: Randomised controlled trial Study grouping: Parallel-group Duration of study: 4 weeks run-in period + 12 weeks
Participants	Baseline Characteristics High-added sugar intake - Number of participants at baseline: 13 - Number lost to follow-up: 0 - Number analysed: 13 - Age, mean: Not reported - Women, %: 42.9 - BMI, mean: 30.6 Low-added sugar intake - Number of participants at baseline: 18 - Number lost to follow-up: 4 - Number analysed: 14 - Age, mean: Not reported - Women, %: 53.8 - BMI, mean: 31.5 Overall - Number of participants at baseline: 31 - Number lost to follow-up: 4 - Number analysed: 27 - Age, mean: Not reported - Women, %: 48.1 - BMI, mean: 31.1 Included criteria: Male and female; BMI greater than 25 kg/m ² ; daily consumption of two or more 22-oz SSBs Excluded criteria: Not reported Pretreatment: Two groups (IHCL < 60 & > 60) which were significantly different at baseline
Interventions	Intervention Characteristics High-added sugar intake

Campos 2015 (Continued)

- *Dose:* Habitual intake of SSBs
- *Type of sugar:* Sucrose
- *Type:* Participants received beverages on a weekly basis. Otherwise, diet and beverage intake ad libitum

Low-added sugar intake

- *Dose:* Replace habitual intake of SSBs with ASBs
- *Type of sugar:* Artificial sweeteners
- *Type:* Same

Outcomes	Secondary outcomes: Systolic blood pressure, diastolic blood pressure, total cholesterol, HDL-cholesterol, triglycerides, fasting plasma glucose	
Identification	Sponsorship source: The Swiss National Foundation for Science the Fondation Raymond Berger pourla recherché sur le diabéte et les maladies métaboliques, Lausanne, Switzerland Country: Switzerland Setting: Not reported Authors name: Vanessa Campos Institution: Department of Physiology, Faculty of Biology and Medicine, University of Lausanne, Lau- sanne Email: luc.tappy@unil.ch Address: Not reported	
Notes		
Risk of bias		
Bias	Authors' judgement	Support for judgement
Random sequence genera- tion (selection bias)	Unclear risk	Judgement Comment: The study stated that randomisation was stratified by sex, but the method was not described in more detail.
Allocation concealment (selection bias)	Unclear risk	Judgement Comment: No information provided
Blinding of participants and personnel (perfor- mance bias) All outcomes	Unclear risk	Judgement Comment: No information provided
Blinding of outcome as- sessment (detection bias) All outcomes	Unclear risk	Judgement Comment: No information provided
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Judgement Comment: Individuals lost to follow-up in ASB group only, no rea- sons for attrition given
Selective reporting (re- porting bias)	Low risk	Judgement Comment: Study protocol registered on clinicaltrials.gov. No iden- tified major deviations between protocol and published article
Other bias	Low risk	Participants returned packages in order to ensure dietary compliance.

Ebbeling 2020

Study characteristics

Methods	Study design: Randomised controlled trial Study grouping: Parallel-group Duration of study: 12 months
Participants	Baseline Characteristics High-added sugar intake • <i>Number of participants at baseline:</i> 67 • <i>Number lost to follow-up:</i> 7 • <i>Number analysed :</i> 60 • <i>Age, mean:</i> 25.9 • <i>Women, %:</i> 40.3 • <i>BMI, mean:</i> 25.8 Low-added sugar intake • <i>Number of participants at baseline:</i> 69 • <i>Number lost to follow-up:</i> 3 • <i>Number analysed :</i> 66 • <i>Age, mean:</i> 27.9 • <i>Women, %:</i> 40.6 • <i>BMI, mean:</i> 26.6 Overall • <i>Number of participants at baseline:</i> 136 • <i>Number lost to follow-up:</i> 10 • <i>Number analysed:</i> 126 • <i>Age, mean:</i> 26.9 • <i>Women, %:</i> 40.4 • <i>BMI, mean:</i> 26.2 Included criteria: Adults aged 18–40 years, consumed at least 1 serving (12 fl oz) per day of SSB and had a BMI between 18.5 to 40.0 Excluded criteria: Fasting blood glucose ≥ 110 mg/dL, physician diagnosis of a major medical or psychiatric illness, chronic use of any medication that could affect ≥ 1 study outcomes, and heavy smoking (> 10 cigarettes/d). Additional exclusion criteria for females included: pregnancy (preceding 12 months) or plans to become pregnant during the study period, lactation (preceding 3 months), and change in hormonal contraceptives (preceding 3 months). Pretreatment: Not reported
Interventions	Intervention Characteristics High-added sugar intake • <i>Dose:</i> Same intake level of sugar-sweetened beverages as determined at baseline • <i>Type of sugar:</i> Sucrose • <i>Type:</i> Home deliveries of free beverages Low-added sugar intake

Ebbeling 2020 (Continued)

- *Dose:* Sugar-sweetened beverages replaced with unsweetened beverages
- *Type of sugar:* None
- *Type:* Home deliveries of free beverages

Outcomes	Secondary outcomes: Systolic blood pressure, diastolic blood pressure, LDL-cholesterol, HDL-cholesterol, fasting plasma glucose	
Identification	Sponsorship source: Grants from National Heart, Lung and Blood Institute; National Institute for Diabetes and Kidney Diseases; the National Center for Research Resources; the Harvard Catalyst Clinical and Translational Science Center; the New Balance Foundation Country: USA Setting: Boston's Children Hospital Authors name: Cara B. Ebbeling Institution: New Balance Foundation Obesity Prevention Center Email: cara.ebbeling@childrens.harvard.edu Address: 300 Longwood Avenue, BCH 3179, Boston, MA 02115	
Notes	The trial consisted of three arms. The SSB and USB arms were used for this review; the ASB arm was not included.	
Risk of bias		
Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Judgement Comment: No details provided
Allocation concealment (selection bias)	Unclear risk	Judgement Comment: No details provided
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement Comment: Study participants were not blinded.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Judgement Comment: Personnel who assessed study outcomes were masked to group assignment.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement Comment: Reasons for dropouts were included and analyses were conducted on dropouts that determined no significant differences from completers of the study.
Selective reporting (reporting bias)	Unclear risk	Judgement Comment: According to the original proposal, insulin secretion was specified as an effect modifier; however, because of scheduling challenges and the need to streamline study visits, insulin secretion was not measured.
Other bias	Low risk	Study personnel made phone calls every two weeks and sent out intervention messages to ensure dietary compliance.

Geidl-Flueck 2021

Study characteristics

Methods	Study design: Randomised controlled trial Study grouping: Parallel-group Duration of study: 7 weeks
Participants	Baseline Characteristics High-added sugar intake • <i>Number of participants at baseline:</i> 24 • <i>Number lost to follow-up:</i> 1 • <i>Number analysed:</i> 23 • <i>Age, mean:</i> Not reported • <i>Women, %:</i> 0 • <i>BMI, mean:</i> 22.9 Low-added sugar intake • <i>Number of participants at baseline:</i> 24 • <i>Number lost to follow-up:</i> 0 • <i>Number analysed:</i> 24 • <i>Age, mean:</i> Not reported • <i>Women, %:</i> 0 • <i>BMI, mean:</i> 21.0 Overall • <i>Number of participants at baseline:</i> 48 • <i>Number lost to follow-up:</i> 1 • <i>Number analysed:</i> 47 • <i>Age, mean:</i> 22.7 • <i>Women, %:</i> 0 • <i>BMI, mean:</i> 21.8 Included criteria: Male sex; age 18-30 years; BMI < 24 Excluded criteria: High SSB consumption (exceeding CHO 60 g/day); more than 3 hours of physical activity per week
Interventions	Intervention Characteristics High-added sugar intake • <i>Dose:</i> 2 dL SSB containing 13.3 g/dL sucrose three times daily • <i>Type of sugar:</i> Sucrose • <i>Type:</i> Participants provided with SSBs Low-added sugar intake • <i>Dose:</i> SSB abstinence • <i>Type of sugar:</i> No sugar or artificial sweeteners • <i>Type:</i> Participants required to abstain from all SSB consumption
Outcomes	Secondary outcomes: systolic blood pressure, diastolic blood pressure

Geidl-Flueck 2021 (Continued)

Identification

Sponsorship source: funded by the Swiss National Science Foundation (No. 144204), Switzerland, Vontobel Foundation, Switzerland, Rhyner Bangerter Foundation, Switzerland, Promedica Foundation, Lichtenstein, Uniscientia Foundation, Lichtenstein, Prof. Otto Beisheim Foundation, Switzerland, Jubiläumsstiftung Swiss Life, Switzerland, Philhuman Foundation, Lichtenstein and the Austrian Federal Ministry of Education (No. BMWFW-10.420/0005-WF/V/3c/2017), Austria

Country: Switzerland

Setting: Not reported

Authors name: Bettina Geidl-Flueck

Institution: Department of Endocrinology, Diabetology and Clinical Nutrition, University Hospital Zurich (USZ) and University of Zurich (UZH), Switzerland

Email: Bettina.geidl@usz.ch

Address: University Hospital Zurich, Raemistrasse 100, 8091 Zurich, Switzerland

Notes

The trial consisted of four arms. The control and sucrose arms were used for this review; the glucose and fructose arms were not included.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Individuals randomised using simple random allocation
Allocation concealment (selection bias)	Low risk	The randomisation method used indicated allocation concealment.
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Double-blinded, SSB provided in coded containers
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	No details provided
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Similar dropout rates across the groups. Reasons for attrition not provided for all participants, therefore, unable to fully judge
Selective reporting (reporting bias)	Low risk	Study protocol registered on clinicaltrials.gov. No identified major deviations between protocol and published article
Other bias	Low risk	Participants had to return empty SSB containers and not consumed SSBs and keep SSB records.

Johnson 2015

Study characteristics

Methods

Study design: Randomised controlled trial

Study grouping: Parallel-group

High versus low-added sugar consumption for the primary prevention of cardiovascular disease (Review)

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Johnson 2015 (Continued)

Duration of study: 8 weeks

Participants	Baseline Characteristics High-added sugar intake • <i>Number of participants at baseline:</i> 31 • <i>Number lost to follow-up:</i> 7 • <i>Number analysed:</i> 24 • <i>Age, mean:</i> 29 • <i>Women, %:</i> 100 • <i>BMI, mean:</i> 43 Low-added sugar intake • <i>Number of participants at baseline:</i> 30 • <i>Number lost to follow-up:</i> 3 • <i>Number analysed:</i> 27 • <i>Age, mean:</i> 29 • <i>Women, %:</i> 100 • <i>BMI, mean:</i> 44 Overall • <i>Number of participants at baseline:</i> 61 • <i>Number lost to follow-up:</i> 10 • <i>Number analysed:</i> 51 • <i>Age, mean:</i> 29.0 • <i>Women, %:</i> 100 • <i>BMI, mean:</i> 43.5 Included criteria: Women; 18-40 years old; morbidly obese (BMI > 40); diagnosed with polycystic ovarian syndrome (PCOS) Excluded criteria: Not reported Pretreatment: There were no significant differences in baseline characteristics in the two groups except for FSH and LH-levels and hirsutism. The withdrawal rate did not differ significantly between groups (23% vs 10%, P = 0.301).
Interventions	Intervention Characteristics High-added sugar intake • <i>Dose:</i> 8 flavoured meal replacement shakes per day and selected vegetables and one fruit. In total, 85 g fructose/day • <i>Type of sugar:</i> Fructose • <i>Type:</i> Participants given detailed instructions on permitted foods and beverages Low-added sugar intake • <i>Dose:</i> Whole grain crisp bread and low-fat, high-protein products for 3 out of 4 meals per day. In total, 17 g fructose/day • <i>Type of sugar:</i> Fructose • <i>Type:</i> Same
Outcomes	Secondary outcomes: systolic blood pressure, diastolic blood pressure, total cholesterol, LDL-cholesterol, HDL-cholesterol, triglycerides, fasting plasma glucose

Johnson 2015 (Continued)

Identification

Sponsorship source: One author was supported by a research fellowship grant from the Norwegian National Advisory Unit on Women's Health, Oslo University Hospital Rikshospitalet.

Country: Norway

Setting: Not stated

Authors name: Line K Johnson

Institution: Morbid Obesity Centre

Email: line.kristin.johnson@siv.no

Address: Vestfold Hospital Trust, PO Box 2168, 3103 Tønsberg

Notes

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Judgement Comment: Randomised in blocks. Sealed pre-numbered containers were opened by the research secretary.
Allocation concealment (selection bias)	Low risk	Judgement Comment: Group allocation was revealed to participant and staff member (registered dietitian or study nurse) after all baseline examinations and measures were completed.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement Comment: Participants and personnel were not blinded.
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	Judgement Comment: No information provided
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Judgement Comment: No significant difference in withdrawal rate ($P = 0.3$). Four participants switched group in the first two weeks of the trial, but were still assessed according to their original group. No reasons for attrition provided
Selective reporting (reporting bias)	Low risk	Judgement Comment: The study protocol was published at clinicaltrials.gov and identifier included in the published article. No major deviations from the study protocol could be identified.
Other bias	Low risk	Study personnel made phone calls every week to ensure dietary compliance.

Lewis 2013

Study characteristics

Methods

Study design: Randomised controlled trial

Study grouping: Cross-over

Duration of study: Two periods of 6 weeks

Lewis 2013 (Continued)

Wash-out period: 4 weeks

Participants	Baseline Characteristics • <i>Number of participants at baseline:</i> 17 • <i>Number lost to follow-up:</i> 4 • <i>Number analysed:</i> 13 • <i>Age, mean:</i> 46.1 • <i>Women, %:</i> 30.8 • <i>BMI, mean:</i> 31.7 Included criteria: Over 18 years of age; BMI 25-35 Excluded criteria: Significant cardiac, renal or hepatic disease; a diagnosis of diabetes; pregnancy and breastfeeding Pretreatment: Not reported
Interventions	Intervention Characteristics High-added sugar intake • <i>Dose:</i> 15% sucrose of total daily energy intake • <i>Type of sugar:</i> Sucrose • <i>Type:</i> Participants were provided with all food and drinks with instructions on preparation and cooking. Low-added sugar intake • <i>Dose:</i> 5% sucrose of total daily energy intake • <i>Type of sugar:</i> Sucrose • <i>Type:</i> Participants were provided with all food and drinks with instructions on preparation and cooking.
Outcomes	Secondary outcomes: systolic blood pressure, diastolic blood pressure, total cholesterol, LDL-cholesterol, HDL-cholesterol, fasting plasma glucose
Identification	Sponsorship source: Sugar Nutrition UK Country: UK Setting: Not stated Authors name: Anthony S. Lewis Institution: Regional Centre for Endocrinology and Diabetes Email: aslewis2000@yahoo.co.uk Address: Royal Victoria Hospital, Grosvenor Road, Belfast, BT12 6BA, Northern Ireland
Notes	
Risk of bias	
Bias	Authors' judgement Support for judgement
Random sequence generation (selection bias)	Low risk Judgement Comment: Subjects were randomised in blocks of four using a random number generator to ensure equal numbers of subjects received high and low-sucrose in the first dietary period.

Lewis 2013 (Continued)

Allocation concealment (selection bias)	Unclear risk	Judgement Comment: No information provided
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Judgement Comment: No information provided
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	Judgement Comment: No information provided
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Judgement Comment: Reasons for attrition were provided, but we were unable to fully judge.
Selective reporting (reporting bias)	Unclear risk	Judgement Comment: All stated outcomes were included in the analyses but no prespecified analysis plan provided
Other bias	Low risk	Study personnel made phone calls every 2-3 days to ensure dietary compliance.

Lowndes 2014a
Study characteristics

Methods	Study design: Randomised controlled trial Study grouping: Parallel-group Duration of study: 10 weeks
Participants	Baseline Characteristics High-added sugar intake (HFCS) - Number of participants at baseline: 78 - Number lost to follow-up: 27 - Number analysed: 51 - Age, mean: 43.4 - Women, %: 55 - BMI, mean: 27.9 Low-added sugar intake (HFCS) - Number of participants at baseline: 83 - Number lost to follow-up: 14 - Number analysed: 69 - Age, mean: 38.9 - Women, %: 61 - BMI, mean: 27.1 High-added sugar intake (sucrose) - Number of participants at baseline: 73 - Number lost to follow-up: 20

Lowndes 2014a (Continued)

- *Number analysed:* 53
- *Age, mean:* 38.9
- *Women, %:* 49
- *BMI, mean:* 26.2

Low-added sugar intake (sucrose)

- *Number of participants at baseline:* 77
- *Number lost to follow-up:* 19
- *Number analysed :* 58
- *Age, mean:* 38.6
- *Women, %:* 45
- *BMI, mean:* 27.6

Overall

- *Number of participants at baseline:* 311
- *Number lost to follow-up:* 80
- *Number analysed:* 231
- *Age, mean:* 40.0
- *Women, %:* 53
- *BMI, mean:* 27.2

Included criteria: Men and women; aged 20–60 years; BMI 23–35 kg/m²

Excluded criteria: Currently enrolled in a commercial weight-loss programme or taking weight-loss medication or supplements; experienced a change in weight of greater than 3% of current body weight in the past 30 days; were currently using tobacco products or had utilised such products within the past year; had uncontrolled hypertension ($\geq 140/90$ mmHg); were diagnosed with type 1 or type 2 diabetes; had thyroid disease and either not taking medication or the dosage changed within the past six months; a history of surgical procedure for weight loss; major surgery within three months of enrolment; a history of heart problems (e.g., bypass surgery, myocardial infarction, etc.) within three months of enrolment; orthopaedic limitations that would interfere with regular physical activity; if they had a history of gastrointestinal disorders or diagnosed eating disorders, a history or the presence of cancer or congestive heart failure; women who were pregnant, lactating or trying to become pregnant; taking any prescription medication for less than three months; any individual consuming ≥ 14 alcoholic drinks per week; a known allergy to SUC or HFCS; participation in another clinical trial within 30 days prior to enrolment; lactose intolerance or any other significant food allergy

Pretreatment: Not reported

Interventions	Intervention Characteristics
	High-added sugar intake (HFCS) • <i>Dose:</i> 30% of calories required for weight maintenance provided by added HFCS in milk • <i>Type of sugar:</i> High-fructose corn syrup • <i>Type:</i> Participants provided with low-fat sweetened milk on a weekly basis Low-added sugar intake (HFCS) • <i>Dose:</i> 8% of calories required for weight maintenance provided by added HFCS in milk • <i>Type of sugar:</i> High-fructose corn syrup • <i>Type:</i> Same High-added sugar intake (sucrose) • <i>Dose:</i> 30% of calories required for weight maintenance provided by added sucrose in milk • <i>Type of sugar:</i> Sucrose • <i>Type:</i> Same

Lowndes 2014a (Continued)

Low-added sugar intake (sucrose)

- *Dose:* 8% of calories required for weight maintenance provided by added sucrose in milk
- *Type of sugar:* Sucrose
- *Type:* Same

Outcomes	Secondary outcomes: Systolic blood pressure, diastolic blood pressure, total cholesterol, HDL-cholesterol, LDL-cholesterol, triglycerides, fasting plasma glucose
Identification	Sponsorship source: Unrestricted grant funding, unclear source Country: USA Setting: Not stated Authors name: Joshua Lowndes Institution: Rippe Lifestyle Institute Email: jlowndes@rippelifestyle.com Address: 215 Celebration Place, Suite 300, Celebration, FL 34747, USA
Notes	The trials consisted of six arms. The 8% sucrose, 8% HFCS, 30% sucrose and 30% HFCS were included in the review; the 18% HFCS and 18% sucrose arms were not included.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Judgement Comment: Randomisation process not described
Allocation concealment (selection bias)	Unclear risk	Judgement Comment: No details provided.
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Judgement Comment: Participants were blinded. Personnel were aware of the percentage of sugar but not if the participant received fructose or sucrose.
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	Judgement Comment: No information provided
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement Comment: Similar dropout rate across all groups. All participants who completed the intervention were analysed. Reasons for attrition provided
Selective reporting (reporting bias)	Unclear risk	Judgement Comment: All outcomes reported but no prespecified analysis plan
Other bias	Low risk	Participants handed in self-reported checklists every week.

Lowndes 2014b

Study characteristics

Lowndes 2014b (Continued)

Methods

Study design: Randomised controlled trial

Study grouping: Parallel-group

Duration of study: 10 weeks

Participants

Baseline Characteristics

High-added sugar intake (sucrose)

- *Number of participants at baseline:* Not reported
- *Number lost to follow-up:* Not reported
- *Number analysed:* 13
- *Age, mean:* 36.48
- *Women, %:* 38
- *BMI, mean:* 29.30

Low-added sugar intake (sucrose)

- *Number of participants at baseline:* Not reported
- *Number lost to follow-up:* Not reported
- *Number analysed:* 18
- *Age, mean:* 41.15
- *Women, %:* 67
- *BMI, mean:* 27.96

High-added sugar intake (HFCS)

- *Number of participants at baseline:* Not reported
- *Number lost to follow-up:* Not reported
- *Number analysed:* 17
- *Age, mean:* 39.33
- *Women, %:* 47
- *BMI, mean:* 29.43

Low-added sugar intake (HFCS)

- *Number of participants at baseline:* Not reported
- *Number lost to follow-up:* Not reported
- *Number analysed:* 17
- *Age, mean:* 39.82
- *Women, %:* 53
- *BMI, mean:* 27.79

Overall

- *Number of participants at baseline:* 86
- *Number lost to follow-up:* 21
- *Number analysed:* 65
- *Age, mean:* 39.12
- *Women, %:* 52
- *BMI, mean:* 28.58

Included criteria: Overweight/obese subjects between the ages of 25 and 60 years of age with Body Mass Index (BMI) 27.0–35.0 kg/m².

Excluded criteria: Current enrolment in any commercial weight-loss programme, prescription medicines or supplements for weight loss, or a greater than five pound change in weight during the past

Lowndes 2014b (Continued)

three months. Individuals with a history of orthopaedic limitations that would interfere with the ability to meet prescribed exercise, a history of heart problems, major surgery within the last three months, clinically diagnosed eating disorders or any gastrointestinal disorder, dietary restrictions or allergies to any component of the diet or which would limit the ability to adhere to dietary requirements of the study were all excluded. Users of tobacco products or individuals consuming more than 14 alcoholic beverages per week were also excluded.

Pretreatment: Not reported

Interventions	Intervention Characteristics High-added sugar intake (sucrose) <i>Dose:</i> 20% of total calories consumed daily <i>Type of sugar:</i> Sucrose <i>Type:</i> Participants provided with eating plan and given milk sweetened with sucrose Low-added sugar intake (sucrose) <i>Dose:</i> 10% of total calories consumed daily <i>Type of sugar:</i> Sucrose <i>Type:</i> Same High-added sugar intake (HFCS) <i>Dose:</i> 20% of total calories consumed daily <i>Type of sugar:</i> HFCS <i>Type:</i> Participants provided with eating plan and given milk sweetened with HFCS Low-added sugar intake (HFCS) <i>Dose:</i> 10% of total calories consumed daily <i>Type of sugar:</i> HFCS <i>Type:</i> Same	
Outcomes	Secondary outcomes: systolic blood pressure, diastolic blood pressure, total cholesterol, LDL-cholesterol, HDL-cholesterol, triglycerides	
Identification	Sponsorship source: Unrestricted grant funding, unclear source. Country: USA Setting: Not reported Authors name: Joshua Lowndes Institution: Rippe Lifestyle Institute Email: jrippe@rippelifestyle.com Address: 215 Celebration Place, Suite 300, Celebration, FL 34747, USA	
Notes		
Risk of bias		
Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Judgement Comment: No details reported

Lowndes 2014b (Continued)

Allocation concealment (selection bias)	Unclear risk	Judgement Comment: No details reported
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Judgement Comment: Participants and personnel were blinded to the type of sugar. Participants were also blinded to the amount of sugar they consumed.
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	No information provided
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Judgement Comment: Reasons for attrition were given but no information on number of dropouts in each group
Selective reporting (reporting bias)	Unclear risk	Judgement Comment: All outcomes reported but no prespecified analysis plan
Other bias	Low risk	Participants handed in self-reported checklists every week.

Madero 2015

Study characteristics

Methods	Study design: Randomised controlled trial Study grouping: Parallel-group Duration of study: 4 weeks (study consisted of two phases: diet and diet + allopurinol. Here, only the first phase was included).
Participants	Baseline Characteristics High-added sugar intake - Number of participants at baseline: 44 - Number lost to follow-up: 6 - Number analysed: 38 - Age, mean: 48 - Women, %: 57.9 - BMI, mean: Not reported Low-added sugar intake - Number of participants at baseline: 34 - Number lost to follow-up: 0 - Number analysed: 34 - Age, mean: 44 - Women, %: 64.7 - BMI, mean: Not reported Overall - Number of participants at baseline: 78 - Number lost to follow-up: 6

Madero 2015 (Continued)

- *Number analysed:* 72
- *Age, mean:* 46
- *Women, %:* 61.3
- *BMI, mean:* Not reported

Included criteria: 18 years or older; overweight (BMI > 25 kg/m²) or obese (BMI > 30 kg/m²); systolic blood pressure > 120 to 140 mmHg and diastolic 80 to 90 mmHg; with a history of high-fructose consumption from sources of added sugar (excluding fruits) of > 70 g/d

Excluded criteria: History of diabetes, chronic kidney disease, liver disease or abnormal liver function tests, haematologic abnormalities, malignancy, taking any medications, or pregnant. Patients were eliminated from the study if during the study period they required any medications, if the BP during the study period became greater than 160/100 mmHg, fasting blood glucose greater than 200 mg/dL, low-density lipoprotein cholesterol greater than 190 mg/dL, triglycerides greater than 500 mg/dL, serum uric acid levels less than 2 mg/dL or greater than 12 mg/dL, became pregnant, had severe disease in the study period (infectious, neoplastic, cardiovascular, and neurologic), or required hospitalisation for medical or surgical causes. In addition, patients were eliminated from the study if they had adverse clinical or biochemically derived from the study medication (allergic reaction, decreased cell blood count, or elevated liver function tests).

Pretreatment: Not reported

Interventions	Intervention Characteristics High-added sugar intake • <i>Dose:</i> Diet as usual with reduced sodium intake • <i>Type of sugar:</i> Fructose • <i>Type:</i> Participants provided with a meal plan and instructed to keep a food diary Low-added sugar intake • <i>Dose:</i> Isocaloric low-fructose and sodium diet • <i>Type of sugar:</i> Fructose • <i>Type:</i> Same
Outcomes	Secondary outcomes: systolic blood pressure, diastolic blood pressure, total cholesterol, triglycerides
Identification	Sponsorship source: Country: Mexico Setting: Instituto Nacional de Cardiología Ignacio Chávez Authors name: Magdalena Madero Institution: Division of Nephrology and Renal Pathophysiology, National Heart Institute of Mexico, Mexico City Email: madero.magdalena@gmail.com Address: Juan Badiano no 1, Col Seccion XVI, Delegacion Tlalpan, Mexico City 14080, Mexico
Notes	
Risk of bias	
Bias	Authors' judgement Support for judgement
Random sequence generation (selection bias)	Low risk Judgement Comment: Randomised by using a random number table

Madero 2015 (Continued)

Allocation concealment (selection bias)	Unclear risk	Judgement Comment: No information provided
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Judgement Comment: No information provided
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Judgement Comment: "To evaluate consistency of measurement, randomized measures were done blinded by a person external to the trial."
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement Comment: More dropouts in the control group, with reasons for attrition provided. However, more individuals in the control group at baseline, so similar number of individuals analysed in both groups
Selective reporting (reporting bias)	Low risk	Judgement Comment: Study protocol published on clinicaltrials.gov; even though the study design had been modified, the stated outcomes remained unchanged.
Other bias	Low risk	Participants were instructed to record their food and beverage intake at least once a week in a food diary.

Maersk 2012
Study characteristics

Methods	Study design: Randomised controlled trial Study grouping: Parallel-group Duration of study: 6 months
Participants	Baseline Characteristics High-added sugar intake - Number of participants at baseline: 15 - Number lost to follow-up: 5 - Number analysed: 10 - Age, mean: 39 - Women, %: 40 - BMI, mean: 31.3 Low-added sugar intake - Number of participants at baseline: 17 - Number lost to follow-up: 4 - Number analysed: 13 - Age, mean: 39 - Women, %: 62 - BMI, mean: 32.2 Overall - Number of participants at baseline: 32

Maersk 2012 (Continued)

- Number lost to follow-up: 9
- Number analysed: 23
- Age, mean: 39
- Women, %: 52.2
- BMI, mean: 31.8

Included criteria: BMI 26-40; aged 20-50; blood pressure < 160/100 mmHg

Excluded criteria: Diabetes

Pretreatment: There was an unequal distribution of sex (i.e. more men in the regular cola group) and a tendency for differences in visceral adipose tissue, skeletal muscle fat, and liver fat ($P < 0.1$) between the groups.

Interventions	Intervention Characteristics High-added sugar intake - Dose: 1 litre Coca-Cola - Type of sugar: Sucrose - Type: Provided with beverages free of charge Low-added sugar intake - Dose: 1 litre still mineral water - Type of sugar: None - Type: Provide with beverages free of charge
Outcomes	Secondary outcomes: total cholesterol, HDL-cholesterol, triglycerides, fasting plasma glucose, adverse events: impaired dental health
Identification	Sponsorship source: Supported by grants from The Danish Council for Strategic Research, The Food Study Group/Danish Ministry of Food, Agriculture and Fisheries, Novo Nordic Foundation, and Clinical Institute at Aarhus University, Denmark. The semi-skim milk was donated by the Danish Dairy Company, Arla Foods, but without any influence on the design, interpretation, or conclusions of the study. Country: Denmark Setting: Aarhus University Hospital and Department of Nutrition, Copenhagen University Authors name: Maria Maersk Institution: Department of Endocrinology and Internal Medicine Email: bjoern.richelsen@aarhus.rm.dk Address: Aarhus University Hospital, Tage Hansensgade 2, DK-8000 Aarhus, Denmark
Notes	The trial consisted of four arms. The sucrose-sweetened regular cola and water arms were used for this review; the milk and aspartame-sweetened diet cola arms were not included.
Risk of bias	
Bias	Authors' judgementSupport for judgement
Random sequence generation (selection bias)	Unclear riskNo information provided
Allocation concealment (selection bias)	Unclear riskJudgement Comment: No information provided

Maersk 2012 (Continued)

Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement Comment: Unblinded
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	Judgement Comment: No information provided
Incomplete outcome data (attrition bias) All outcomes	High risk	Judgement Comment: 9/32 = 28.1% dropout rate, similar proportion in both groups. Unclear reasons for dropout
Selective reporting (reporting bias)	Unclear risk	Judgement Comment: All stated outcomes were included in the analyses but no prespecified analysis plan provided
Other bias	Low risk	Participants were asked to bring empty bottles or cartons to the research centre for counting every 3–4 weeks to ensure dietary compliance. They also filled out a 7-day dietary record at baseline, 3 months and at follow-up.

Maki 2015

Study characteristics

Methods	Study design: Randomised controlled trial Study grouping: Cross-over Duration of study: 2 periods of 6 weeks each Wash-out period: 2 weeks
Participants	Baseline Characteristics - Number of participants at baseline: 43 - Number lost to follow-up: 9 - Number analysed: 34 - Age, mean: 53.8 - Women, %: 50 - BMI, mean: 32.2 Included criteria: Men and women; 18–74 years old; with BMI < 45.0 kg/m²; waist circumference of > 83.8 cm (33.0 inches) for women and > 91.4 cm (36.0 inches) for men; self-reported use of high-sugar beverages; high risk of the development of T2DM; willing to maintain a stable body weight; maintain their habitual diet; limit non-study-related dairy product intake Excluded criteria: Uncontrolled hypertension (systolic blood pressure ≥160 mmHg or diastolic blood pressure ≥100 mmHg); abnormal laboratory test results of clinical importance (at the discretion of the investigator), including TGs ≥400 mg/dL, fasting creatinine ≥1.5 mg/dL, and fasting blood glucose ≥126 mg/dL; a history or presence of clinically important cardiac (including coronary artery disease), renal, hepatic, endocrine (including type 1 diabetes mellitus or T2DM), pulmonary, biliary, gastrointestinal, pancreatic, or neurologic disorders; a recent history or presence of cancer, except non-melanoma skin cancer; weight loss or gain of > 4.5 kg in the month before screening or extreme dietary habits (e.g. very-low carbohydrate diet); and a history within the last 12 mo or strong potential for alcohol abuse (> 14 servings/wk) or substance abuse; Individuals who reported habitual consumption of > 4 servings/d of dairy foods and beverages; use of medications or supplements known to alter carbohydrate or lipid

Maki 2015 (Continued)

metabolism; using weight-loss drugs or programmes within 4 wks before the screening visit; antibiotics used within 4 wks before the screening visit

Pretreatment: Not reported

Interventions	Intervention Characteristics High-added sugar intake • <i>Dose:</i> 710 mL/d of non-diet soda and 108 g/d of nondairy pudding containing 39 g and 14 g of sugar, respectively • <i>Type of sugar:</i> Sugar-sweetened products • <i>Type:</i> Participants provided with study products, otherwise diet ad libitum Low-added sugar intake • <i>Dose:</i> 474 mL/d of 2% milk and 170 g/d low-fat yogurt with no added sugar containing 12 g and 14 g of sugar respectively • <i>Type of sugar:</i> None • <i>Type:</i> Same
Outcomes	Secondary outcomes: total cholesterol, LDL-cholesterol, HDL-cholesterol, triglycerides, fasting plasma glucose
Identification	Sponsorship source: Dairy Research Institute/National Dairy Council (Rosemont, IL) Country: USA Setting: Not stated Comments: Authors name: Kevin C Maki Institution: Biofortis Clinical Research, Addison, IL and Midwest Center for Metabolic and Cardiovascular Research, Glen Ellyn, IL Email: kmaki@mc-mcr.com Address: Not reported
Notes	
Risk of bias	
Bias	Authors' judgement Support for judgement
Random sequence generation (selection bias)	Low risk Judgement Comment: A randomisation list was created with the use of SAS with the seed number recorded.
Allocation concealment (selection bias)	Low risk Judgement Comment: A series of envelopes were created, with each envelope showing the subject's treatment sequence (dairy products then SSPs or SSPs then dairy products) and opened by the study co-ordinator at the time of random assignment.
Blinding of participants and personnel (performance bias) All outcomes	High risk Judgement Comment: Participants and personnel not blinded

Maki 2015 (Continued)

Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	Judgement Comment: No information provided
Incomplete outcome data (attrition bias) All outcomes	High risk	Judgement Comment: Unequal number of individuals lost to follow-up depending on randomisation sequence (8 vs. 2) and only nonspecific reasons for attrition provided
Selective reporting (reporting bias)	Unclear risk	Judgement Comment: All outcomes reported but no prespecified analysis plan
Other bias	Low risk	Dietary compliance was based on returned food items.

Mann 1972
Study characteristics

Methods	Study design: Randomised controlled trial Study grouping: Parallel-group Duration of study: 22 weeks
Participants	Baseline Characteristics High-added sugar intake - Number of participants at baseline: 17 - Number lost to follow-up: 0 - Number analysed: 17 - Age, mean: Not reported - Women, %: 0 - BMI, mean: Not reported Low-added sugar intake - Number of participants at baseline: 17 - Number lost to follow-up: 0 - Number analysed: 17 - Age, mean: Not reported - Women, %: 0 - BMI, mean: Not reported Overall - Number of participants at baseline: 34 - Number lost to follow-up: 0 - Number analysed: 34 - Age, mean: Not reported - Women, %: 0 - BMI, mean: Not reported Included criteria: Male, healthy office workers 35-53 years old Excluded criteria: Not reported

Mann 1972 (Continued)

Pretreatment: Not reported

Interventions	Intervention Characteristics High-added sugar intake <i>Dose:</i> Continue with usual diet <i>Type of sugar:</i> Usual diet <i>Type:</i> Same Low-added sugar intake <i>Dose:</i> Cut out foods containing sucrose completely <i>Type of sugar:</i> No sucrose <i>Type:</i> Given list of dietary instructions and asked to keep diet records throughout experiment
Outcomes	Secondary outcomes: triglycerides
Identification	Sponsorship source: The South African Sugar Association Country: South Africa Setting: Insurance company Authors name: J. I. MANN Institution: Department of Medicine, University of Cape Town Email: Not reported Address: Not reported
Notes	The trial consisted of three arms. The low-sugar and usual-diet arms were used for this review; the reduced starch arm was not included.
Risk of bias	
Bias	Authors' judgement Support for judgement
Random sequence generation (selection bias)	Unclear risk Judgement Comment: No information provided
Allocation concealment (selection bias)	Unclear risk Judgement Comment: No information provided
Blinding of participants and personnel (performance bias) All outcomes	High risk Judgement Comment: Participants unblinded
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk Judgement Comment: No information provided
Incomplete outcome data (attrition bias) All outcomes	Unclear risk Judgement Comment: Only one dropout during study, but no further details provided to allow us to make a full assessment
Selective reporting (reporting bias)	Unclear risk Judgement Comment: All stated outcomes were included in the analyses but no prespecified analysis plan provided

Mann 1972 (Continued)

Other bias	Low risk	Participants were asked to fill out 3-day diet records twice during the experimental period.
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Markey 2016
Study characteristics

Methods	Study design: Randomised controlled trial Study grouping: Cross-over Duration of study: Two periods of 8 weeks each Wash-out period: 4 weeks
Participants	Baseline Characteristics • <i>Number of participants at baseline:</i> 56 • <i>Number lost to follow-up:</i> 6 • <i>Number analysed:</i> 50 • <i>Age, mean:</i> 31.6 • <i>Women, %:</i> 68 • <i>BMI, mean:</i> 24 Included criteria: Men and women aged 20–49 years who were not underweight or obese (BMI between 18.5 and 30 kg/m²). Excluded criteria: Clinical diagnosis of diabetes, CVD, anti-inflammatory or hypertensive medication, smoking, excessive alcohol consumption (> 21 units per week for men and > 14 units per week for women), participation in regular vigorous exercise or fitness training (≥ 20 min × 3 times/week), pregnancy or lactating Pretreatment: Not reported
Interventions	Intervention Characteristics High-added sugar intake • <i>Dose:</i> Participants exchanged at least one beverage and at least one food portion from habitual diet with equivalent sugar-containing products • <i>Type of sugar:</i> Non-milk extrinsic sugars • <i>Type:</i> Participants given study product supplies at beginning of each study period Low-added sugar intake • <i>Dose:</i> Participants exchanged at least one beverage and at least one food portion from habitual diet with equivalent sugar-reformulated products • <i>Type of sugar:</i> Reformulated, sugar-reduced foods and beverages • <i>Type:</i> Same
Outcomes	Secondary outcomes: systolic blood pressure, diastolic blood pressure, total cholesterol, LDL-cholesterol, HDL-cholesterol, triglycerides, fasting plasma glucose
Identification	Sponsorship source: Sugar Nutrition UK Country: UK Setting: Reading

Markey 2016 (Continued)

Authors name: Oonagh Markey

Institution: Hugh Sinclair Unit of Human Nutrition, Department of Food and Nutritional Sciences, University of Reading

Email: j.a.lovegrove@reading.ac.uk

Address: Hugh Sinclair Unit of Human Nutrition, Department of Food and Nutritional Sciences, University of Reading, Reading, Berkshire RG6 6AP, UK

Notes

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Judgement Comment: Stated that a computer-based minimisation procedure was used to randomly allocate participants
Allocation concealment (selection bias)	Unclear risk	Judgement Comment: No information provided
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Judgement Comment: Study products were blinded to the participants. An investigator who was not involved with assessing study outcomes relabelled products so that these looked identical.
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	Judgement Comment: No information provided
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement Comment: Reasons for attrition given and most participants completed both interventions.
Selective reporting (reporting bias)	Low risk	Judgement Comment: Study protocol was published on clinicaltrials.gov and no deviations between the protocol and the published article were identified.
Other bias	Low risk	Participants were asked to fill out a daily dietary compliance sheet and were asked to return unopened leftover products to the Nutrition Unit.

Njike 2011

Study characteristics

Methods	Study design: Randomised controlled trial Study grouping: Cross-over Duration of study: 3 periods of 6 weeks each Wash-out period: 4 weeks
Participants	Baseline Characteristics Overall Number of participants at baseline: 44

Njike 2011 (Continued)

- Number lost to follow-up: 7
- Number analysed: 37
- Age, mean: 52.4
- Women, %: 86
- BMI, mean: 30.4

Included criteria: BMI 25-35; waist circumference > 100 cm for men and > 80 cm for women; non-smoker

Excluded criteria: Current eating disorder; diagnosed coronary artery disease; diabetes mellitus; sleep apnoea; pregnancy; insulin or glucose sensitising medication use; restricted diet; allergy to cocoa; inability to comply with the protocol

Pretreatment: Not reported

Interventions	Intervention Characteristics High-added sugar intake • <i>Dose:</i> 91 grams of sugar/day • <i>Type of sugar:</i> Sucrose • <i>Type:</i> Participants asked to drink two beverages daily containing sugar-sweetened cocoa powder (11 g per beverage) and hot water (8 oz per beverage) Low-added sugar intake • <i>Dose:</i> 0 grams of sugar/day • <i>Type of sugar:</i> None • <i>Type:</i> Participants asked to drink two beverages daily containing sugar-free cocoa powder (11 g per beverage) and hot water (8 oz per beverage)	
Outcomes	Secondary outcomes: systolic blood pressure, diastolic blood pressure, total cholesterol, LDL-cholesterol, HDL-cholesterol, triglycerides, fasting plasma glucose	
Identification	Sponsorship source: The Hershey Company Country: USA Setting: Naugatuck Valley, Connecticut Authors name: Valentine Yanchou Njike Institution: Prevention Research Center, Yale University School of Medicine, Derby, CT Email: katzdl@pol.net, shelli.larovera@yalegriffinprc.org Address: Yale-Griffin Prevention Research Center, 130 Division Street, Derby, CT 06418, US	
Notes	The trial consisted of three arms. The sugar-free cocoa and sugar-sweetened cocoa arms were used in the review; the placebo hot liquid arm was not included.	
Risk of bias		
Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Judgement Comment: Randomised by using a computer-generated random number sequence
Allocation concealment (selection bias)	Unclear risk	Judgement Comment: No information provided

Njike 2011 (Continued)

Blinding of participants and personnel (performance bias) All outcomes	Low risk	Judgement Comment: Participants were blinded to the composition of each treatment.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Judgement Comment: The ultrasonographer and the data analyst were unaware of the treatment assignment.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement Comment: Relatively few individuals lost to follow-up and reasons for attrition provided and evenly distributed between the groups
Selective reporting (reporting bias)	Unclear risk	Judgement Comment: All stated outcomes were included in the analyses but no prespecified analysis plan provided
Other bias	Low risk	Participants were asked to complete a 3-day food diary for each of the intervention periods as well as the four-week washout period to assess dietary patterns throughout the study period.

Raben 2002

Study characteristics

Methods	Study design: Randomised controlled trial Study grouping: Parallel-group Duration of study: 10 weeks
Participants	Baseline Characteristics High-added sugar intake - Number of participants at baseline: - Number lost to follow-up: - Number analysed: 21 - Age, mean: 33.3 - Women, %: Not reported - BMI, mean: 28.0 Low-added sugar intake - Number of participants at baseline: - Number lost to follow-up: - Number analysed: 20 - Age, mean: 37.1 - Women, %: Not reported - BMI, mean: 27.6 Overall - Number of participants at baseline: 42 - Number lost to follow-up: 1 - Number analysed: 41 - Age, mean: 35.2

Raben 2002 (Continued)

- *Women, %:* 85.4
- *BMI, mean:* 27.8

Included criteria: 20–50 y old; overweight [BMI of 25–30 or > 10% overweight according to weight-and-height tables]; healthy; not dieting; not pregnant or lactating

Excluded criteria: Not stated

Pretreatment: Not reported

Interventions	Intervention Characteristics High-added sugar intake • <i>Dose:</i> Approximately 2 g sucrose per kilo bodyweight daily • <i>Type of sugar:</i> Sucrose • <i>Type:</i> Participants provided with soft drinks, fruit juices and solid foods (e.g. yoghurt and marmalade) sweetened with sucrose. Otherwise, diet ad libitum Low-added sugar intake • <i>Dose:</i> Approximately 2 g artificial sweeteners per kilo bodyweight daily • <i>Type of sugar:</i> Artificial sweeteners • <i>Type:</i> Participants provided with soft drinks, fruit juices and solid foods (e.g. yoghurt and marmalade) sweetened with artificial sweeteners. Otherwise, diet ad libitum	
Outcomes	Secondary outcomes: systolic blood pressure, diastolic blood pressure, total cholesterol, HDL-cholesterol, triglycerides, fasting plasma glucose	
Identification	Sponsorship source: Supported by the Danish Research and Development Programme for Food Technology 1990–1994 and Danisco Sugar. Coca Cola (Nordic and Eurasia Divisions) provided soft drinks for the study. Country: Denmark Setting: Not stated Authors name: Anne Raben Institution: Department of Human Nutrition, Faculty of LIFE sciences, University of Copenhagen Email: ara@kvl.dk Address: Department of Human Nutrition, Faculty of LIFE sciences, University of Copenhagen, Rolighedsvej 30, 1958 Frederiksberg C, Denmark	
Notes	Glucose, cholesterol and triglycerides only reported for a subset of the study population (n = 23)	
Risk of bias		
Bias	Authors' judgement Support for judgement	
Random sequence generation (selection bias)	Unclear risk	Judgement Comment: Randomised, but no details provided
Allocation concealment (selection bias)	Unclear risk	Judgement Comment: Not specified if the allocation sequence was concealed
Blinding of participants and personnel (performance bias)	Unclear risk	Judgement Comment: Participants blinded but no information provided if personnel were blinded

Raben 2002 (Continued)

All outcomes

Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	Judgement Comment: No information provided
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Judgement Comment: No information provided on loss to follow-up and potential reasons for attrition
Selective reporting (reporting bias)	Unclear risk	Judgement Comment: All stated outcomes were included in the analyses but no prespecified analysis plan provided
Other bias	Low risk	Participants were asked to fill out 7-day dietary records at week 0, 5, and 10.

Sanchez-Delgado 2021
Study characteristics

Methods	Study design: Randomised controlled trial Study grouping: Parallel-group Duration of study: 6 weeks
Participants	Baseline Characteristics High-added sugar intake • Number of participants at baseline: 14 • Number lost to follow-up: 2 • Number analysed: 12 • Age, mean: 22.3 • Women, %: 75 • BMI, mean: 22.5 Low-added sugar intake • Number of participants at baseline: 14 • Number lost to follow-up: 1 • Number analysed: 13 • Age, mean: 23.9 • Women, %: 62 • BMI, mean: 21.8 Overall • Number of participants at baseline: 28 • Number lost to follow-up: 3 • Number analysed: 25 • Age, mean: 23.1 • Women, %: 68 • BMI, mean: 22.2 Included criteria: age 18-30 years; BMI 18.5-24.9

Sanchez-Delgado 2021 (Continued)

Excluded criteria: Hypertension; hyperglycaemia; current treatment to reduce weight; current hormonal therapy

Interventions	Intervention Characteristics High-added sugar intake • <i>Dose:</i> Eight 5 g packets/day • <i>Type of sugar:</i> Sucrose • <i>Type:</i> Participants provided with packets and asked to sweeten beverages and foods Low-added sugar intake • <i>Dose:</i> Four 1 g packet/day (each packet contains 0.025 g of steviol glycosides) • <i>Type of sugar:</i> Steviol glycosides • <i>Type:</i> Same
Outcomes	Secondary outcomes: total cholesterol, triglycerides, fasting glucose
Identification	Sponsorship source: No specific grant from any funding agency, commercial or not-for-profit sectors Country: Mexico Setting: Not reported Authors name: Marcela Sanchez-Delgado Institution: Laboratorio de Neuroquímica, Facultad de Medicina, Universidad Autónoma del Estado de México, México Email: icontreras@uaemex.mx Address: Colonia Moderna de la Cruz Toluca, México
Notes	The trial consisted of three arms. The sucrose and steviol glycosides arms were used for this review; the sucralose arm was not included.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomised, but no further details provided. Therefore unable to fully judge
Allocation concealment (selection bias)	Unclear risk	No information provided
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	No information provided
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	No information provided
Incomplete outcome data (attrition bias) All outcomes	Low risk	Low dropout rate, 4 out of 42 individuals. Reasons for attrition provided

Sanchez-Delgado 2021 (Continued)

Selective reporting (reporting bias)	Unclear risk	The trial did not refer to any study protocol, therefore, unable to fully judge
Other bias	Low risk	Participants were required to visit the study facilities once a week and to maintain a sweetener consumption diary.

Smith 1996

Study characteristics

Methods	Study design: Randomised controlled trial Study grouping: Parallel-group Duration of study: 9 months
Participants	Baseline Characteristics High-added sugar intake - Number of participants at baseline: - Number lost to follow-up: - Number analysed: 10 - Age, mean: 50 - Women, %: Not reported - BMI, mean: Not reported Low-added sugar intake - Number of participants at baseline: - Number lost to follow-up: - Number analysed: 22 - Age, mean: 53 - Women, %: Not reported - BMI, mean: Not reported Overall - Number of participants at baseline: 47 - Number lost to follow-up: 15 - Number analysed: 32 - Age, mean: 51.5 - Women, %: Not reported - BMI, mean: Not reported Included criteria: General good health; a fasting plasma triglyceride level > 2.0 mmol/L; total cholesterol level < 6.5 mmol/L; no other metabolic derangement; a willingness to take part in a long-term dietary study Excluded criteria: Not provided Pretreatment: Not reported
Interventions	Intervention Characteristics High-added sugar intake

Smith 1996 (Continued)

- *Dose:* Usual diet
- *Type of sugar:* Sucrose
- *Type:* Participants asked to consumer their usual diet

Low-added sugar intake

- *Dose:* Added sugar replaced with starch
- *Type of sugar:* None
- *Type:* Extrinsic sucrose removed and its energy content replaced with high starch foods such as bread and pasta

Outcomes	Secondary outcomes: total cholesterol, HDL-cholesterol, triglycerides
Identification	Sponsorship source: Country: New Zealand Setting: Dunedin Authors name: JB Smith Institution: Department of Human Nutrition, University of Otago, Dunedin, New Zealand Email: Not provided Address: Not provided
Notes	The included results were deciphered by visually inspecting plots since no numbers were included in the study results.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Judgement Comment: The participants were randomly allocated to study and control groups using a system of random numbers designed to allocate twice as many subjects to the intervention group as were allocated to the control group.
Allocation concealment (selection bias)	Unclear risk	Judgement Comment: No information provided
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement Comment: Participants unblinded
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	Judgement Comment: No information provided
Incomplete outcome data (attrition bias) All outcomes	High risk	Judgement Comment: 15/37 = 40.5% dropout rate. Proportion between groups not described, however compliance was worse in the intervention group. Reasons for attrition not provided in sufficient detail
Selective reporting (reporting bias)	Unclear risk	Judgement Comment: All outcomes reported but no prespecified analysis plan
Other bias	Low risk	Dietary compliance was assessed by interviews each month.

Surwit 1997

Study characteristics

Methods	Study design: Randomised controlled trial Study grouping: Parallel-group Duration of study: 6 weeks
Participants	Baseline Characteristics High-added sugar intake • <i>Number of participants at baseline:</i> 28 • <i>Number lost to follow-up:</i> 8 • <i>Number analysed:</i> 20 • <i>Age, mean:</i> 40.6 • <i>Women, %:</i> 100 • <i>BMI, mean:</i> 35.93 Low-added sugar intake • <i>Number of participants at baseline:</i> 24 • <i>Number lost to follow-up:</i> 2 • <i>Number analysed:</i> 22 • <i>Age, mean:</i> 40.3 • <i>Women, %:</i> 100 • <i>BMI, mean:</i> 34.93 Overall • <i>Number of participants at baseline:</i> 52 • <i>Number lost to follow-up:</i> 10 • <i>Number analysed:</i> 42 • <i>Age, mean:</i> 40.5 • <i>Women, %:</i> 100 • <i>BMI, mean:</i> 35.43 Included criteria: Women 130-200% of their ideal body weights Excluded criteria: Used any drug that affected the autonomic nervous system or metabolism (including nicotine) or any psychotropic agent; if they had a history of significant cardiopulmonary, neurologic, gastrointestinal, or endocrinologic illness; or if they participated in a regular exercise programme Pretreatment: Not reported
Interventions	Intervention Characteristics High-added sugar intake • <i>Dose:</i> Diet containing about 4606 kJ/d; high in starch, total sugars and sucrose • <i>Type of sugar:</i> Sucrose • <i>Type:</i> Participants provided with all meals and snacks Low-added sugar intake • <i>Dose:</i> Diet containing about 4606 kJ/d; low in starch, total sugars and sucrose • <i>Type of sugar:</i> Sucrose

Surwit 1997 (Continued)

- *Type:* Same

Outcomes	Secondary outcomes: systolic blood pressure, diastolic blood pressure, total cholesterol, LDL-cholesterol, HDL-cholesterol, triglycerides, fasting plasma glucose
Identification	Sponsorship source: The National Institute Of Mental Health, The Sugar Association and the Kellogg Company Country: USA Setting: Not reported Authors name: Richard S Surwit Institution: From the Department of Psychiatry and Behavioral Sciences, the Department of Medicine, and the Sarah W Stedman Nutrition Center, Duke University Medical Center, Durham Email: surwi001@mc.duke.edu Address: Department of Psychiatry and Behavioral Sciences, Box 3842, Duke University Medical Center, Durham, NC 27710

Notes

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Judgement Comment: No details reported
Allocation concealment (selection bias)	Unclear risk	Judgement Comment: No details reported
Blinding of participants and personnel (performance bias) All outcomes	Unclear risk	Judgement Comment: No details reported
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	Judgement Comment: No details reported
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Judgement Comment: Number and reasons for dropouts were provided, but higher proportion in high-sucrose group
Selective reporting (reporting bias)	Unclear risk	Judgement Comment: All outcomes reported but unable to fully judge
Other bias	Low risk	Participants were asked to fill out a one-page daily diary to document deviation from the study diets, hunger, health problems, or concerns. A staff monitor was present daily during dinner to review the diary, record blood pressure, answer questions, and give support.

Umpleby 2017

Study characteristics

Methods	Study design: Randomised controlled trial Study grouping: Cross-over Duration of study: 2 periods of 12 weeks each Wash-out period: 4 weeks
Participants	Baseline Characteristics NAFLD: • <i>Number of participants at baseline:</i> Not reported • <i>Number lost to follow-up:</i> Not reported • <i>Number analysed:</i> 11 • <i>Age, mean:</i> 59 • <i>Women, %:</i> 0 • <i>BMI, mean:</i> 28.9 Controls: • <i>Number of participants at baseline:</i> Not reported • <i>Number lost to follow-up:</i> Not reported • <i>Number analysed:</i> 14 • <i>Age, mean:</i> 54 • <i>Women, %:</i> 0 • <i>BMI, mean:</i> 28.4 Included criteria: Men (aged 40–65 years, BMI 25–30) at increased cardio-metabolic risk Excluded criteria: Diabetes; any medical condition other than NAFLD; lipid-lowering medication; unstable weight in the preceding 3 months; intake of alcohol exceeding 20 g/day Pretreatment: Not reported
Interventions	Intervention Characteristics High-added sugar intake • <i>Dose:</i> Sugar intake corresponding to the upper 2.5th percentile of intake for the UK population, equal to 26% sugar intake of daily calories • <i>Type of sugar:</i> Non-milk extrinsic sugars • <i>Type:</i> Participants requested to fill in diet diaries. Regular visits to the homes of the participants to assess compliance Low-added sugar intake • <i>Dose:</i> Low intake corresponding to the lower 2.5th percentile of intake for the UK population, equal to 6% sugar intake of daily calories • <i>Type of sugar:</i> Non-milk extrinsic sugars • <i>Type:</i> Same
Outcomes	Secondary outcomes: total cholesterol, LDL-cholesterol, HDL-cholesterol, triglycerides, fasting plasma glucose
Identification	Sponsorship source: UK government grant from the Biological Biotechnology Scientific Research Council [Grant no. BB/G009899/1]; University of Surrey PhD scholarship (to A.M.); Medical Research

Umpleby 2017 (Continued)

Council (body composition measurements); and infrastructure support from the National Institute of Health Research at the Cambridge Biomedical Research Centre.

Country: UK

Setting: Not stated

Authors name: A. Margot Umpleby

Institution: Faculty of Health and Medical Sciences, University of Surrey, Guildford, U.K

Email: b.griffin@surrey.ac.uk

Address: Not reported

Notes

Since NAFLD is a medical condition we did not feel that it was appropriate to combine the groups. Therefore, the results were presented separately for the controls and NAFLD individuals.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Judgement Comment: Participants randomised using a computer-generated sequence of treatments in sealed envelopes
Allocation concealment (selection bias)	Low risk	Judgement Comment: Computer-generated sequence of treatments in sealed envelopes
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement Comment: Participants unblinded
Blinding of outcome assessment (detection bias) All outcomes	Unclear risk	Judgement Comment: No information provided
Incomplete outcome data (attrition bias) All outcomes	High risk	Judgement Comment: Only information on how many participants completed the study. Number of people lost to follow-up (if any) or reasons for attrition were not included.
Selective reporting (reporting bias)	Low risk	Judgement Comment: Trial registered on clinicaltrials.gov, and the outcomes included in the protocol were reported in the published article.
Other bias	Low risk	Self-recorded dietary intakes were monitored by regular visits to the homes of participants to ensure dietary compliance.

Vazquez-Duran 2016

Study characteristics

Methods	Study design: Randomised controlled trial Study grouping: Parallel-group Duration of study: 6 months
Participants	Baseline Characteristics

Vazquez-Duran 2016 (Continued)

High-added sugar intake

- *Number of participants at baseline:* 49
- *Number lost to follow-up:* 0
- *Number analysed:* 49
- *Age, mean:* 21.98
- *Women, %:* 83.7
- *BMI, mean:* 26.00

Low-added sugar intake

- *Number of participants at baseline:* 49
- *Number lost to follow-up:* 1
- *Number analysed:* 48
- *Age, mean:* 22.55
- *Women, %:* 79.6
- *BMI, mean:* 27.27

Overall

- *Number of participants at baseline:* 98
- *Number lost to follow-up:* 1
- *Number analysed:* 97
- *Age, mean:* 21.99
- *Women, %:* 81
- *BMI, mean:* 26.24

Included criteria: Nursing students; male and female; > 18 years old; consumed < 12 ounces of sweetened beverages/day

Excluded criteria: Maintaining a special diet; arterial hypertension diagnosed with or without use of antihypertensive medications; hyperthyroidism or hypothyroidism; use of weight loss medications; diabetes mellitus; cardiovascular or psychiatric diseases or hormonal therapy; participated in vigorous physical activity for more than one hour/day.

Pretreatment: Not reported

Interventions	Intervention Characteristics High-added sugar intake • <i>Dose:</i> No modification in beverage consumption • <i>Type of sugar:</i> Sugar sweetened beverages • <i>Type:</i> Given individualised isocaloric diets, an online questionnaire filled in once a week and email reminders Low-added sugar intake • <i>Dose:</i> No sweetened beverages were permitted • <i>Type of sugar:</i> No sweetened beverages • <i>Type:</i> Same
Outcomes	Secondary outcomes: systolic blood pressure, diastolic blood pressure
Identification	Sponsorship source: Instituto Nacional de Ciencias Medicas y Nutricion Salvador Zubiran Country: Mexico Setting: Not stated

Vazquez-Duran 2016 (Continued)

Authors name: Marisela Vázquez-Durán

Institution: Department of Clinical Nutrition

Email: cam7125@gmail.com

Address: Department of Clinical Nutrition. Instituto Nacional de Ciencias Médicas y Nutrición. Salvador Zubirán. Vasco de Quiroga No. 15 Col. Sección XVI, Tlalpan. CP 14000 México. D.F.

Notes The trial consisted of three arms. The without-sugar-sweetened beverages and caloric and non-caloric beverages arms were included in the review; the with non-calorie-sweetened beverages arm was not included.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Judgement Comment: Participants randomly assigned using a sequential series of numbers computer-generated to one of the three treatment groups
Allocation concealment (selection bias)	Low risk	Judgement Comment: One of the researchers of the present study who did not have contact with participants performed the randomisation.
Blinding of participants and personnel (performance bias) All outcomes	High risk	Judgement Comment: Participants and nutritionist who prescribed the diets were not blinded.
Blinding of outcome assessment (detection bias) All outcomes	Low risk	Judgement Comment: The individuals who performed the clinical evaluations were blinded.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Judgement Comment: Low loss to follow-up and similar population size across groups at the end of the study
Selective reporting (reporting bias)	Low risk	Judgement Comment: Prespecified analysis plan available on clinicaltrials.gov, and was not changed
Other bias	Low risk	Participants were asked to fill out a 24-hour dietary record once a week. Each student received reminders via email about healthy eating at the beginning and at three and six months of the intervention.

Characteristics of excluded studies [ordered by study ID]

Study	Reason for exclusion
ACTRN12612000874819	Wrong study design
ACTRN12614000431628	Short-term (< 4 weeks)
ACTRN12615001321538	Wrong comparator
ACTRN12616000840482	Short-term (< 4 weeks)
ACTRN12616001352493	Wrong study design

Study	Reason for exclusion
ACTRN12617000319370	Wrong comparator
ACTRN12618000322235	Wrong study design
ACTRN12618001758291	Wrong comparator
Aeberli 2011	Short-term (< 4 weeks)
Aeberli 2012	Short-term (< 4 weeks)
Ahmad 2020b	Wrong study design
Al Dujaili 2017	Short-term (< 4 weeks)
Al Tamimi 2020	Wrong intervention
Al Waili 2004	Short-term (< 4 weeks)
Anderson 2014	Wrong intervention
Angelopoulos 2015	Wrong comparator
Angelopoulos 2016	Wrong comparator
Angelopoulos 2016a	Wrong comparator
Bachlechner 2017	Short-term (< 4 weeks)
Bantle 2000	Wrong comparator
Bonora 1999	Wrong study design
Brandes 1981	Wrong study design
Brown 2011	Short-term (< 4 weeks)
Bruun 2015	No relevant outcome for present review measured
Brynes 2003	Short-term (< 4 weeks)
Campos 2017	No relevant outcome for present review measured
Chen 2009	Wrong intervention
Chen 2010	Wrong intervention
Chew 2019	Wrong intervention
Clayton 2019	Wrong intervention
Creighton 2014	Wrong intervention
Davila 2016	Short-term (< 4 weeks)
Derakhshandeh-Rishehri 2014	Wrong intervention

Study	Reason for exclusion
Dhillon 2018	Wrong intervention
Dominguez Coello 2017	Wrong intervention
Feig 2010	Wrong intervention
Geidl 2018	No relevant outcome for present review measured
Genta 2009	Wrong intervention
Gonzalez-Granda 2018	Short-term (< 4 weeks)
Gostner 2005	Wrong study design
Grande 1974	Short-term (< 4 weeks)
Hallfrisch 1983	Wrong study design
Hidvegi 2016	Wrong study design
Higgins 2018	Wrong intervention
Higgins 2019	No relevant outcome for present review measured
Hochuli 2011	Short-term (< 4 weeks)
Hollenbeck 1993	Wrong study design
Houston 2015	Wrong study design
Hudgins 1998	Short-term (< 4 weeks)
Huttunen 1976	Wrong study design, not an RCT
ISRCTN11540234	Wrong intervention
ISRCTN41579277	Wrong intervention
ISRCTN57611296	Wrong intervention
ISRCTN90960030	Wrong intervention
Juturu 2011	Wrong intervention
Kuzma 2019	Short-term (< 4 weeks)
Lambert 2018	Wrong intervention
Lowndes 2012	Wrong intervention
Madero 2011	Wrong intervention
Madjd 2018	Wrong comparator
Majid 2013	Wrong intervention

Study	Reason for exclusion
Meng 2018	Wrong comparator
NCT00374218	Wrong study design
NCT00525694	Short-term (< 4 weeks)
NCT00868673	Wrong intervention
NCT01364337	Wrong intervention
NCT01394380	No relevant outcome for present review measured
NCT01645995	No relevant outcome for present review measured
NCT01683929	Short-term (< 4 weeks)
NCT01790984	Wrong comparator
NCT02252952	No relevant outcomes for current review measured
NCT02347267	Wrong comparator
NCT02911753	Wrong intervention
NCT03527277	Wrong comparator
NCT03868683	Wrong comparator
Nickols Richardson 2014	Wrong intervention
Nikkila 1972	Short-term (< 4 weeks)
Okuno 2010	Wrong comparator
Padilla Camberos 2018	Wrong intervention
Ramirez Velez 2016	Wrong study design
Rasad 2018	Wrong comparator
Reid 2013	No relevant outcome for present review measured
Reiser 1981	Wrong study design
Reiser 1989	Wrong study design
Roberts 1974	Wrong study design
Swanson 1992	Wrong intervention
Tate 2012	Wrong comparator
UMIN000016456	Wrong intervention
UMIN000016586	Wrong intervention

Study	Reason for exclusion
UMIN000018120	Short-term (< 4 weeks)
UMIN000022168	Wrong study design
Zaveri 2009	Wrong intervention

RCT:

Characteristics of studies awaiting classification *[ordered by study ID]*

[ACTRN12610000106033](#)

Methods	Randomised controlled study
Participants	120 healthy males, ages 20-55
Interventions	Sugar-sweetened beverages vs non-caloric beverages for 8 weeks
Outcomes	Body weight; plasma total and high-density lipoprotein cholesterol; plasma triacylglycerol
Notes	

[Crutchley 2013](#)

Methods	Randomised controlled study
Participants	Relatively insulin-resistant population, number not stated
Interventions	Sugar-sweetened beverages vs non-sweetened beverages for 8 weeks
Outcomes	Fasting glucose; triglyceride; HDL-cholesterol; weight; BMI; blood pressure
Notes	

[Kassi 2016](#)

Methods	Randomised controlled trial
Participants	38 men and women with metabolic syndrome, aged 47.3 +/- 10.3 years
Interventions	Sweet of choice vs Stevia snack for 16 weeks
Outcomes	Systolic blood pressure; diastolic blood pressure; fasting glucose
Notes	

UMIN000014317

Methods	Randomised controlled study
Participants	44 healthy men and women of ages 20-59 years
Interventions	Placebo vs (unknown) food for 12 weeks
Outcomes	Plasma glucose; triglyceride
Notes	

UMIN000026291

Methods	Randomised controlled trial
Participants	70 healthy men and women of ages 20-65
Interventions	Placebo vs (unknown) food for 12 weeks
Outcomes	Body weight; total cholesterol; HDL-cholesterol; LDL-cholesterol; fasting blood glucose
Notes	

UMIN000031206

Methods	Randomised controlled trial
Participants	24 healthy men and women of ages 20-65 years
Interventions	Placebo vs (unknown) test food for 4 weeks
Outcomes	Body weight; blood pressure, blood tests (not described in detail)
Notes	

BMI:

HDL:

LDL:

vs:

Characteristics of ongoing studies *[ordered by study ID]*

Boakes 2015

Study name	Effects of sugar-sweetened drinks on psychological and metabolic outcomes
Methods	RCT, parallel
Participants	153
Interventions	Sugar-sweetened drink vs water, 4.5 L per week

Boakes 2015 (Continued)

Outcomes	Primary outcome: Waist circumference. Secondary outcome: fasting blood glucose, triglyceride, cholesterol, blood pressure, BMI, OGTT, Go/No go test for response inhibition, sweetness preference test, total body fat as measured by Bio-electrical Impedance Analysis
Starting date	25/09/2015
Contact information	Prof Robert A Boakes, School of Psychology, University of Sydney, Sydney NSW 2006 Australia
Notes	Duration of intervention: 12 weeks

Flores-Muñoz 2018

Study name	Metabolism effects of artificially sweetened beverages restriction
Methods	RCT, parallel
Participants	45
Interventions	Experimental: artificially sweetened beverages Control group: no changes in their alimentary habits
Outcomes	Primary: • Glucose in mg/dL • Triglycerides in mg/dL • Insulin • Cholesterol in mg/dL Secondary: • Weight in kilograms • Body fat percentage • Portions consumed of artificially sweetened beverages (1 portion = 250 mL)
Starting date	February 1, 2017
Contact information	Paulina E Viveros-Watty, pauwatty@gmail.com
Notes	Inclusion Criteria: • BMI between 25 kg/m² to 36 kg/m², with low-intensity physical activities and consumers of at least 3 portions of artificially sweetened beverages per week Exclusion Criteria: • students with hyper or hypothyroidism, pregnant women and those who performed high to moderate physical activities

Gonzalez 2018

Study name	Dietary carbohydrate manipulation and energy balance
Methods	RCT, parallel assignment

Gonzalez 2018 (Continued)

Participants	60
Interventions	- Active Comparator: Diet consisting of 50% carbohydrate (20% sugar), 15% protein, 35% fat - Experimental: Diet consisting of 50% carbohydrate (< 5% sugar), 15% protein, 35% fat - Experimental: Diet consisting of < 8% carbohydrate (< 5% sugar), 15% protein, > 77% fat
Outcomes	Primary: Physical activity energy expenditure (kJ/day or kcal/day) Secondary: Time spent in different physical activity intensities (MET categories) (minutes), energy expended in different physical activity intensities (MET categories) (kJ or kcal), step count, energy intake and dietary macronutrient composition, weight, bone mineral density, bone mineral content, fat body mass, lean body mass, android fat mass, gynoid fat mass, resting metabolic rate, substrate oxidation, waist and hip circumference, fasting metabolite/hormone profile, postprandial metabolite/hormone profile, blood pressure, food preference ratings, subjective appetite and mood ratings, interstitial glucose concentrations, adipose tissue gene expression, adipose tissue protein expression, muscle tissue gene expression, muscle tissue protein expression, muscle glycogen concentrations, protein glycation, gut microbiome characterisation, metabolomics, faecal energy density, urinary beta-hydroxybutyrate concentrations, urine urea nitrogen excretion, genetic variation
Starting date	October 1, 2018
Contact information	
Notes	Inclusion Criteria: - Body mass index 18.5-29.9 kg/m² - Age 18-65 years - Able and willing to provide informed consent and safely comply with study procedures - Females to maintain record of regular menstrual cycle phase/contraceptive use - No anticipated changes in physical activity during the first 4 weeks of the study (e.g. holidays or training programmes) Exclusion Criteria: - Any reported condition or behaviour deemed either to pose undue personal risk to the participant or introduce bias - Any diagnosed metabolic disease (e.g. type 1 or type 2 diabetes) - Any reported use of substances which may pose undue personal risk to the participants or introduce bias into the experiment - Lifestyle not conforming to standard sleep-wake cycle (e.g. shift worker) - Any reported recent (< 6 months) change in body mass ($\pm$ 3%) - Use of antibiotic medication in the last 3 months - Use of prebiotic or probiotic products in the last month

Havel 2016

Study name	Adverse metabolic effects of dietary sugar
Methods	RCT, parallel
Participants	72
Interventions	Interventions:

Havel 2016 (Continued)

	• Experimental: HFCS-EB • Placebo Comparator: Asp-EB • Experimental: HFCS-AL • Placebo Comparator: Asp-AL
Outcomes	Primary outcomes: Change of de novo lipogenesis: palmitate tracer-to-tracee ratios by gas chromatography-mass spectrometry Secondary outcomes: • Change of endogenous glucose production measured by standard dilution techniques • Change of whole body insulin sensitivity • Change of liver lipids • Change of fat oxidation • Change of very-low density lipoprotein (VLDL)-triglyceride (TG) kinetics
Starting date	February 2016
Contact information	Kimber L Stanhope, klstanhope@ucdavis.edu
Notes	Inclusion Criteria: • 18-40 years • BMI 22-28 kg/m² • Self-reported stable body weight during the prior six months Exclusion criteria: • Fasting glucose > 105 mg/dL • Evidence of liver disorder [AST (Aspartate Aminotransferase) or ALT (Alanine Aminotransferase) > 200% upper limit of normal range] • Evidence of kidney disorder (> 2.0 mg/dL creatinine) • Evidence of thyroid disorder (out of normal range) • Systolic blood pressure consistently over 140 mmHg (mercury) or diastolic blood pressure over 90 mmHg • Triglycerides > 200 mg/dL • LDL-C > 130 mg/dL in combination with Chol:HDL > 4 • Haemoglobin < 8.5 g/dL • Pregnant or lactating women • Any other condition that, in the opinion of the investigators, would put the subject at risk • Current, prior (within 12 months), or anticipated use of any hypolipidaemic or anti-diabetic agents • Use of thyroid, antihypertensive, antidepressant, weight-loss medications or any other medication which, in the opinion of the investigator, may confound study results • Use of tobacco • Strenuous exerciser (> 3.5 hours/week at a level more vigorous than walking) • Surgery for weight loss • Diet exclusions: Food allergies, special dietary restrictions, routine consumption of fewer than 3 meals/day, routine ingestion of more than 2 sugar-sweetened beverages or 1 alcoholic beverage/day, unwillingness to consume any food on study menu • Hydrogen concentration in breath sample following consumption of HFCS-beverage during screening > 50 ppm • Veins that are assessed by the CCRC (Clinical Research Center) R.N.s as being unsuitable for long-term infusions and multiple blood draws from a catheter • Pre-existing claustrophobia or metal implants that preclude MRI

Raben 2019

Study name	Sweeteners and sweetness enhancers: prolonged effects on health obesity and safety
Methods	RCT, parallel
Participants	330 healthy participants
Interventions	Sugar vs sweeteners
Outcomes	Glucose; blood pressure
Starting date	2019
Contact information	
Notes	

Sievenpiper 2018

Study name	Strategies To OPpose Sugars with NOn-nutritive sweeteners Or Water (STOP Sugars NOW) trial
Methods	RCT, parallel
Participants	81
Interventions	Experimental: Non-nutritive sweetened beverage (NSB) Experimental: Water Active Comparator: Sugar-sweetened beverage (SSB)
Outcomes	Primary: - Gut microbiome composition measured by 16S rRNA gene sequencing 75g OGTT-derived plasma glucose iAUC Secondary: - Change in waist circumference - Change in body weight - Change in fasting plasma glucose 75 g OGTT derived 2-hour plasma glucose 75 g OGTT derived Matsuda whole body insulin sensitivity index
Starting date	May 31, 2018
Contact information	
Notes	Inclusion Criteria: - Healthy, adult (age, 18-75 years) men and non-pregnant women; - Overweight or obese (BMI > 23 kg/m² for Asian individuals and > 25 kg/m² other individuals); - High-waist circumference (> 94 cm in men, > 80 cm in women in Europid, Sub-Saharan African, Eastern Mediterranean, and Middle Eastern individuals; > 90 cm in men and > 80 cm in women for South Asian, Chinese, Japanese, and South and Central American individuals);

Sievenpiper 2018 (Continued)

- Currently report drinking SSBs regularly (≥ 1 serving daily);
- Have a primary care physician;
- Nonsmoker;
- Free of any diseases or illnesses;
- Not regularly taking any medications that have a clinically relevant effect on the primary outcomes, as deemed inappropriate by investigators.

Exclusion Criteria:

- Age < 18 or > 75 years;
- BMI < 23 kg/m² for Asian individuals and < 25 kg/m² other individuals;
- Waist circumference < 94 cm in men, < 80 cm in women in Europid, Sub-Saharan African, Eastern Mediterranean, and Middle Eastern individuals; < 90 cm in men and < 80 cm in women for South Asian, Chinese, Japanese, and South and Central American individuals;
- Not regularly drinking SSBs (≥ 1 serving per day);
- Pregnant or breastfeeding females, or women planning on becoming pregnant throughout study duration;
- Regular medication use that has a clinically relevant effect on the primary outcomes, as deemed inappropriate by investigators;
- Antibiotic use in the last 3 months;
- Complementary or alternative medicine (CAM) use as deemed inappropriate by investigators;
- Self-reported diabetes;
- Self-reported uncontrolled hypertension (or systolic blood pressure (BP) ≥ 160 mmHg or diastolic BP ≥ 100 mmHg);
- Self-reported polycystic ovarian syndrome;
- Self-reported cardiovascular disease;
- Self-reported gastrointestinal disease;
- Previous bariatric surgery;
- Self-reported liver disease;
- Self-reported uncontrolled hyperthyroidism or hypothyroidism;
- Self-reported kidney disease;
- Self-reported chronic infection;
- Self-reported lung disease;
- Self-reported cancer/malignancy;
- Self-reported schizophrenia spectrum and other psychotic disorders, bipolar and related disorders, and dissociative disorders;
- Major surgery in the last 6 months;
- Other major illness or health-related incidence within the last 6 months;
- Smoker;
- Regular recreational drug users;
- Heavy alcohol use (> 3 drinks/day);
- Do not have a primary care physician;
- Participation in any trials within the last 6 months or for the duration of this study;
- Individuals planning on making dietary or physical activity changes throughout study duration;
- If participating in MRI portion of study: any condition or circumstance which would prevent the participant from having an MRI (e.g. having prostheses or metal implants, tattoos, or claustrophobia).

Tobias 2020

Study name	Health effects of substituting sugary beverages with artificially-sweetened beverages or water (SUB-POP)
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Tobias 2020 (Continued)

Methods	RCT, parallel
Participants	270 healthy participants, 20-69 years old
Interventions	SSB vs aspartame SSB
Outcomes	Weight; fP-glucose; triglycerides; HDL-C; LDL-C; systolic blood pressure; diastolic blood pressure
Starting date	2020
Contact information	
Notes	

Vohl 2017

Study name	Effect of non-nutritive sweeteners of high-sugar sweetened beverages on metabolic health and gut microbiome
Methods	RCT, parallel
Participants	66
Interventions	- Active Comparator: 710 mL of regular soft drinks, taken daily for 10 weeks - Experimental: 710 mL of diet soft drinks, taken daily for 10 weeks - Experimental: 710 mL of stevia-sweetened soft drinks, taken daily for 10 weeks
Outcomes	Primary: Changes in metabolic syndrome parameters including insulin/glucose homeostasis and lipid/lipoprotein metabolism in sugar-sweetened beverages (SSB) consumers Secondary: - Changes in intestinal homeostasis of SSB consumers following regular, diet or stevia-sweetened beverages intakes for 10 weeks [time frame: 4-6 months] - Changes in gut microbiota composition of SSB consumers following regular, diet or stevia-sweetened beverages intakes for 10 weeks [time frame: 4-6 months]
Starting date	October 18, 2017
Contact information	Marie-Claude Vohl, marie-claude.vohl@fsaa.ulaval.ca
Notes	

ALT:
 Asp(-AL)(-EB):
 AST:
 BMI:
 BP:
 CAM:
 CCRC:
 Chol:
 fP:
 HDL(-C):
 HFCS(-AL)(-EB):
 iAUC:

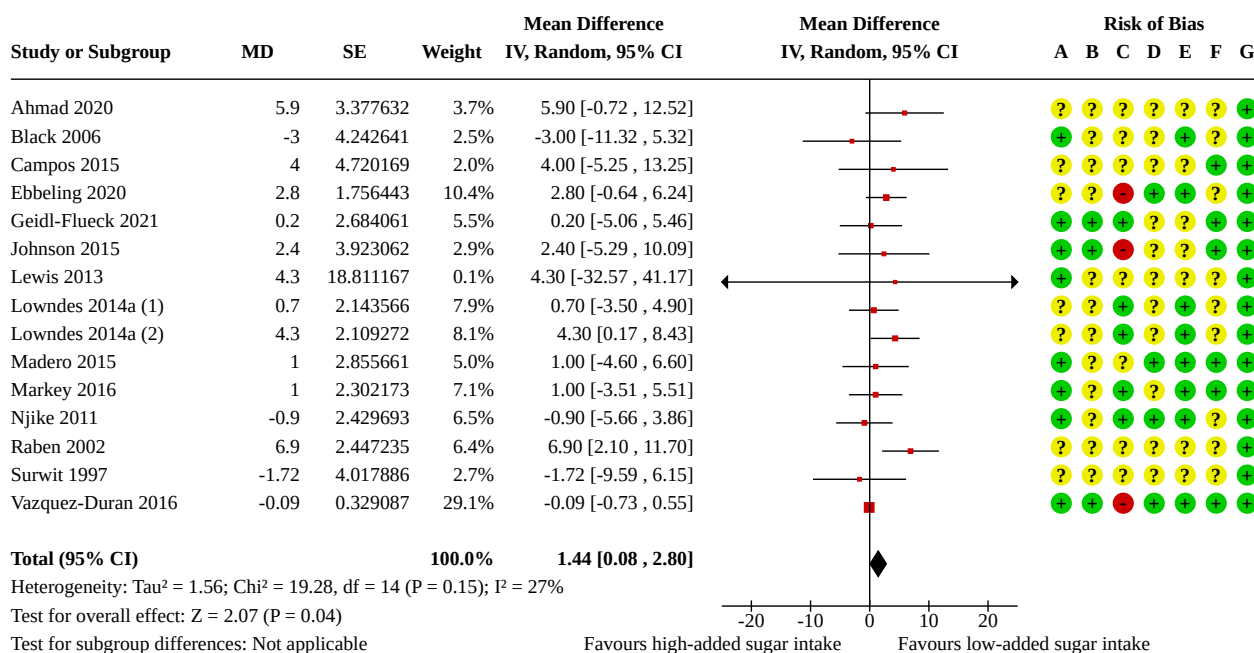
kcal:
 kJ:
 LDL(-C):
 MET:
 MRI:
 NSB:
 OGTT:
 RCT:
 R.N.s:
 rRNA:
 SSB:
 TG:
 VLDL:

DATA AND ANALYSES

Comparison 1. Effects of high vs low-added sugar consumption

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1.1 Systolic blood pressure (mmHg)	14		Mean Difference (IV, Random, 95% CI)	1.44 [0.08, 2.80]
1.2 Diastolic blood pressure (mmHg)	14		Mean Difference (IV, Random, 95% CI)	1.52 [0.67, 2.37]
1.3 Total cholesterol (mmol/L)	16		Mean Difference (IV, Random, 95% CI)	0.11 [0.01, 0.21]
1.4 LDL-cholesterol (mmol/L)	12		Mean Difference (IV, Random, 95% CI)	0.01 [-0.08, 0.10]
1.5 HDL-cholesterol (mmol/L)	16		Mean Difference (IV, Random, 95% CI)	0.01 [-0.02, 0.04]
1.6 Triglycerides (mmol/L)	14		Mean Difference (IV, Random, 95% CI)	0.10 [0.03, 0.17]
1.7 Fasting plasma glucose (mmol/L)	14		Mean Difference (IV, Random, 95% CI)	0.06 [-0.01, 0.13]

Analysis 1.1. Comparison 1: Effects of high vs low-added sugar consumption, Outcome 1: Systolic blood pressure (mmHg)



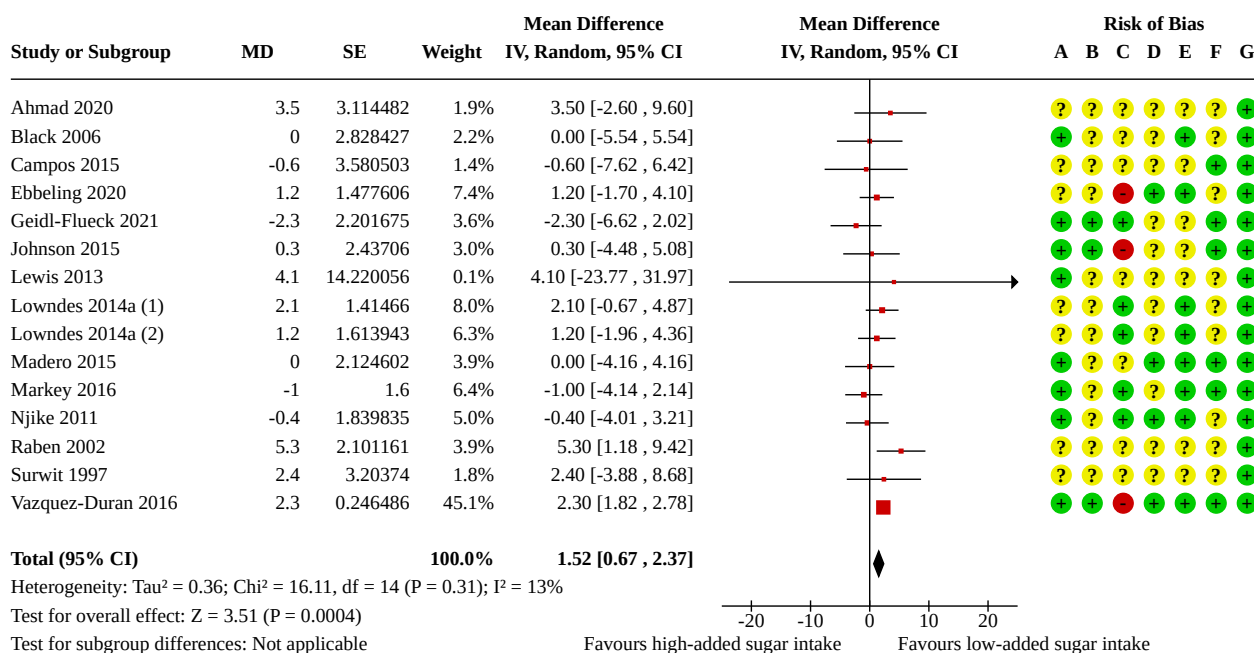
Footnotes

- (1) Shows the two arms of the intervention that compares high and low HFCS intake.
 (2) Shows the two arms of the intervention that compares high and low sucrose intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
 (B) Allocation concealment (selection bias)
 (C) Blinding of participants and personnel (performance bias)
 (D) Blinding of outcome assessment (detection bias)
 (E) Incomplete outcome data (attrition bias)
 (F) Selective reporting (reporting bias)
 (G) Other bias

Analysis 1.2. Comparison 1: Effects of high vs low-added sugar consumption, Outcome 2: Diastolic blood pressure (mmHg)



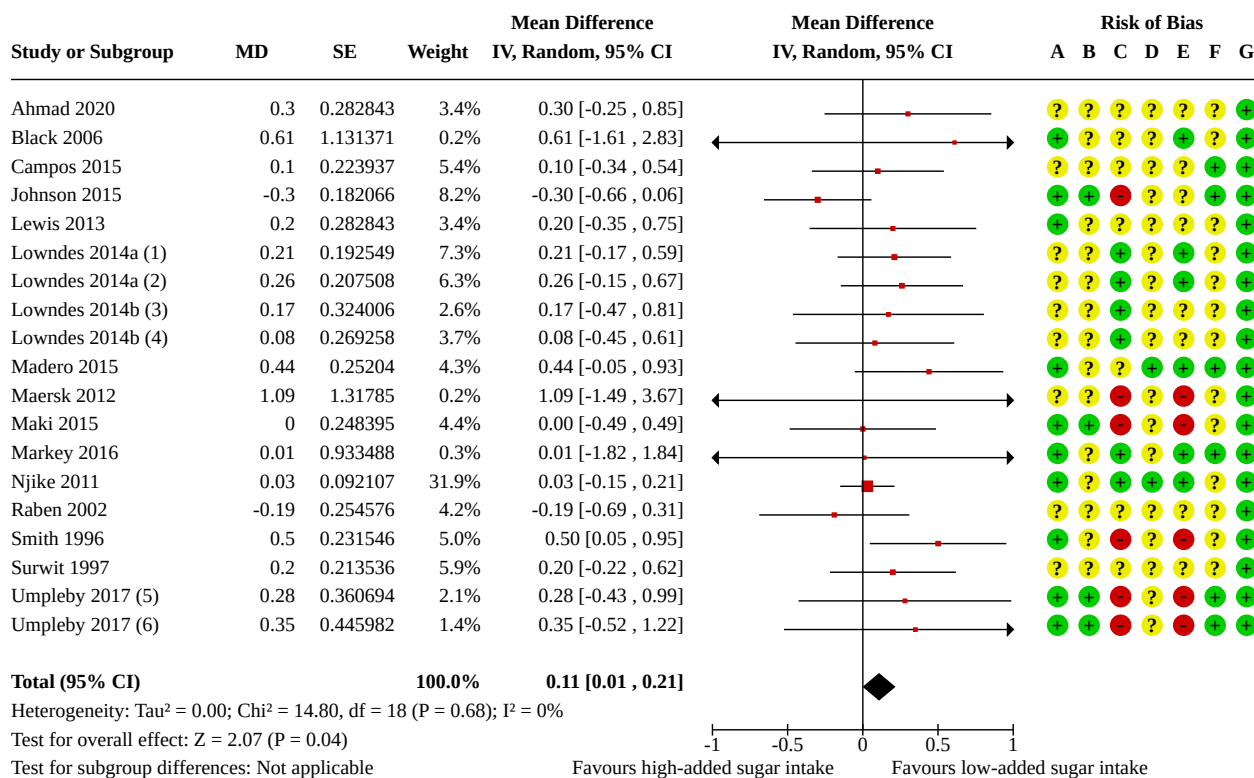
Footnotes

- (1) Shows the two arms of the intervention that compares high and low HFCS intake.
 (2) Shows the two arms of the intervention that compares high and low sucrose intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
 (B) Allocation concealment (selection bias)
 (C) Blinding of participants and personnel (performance bias)
 (D) Blinding of outcome assessment (detection bias)
 (E) Incomplete outcome data (attrition bias)
 (F) Selective reporting (reporting bias)
 (G) Other bias

Analysis 1.3. Comparison 1: Effects of high vs low-added sugar consumption, Outcome 3: Total cholesterol (mmol/L)



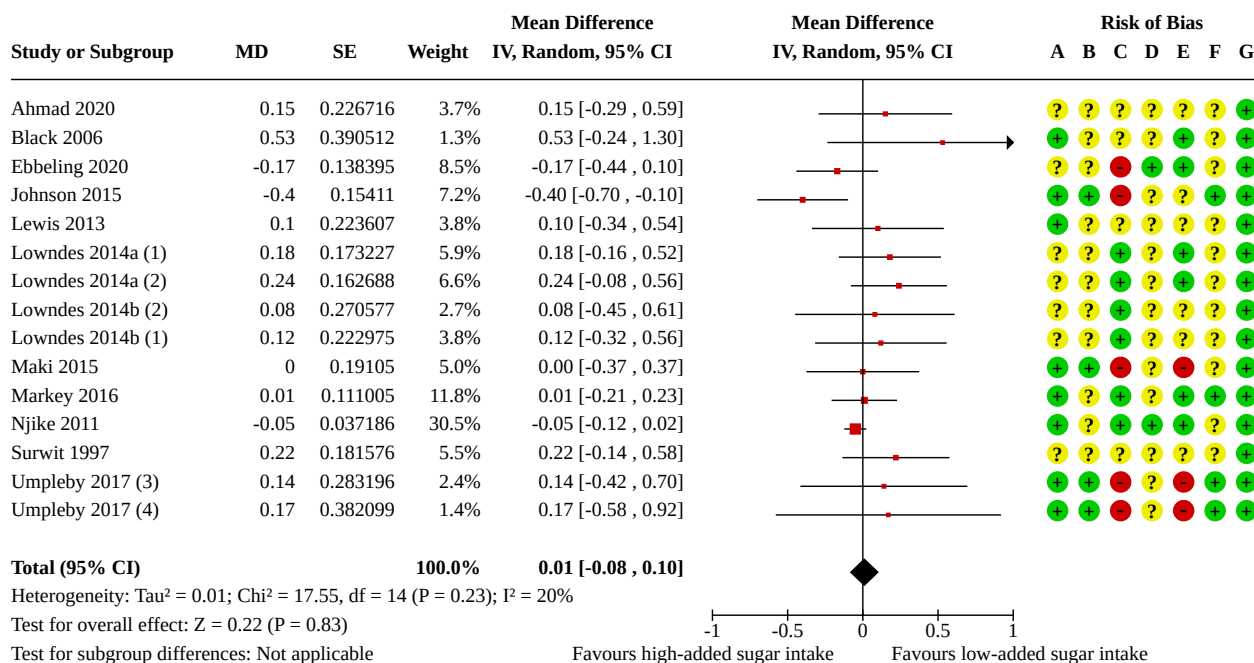
Footnotes

- (1) Shows the two arms of the intervention that compares high and low HFCS intake
- (2) Shows the two arms of the intervention that compares high and low sucrose intake
- (3) Shows the two arms of the intervention that compares high and low HFCS intake.
- (4) Shows the two arms of the intervention that compares high and low sucrose intake.
- (5) Includes only individuals not diagnosed with NAFLD, the controls.
- (6) Includes only individuals diagnosed with NAFLD.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 1.4. Comparison 1: Effects of high vs low-added sugar consumption, Outcome 4: LDL-cholesterol (mmol/L)



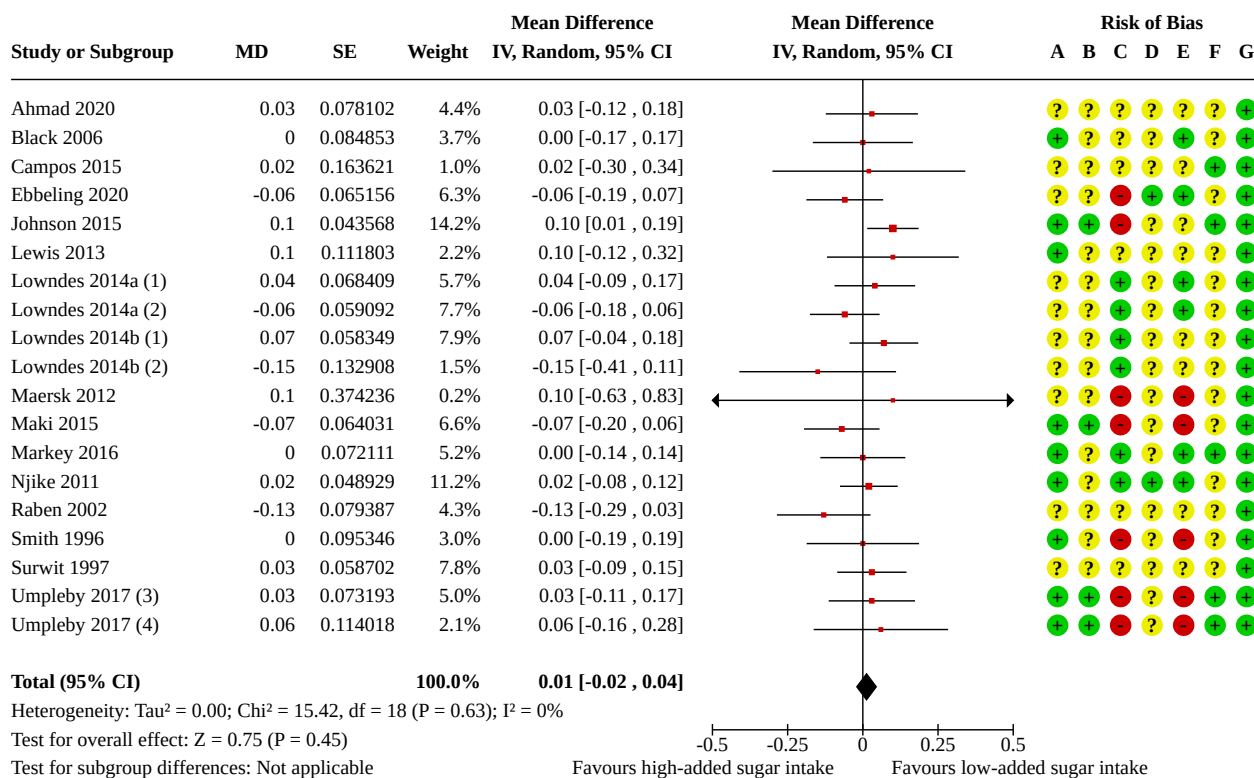
Footnotes

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
- (2) Shows the two arms of the intervention that compares high and low HFCS intake.
- (3) Includes only individuals not diagnosed with NAFLD, the controls.
- (4) Includes only individuals diagnosed with NAFLD.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 1.5. Comparison 1: Effects of high vs low-added sugar consumption, Outcome 5: HDL-cholesterol (mmol/L)



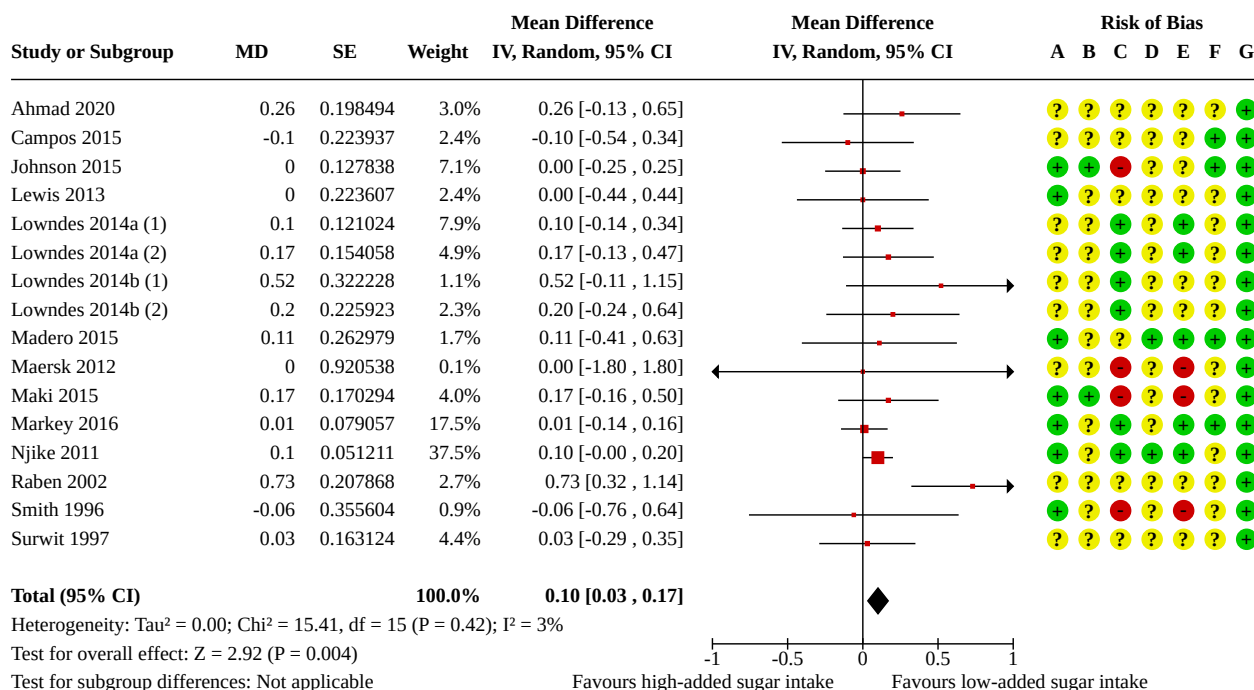
Footnotes

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
- (2) Shows the two arms of the intervention that compares high and low HFCS intake.
- (3) Includes only individuals not diagnosed with NAFLD, the controls.
- (4) Includes only individuals diagnosed with NAFLD.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 1.6. Comparison 1: Effects of high vs low-added sugar consumption, Outcome 6: Triglycerides (mmol/L)



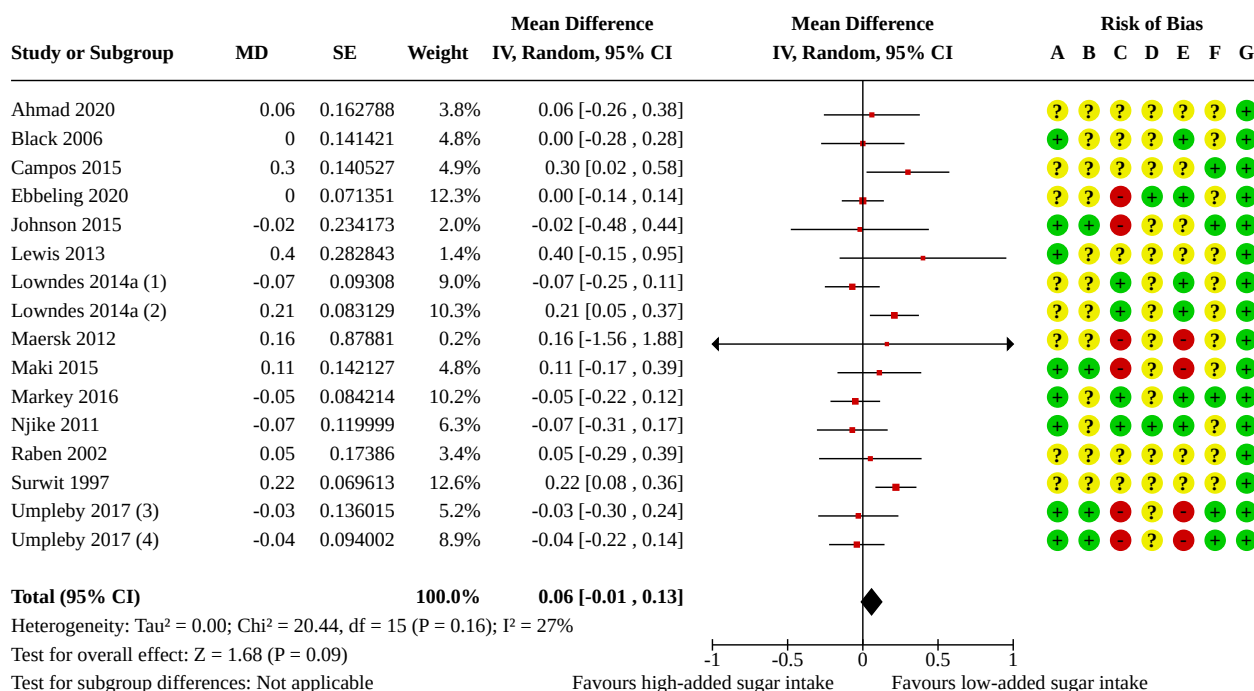
Footnotes

- (1) Shows the two arms of the intervention that compares high and low HFCS intake.
(2) Shows the two arms of the intervention that compares high and low sucrose intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Analysis 1.7. Comparison 1: Effects of high vs low-added sugar consumption, Outcome 7: Fasting plasma glucose (mmol/L)



Footnotes

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
- (2) Shows the two arms of the intervention that compares high and low HFCS intake.
- (3) Includes only individuals not diagnosed with NAFLD, the controls.
- (4) Includes only individuals diagnosed with NAFLD.

Risk of bias legend

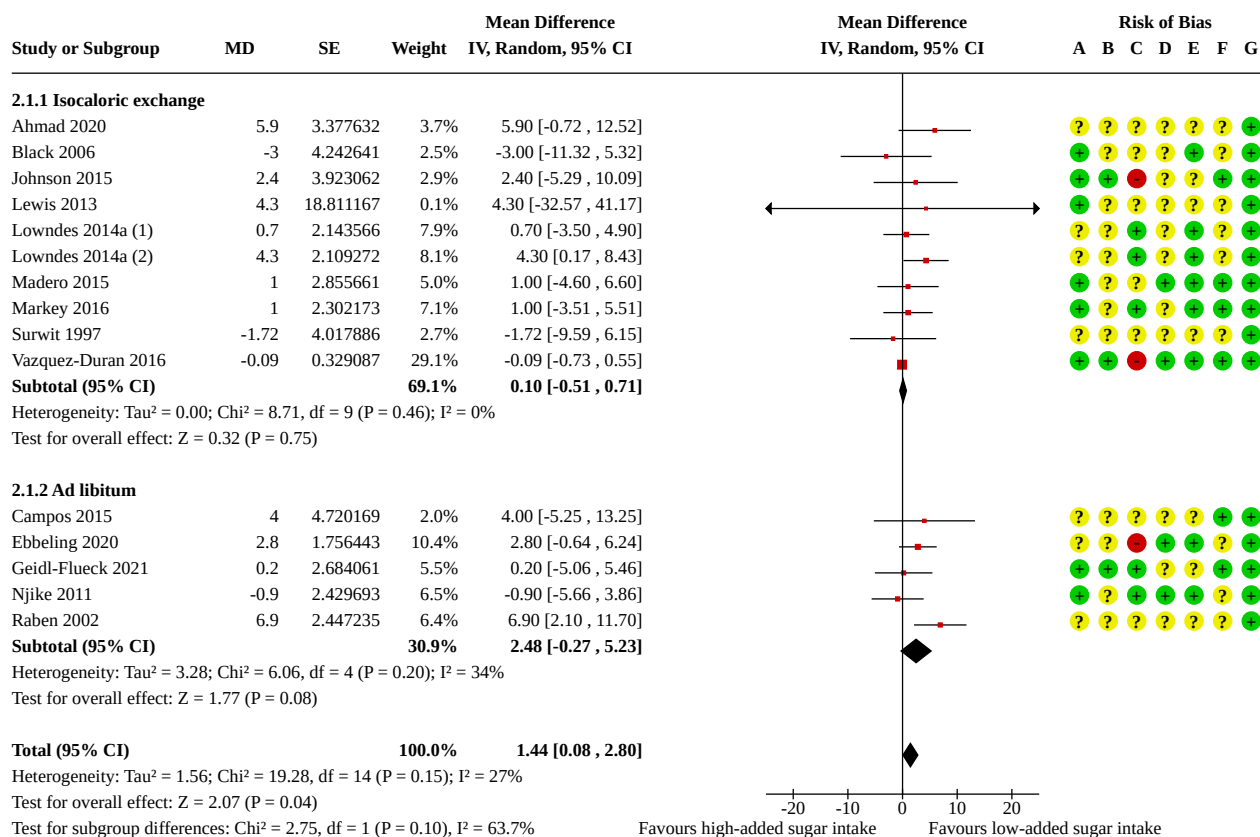
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Comparison 2. Subgroup analysis - Isocaloric intake or ad libitum

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
2.1 Systolic blood pressure (mmHg)	14		Mean Difference (IV, Random, 95% CI)	1.44 [0.08, 2.80]
2.1.1 Isocaloric exchange	9		Mean Difference (IV, Random, 95% CI)	0.10 [-0.51, 0.71]
2.1.2 Ad libitum	5		Mean Difference (IV, Random, 95% CI)	2.48 [-0.27, 5.23]
2.2 Diastolic blood pressure (mmHg)	14		Mean Difference (IV, Random, 95% CI)	1.52 [0.67, 2.37]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
2.2.1 Isocaloric exchange	9		Mean Difference (IV, Random, 95% CI)	2.15 [1.69, 2.60]
2.2.2 Ad libitum	5		Mean Difference (IV, Random, 95% CI)	0.81 [-1.64, 3.27]
2.3 Total cholesterol (mmol/L)	16		Mean Difference (IV, Random, 95% CI)	0.11 [0.01, 0.21]
2.3.1 Isocaloric exchange	11		Mean Difference (IV, Random, 95% CI)	0.14 [-0.00, 0.28]
2.3.2 Ad libitum	5		Mean Difference (IV, Random, 95% CI)	0.10 [-0.11, 0.31]
2.4 LDL-cholesterol (mmol/L)	12		Mean Difference (IV, Random, 95% CI)	0.01 [-0.08, 0.10]
2.4.1 Isocaloric exchange	10		Mean Difference (IV, Random, 95% CI)	0.06 [-0.05, 0.18]
2.4.2 Ad libitum	2		Mean Difference (IV, Random, 95% CI)	-0.06 [-0.13, 0.01]
2.5 HDL-cholesterol (mmol/L)	16		Mean Difference (IV, Random, 95% CI)	0.01 [-0.02, 0.04]
2.5.1 Isocaloric exchange	10		Mean Difference (IV, Random, 95% CI)	0.03 [-0.01, 0.06]
2.5.2 Ad libitum	6		Mean Difference (IV, Random, 95% CI)	-0.03 [-0.09, 0.04]
2.6 Triglycerides (mmol/L)	14		Mean Difference (IV, Random, 95% CI)	0.10 [0.03, 0.17]
2.6.1 Isocaloric exchange	9		Mean Difference (IV, Random, 95% CI)	0.08 [-0.01, 0.17]
2.6.2 Ad libitum	5		Mean Difference (IV, Random, 95% CI)	0.18 [-0.13, 0.49]
2.7 Fasting plasma glucose (mmol/L)	14		Mean Difference (IV, Random, 95% CI)	0.06 [-0.01, 0.13]
2.7.1 Isocaloric exchange	9		Mean Difference (IV, Random, 95% CI)	0.06 [-0.02, 0.15]
2.7.2 Ad libitum	5		Mean Difference (IV, Random, 95% CI)	0.04 [-0.08, 0.16]

Analysis 2.1. Comparison 2: Subgroup analysis - Isocaloric intake or ad libitum, Outcome 1: Systolic blood pressure (mmHg)



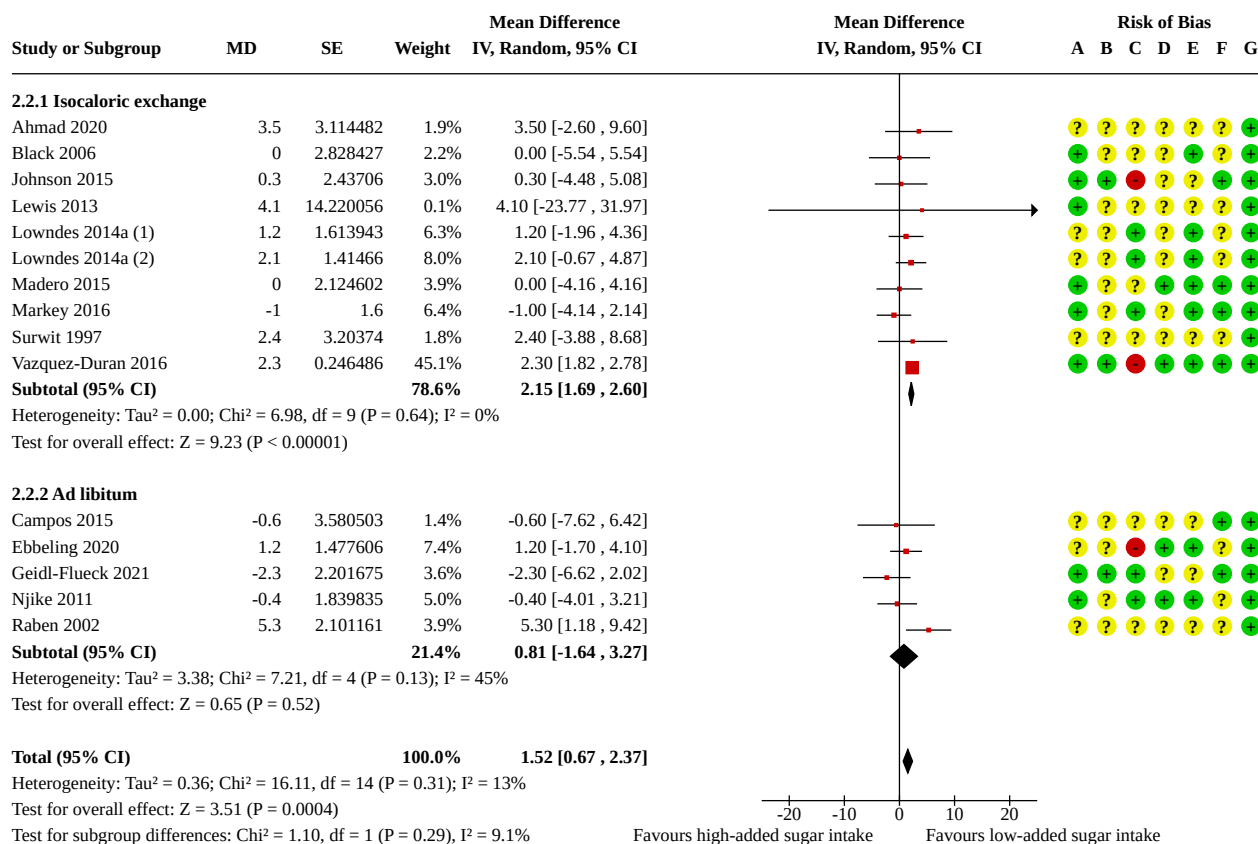
Footnotes

- (1) Shows the two arms of the intervention that compares high and low HFCS intake.
(2) Shows the two arms of the intervention that compares high and low sucrose intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Analysis 2.2. Comparison 2: Subgroup analysis - Isocaloric intake or ad libitum, Outcome 2: Diastolic blood pressure (mmHg)



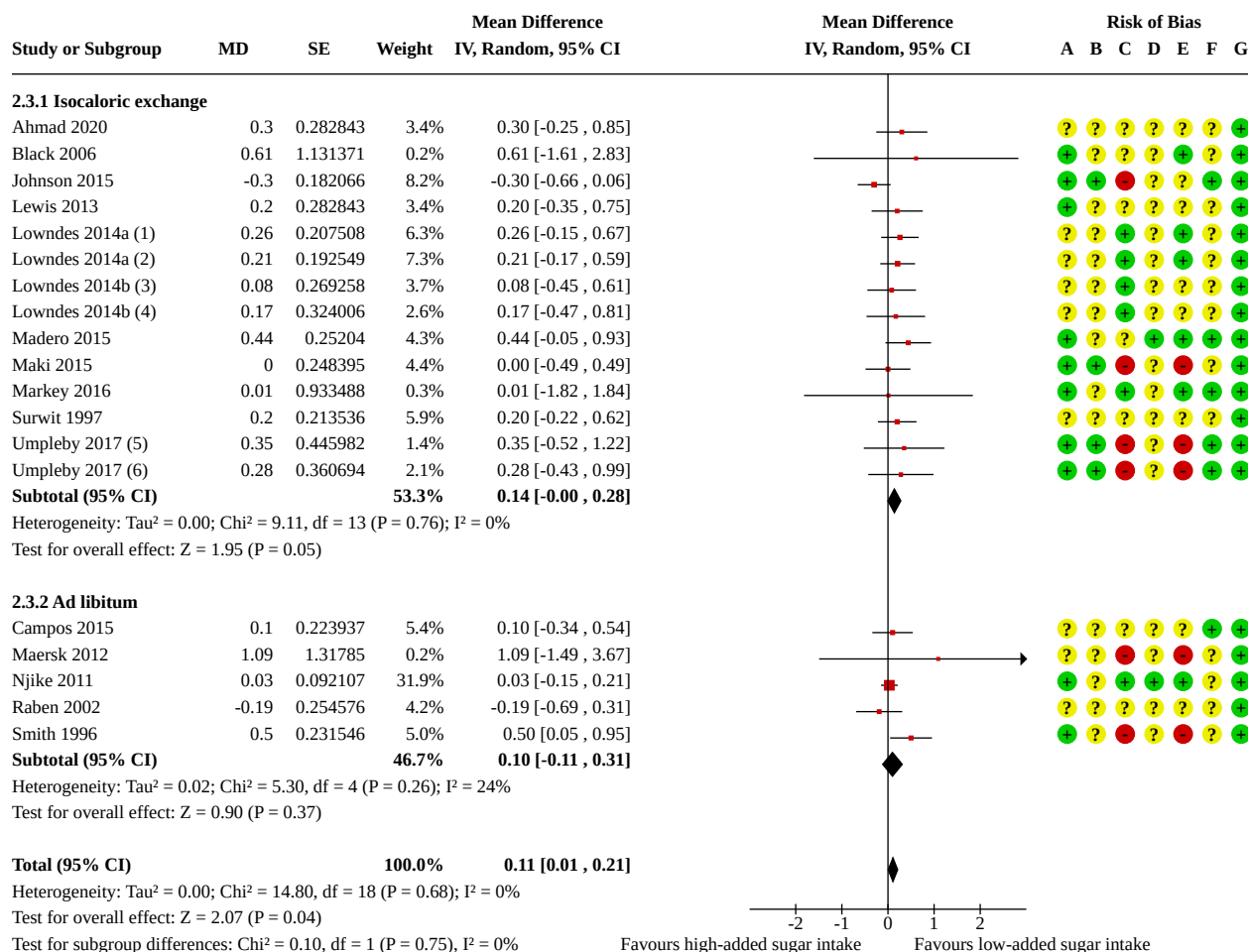
Footnotes

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
(2) Shows the two arms of the intervention that compares high and low HFCS intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Analysis 2.3. Comparison 2: Subgroup analysis - Isocaloric intake or ad libitum, Outcome 3: Total cholesterol (mmol/L)



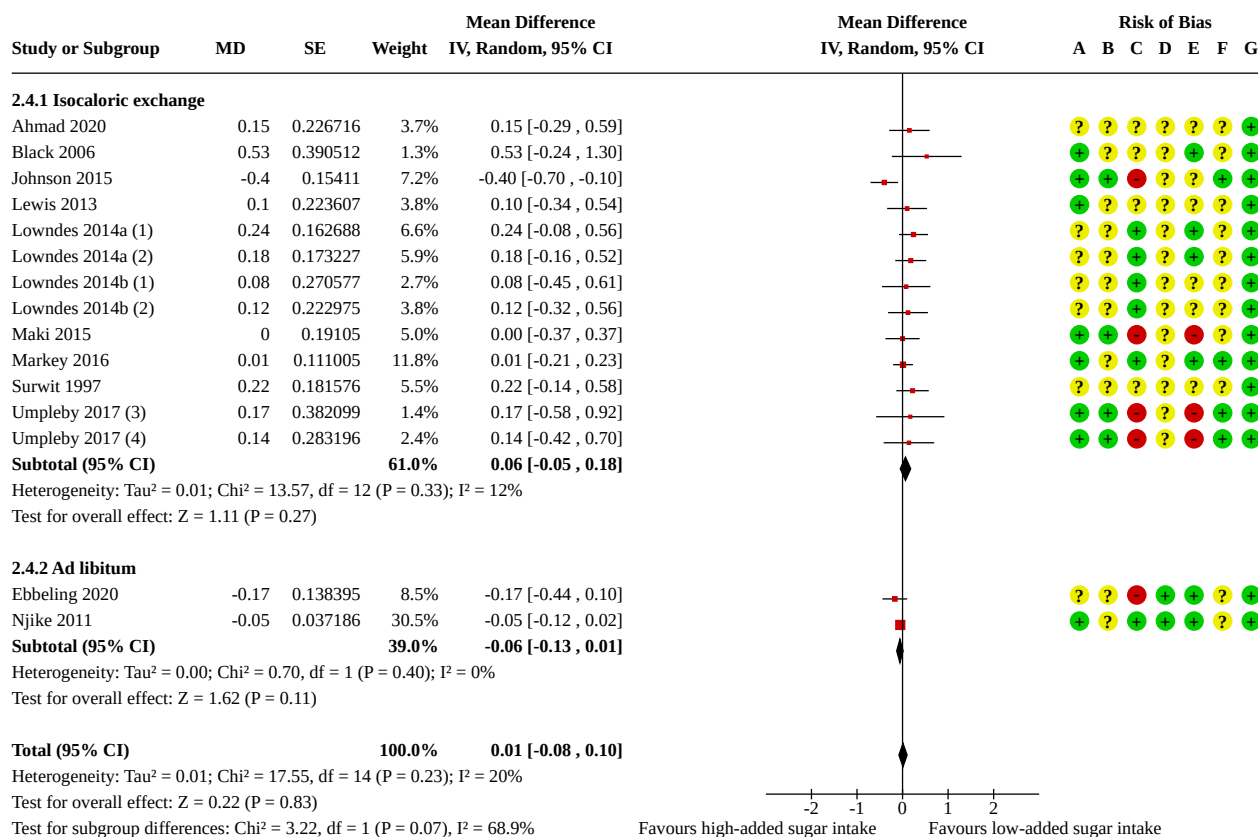
Footnotes

- (1) Shows the two arms of the intervention that compares high and low sucrose intake
- (2) Shows the two arms of the intervention that compares high and low HFCS intake
- (3) Shows the two arms of the intervention that compares high and low sucrose intake.
- (4) Shows the two arms of the intervention that compares high and low HFCS intake.
- (5) Includes the individuals diagnosed with NAFLD.
- (6) Includes the individuals not diagnosed with NAFLD, the controls.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 2.4. Comparison 2: Subgroup analysis - Isocaloric intake or ad libitum, Outcome 4: LDL-cholesterol (mmol/L)

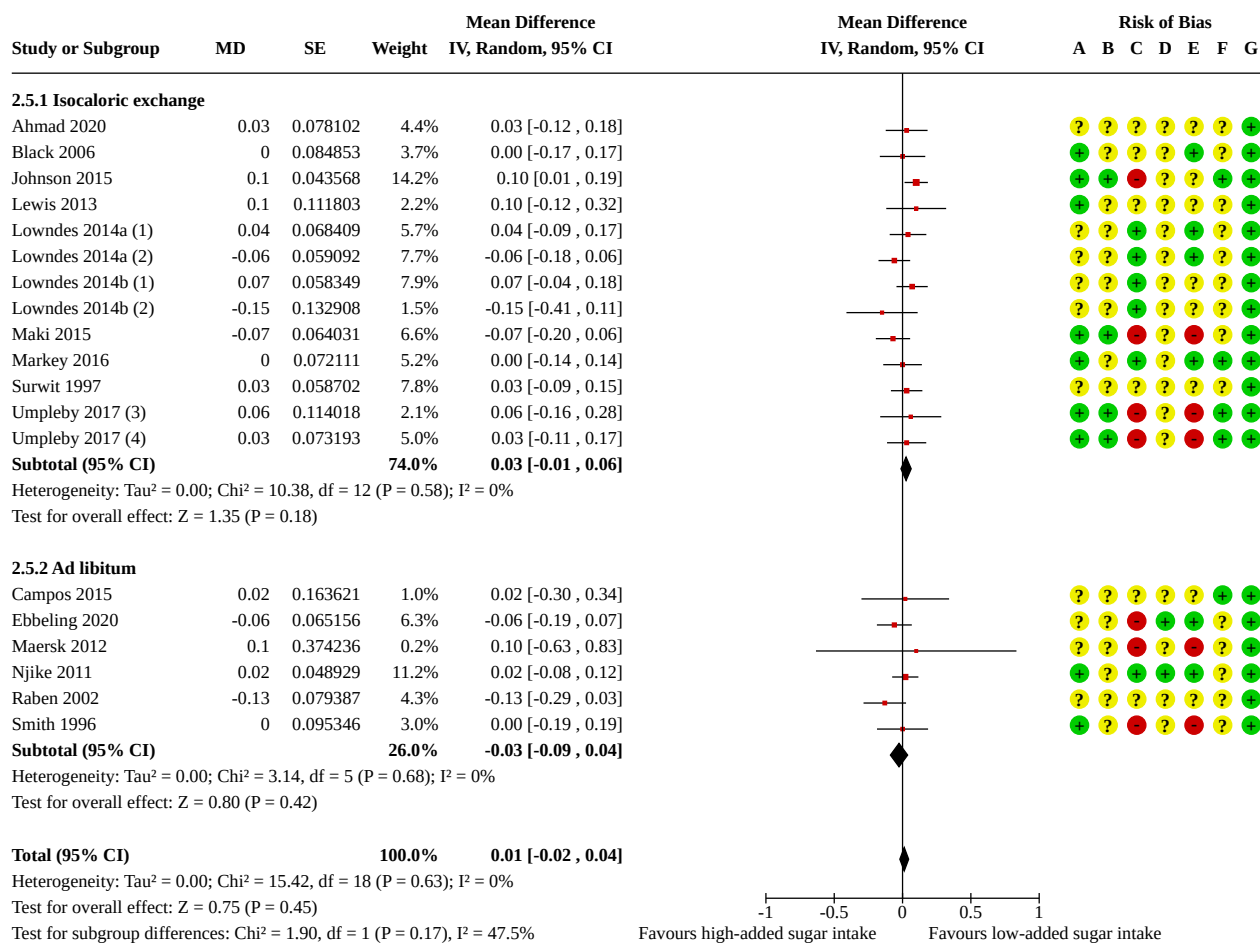


Footnotes

- (1) Shows the two arms of the intervention that compares high and low HFCS intake.
- (2) Shows the two arms of the intervention that compares high and low sucrose intake.
- (3) Includes the individuals diagnosed with NAFLD.
- (4) Includes the individuals not diagnosed with NAFLD, the controls.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

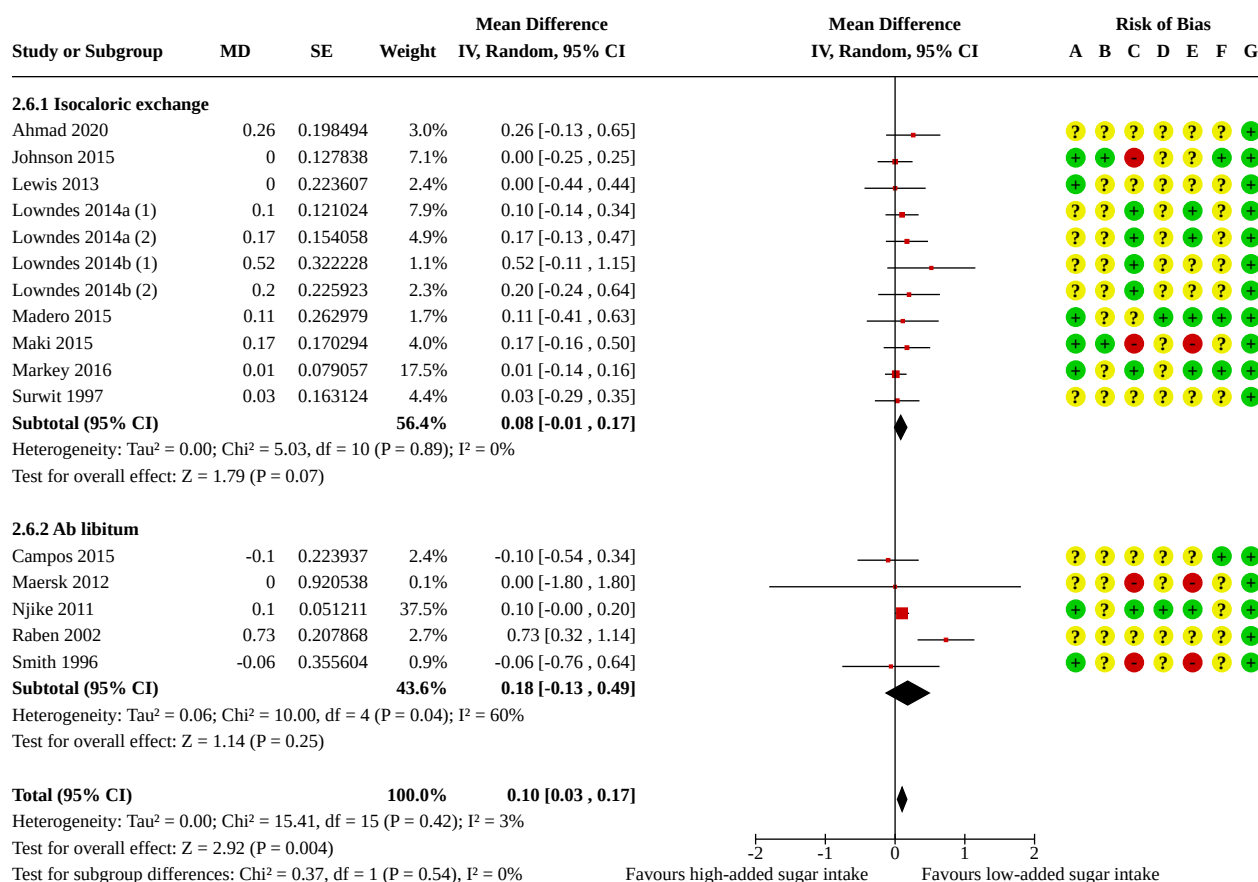
Analysis 2.5. Comparison 2: Subgroup analysis - Isocaloric intake or ad libitum, Outcome 5: HDL-cholesterol (mmol/L)**Footnotes**

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
- (2) Shows the two arms of the intervention that compares high and low HFCS intake.
- (3) Includes the individuals diagnosed with NAFLD.
- (4) Includes the individuals not diagnosed with NAFLD, the controls.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 2.6. Comparison 2: Subgroup analysis - Isocaloric intake or ad libitum, Outcome 6: Triglycerides (mmol/L)

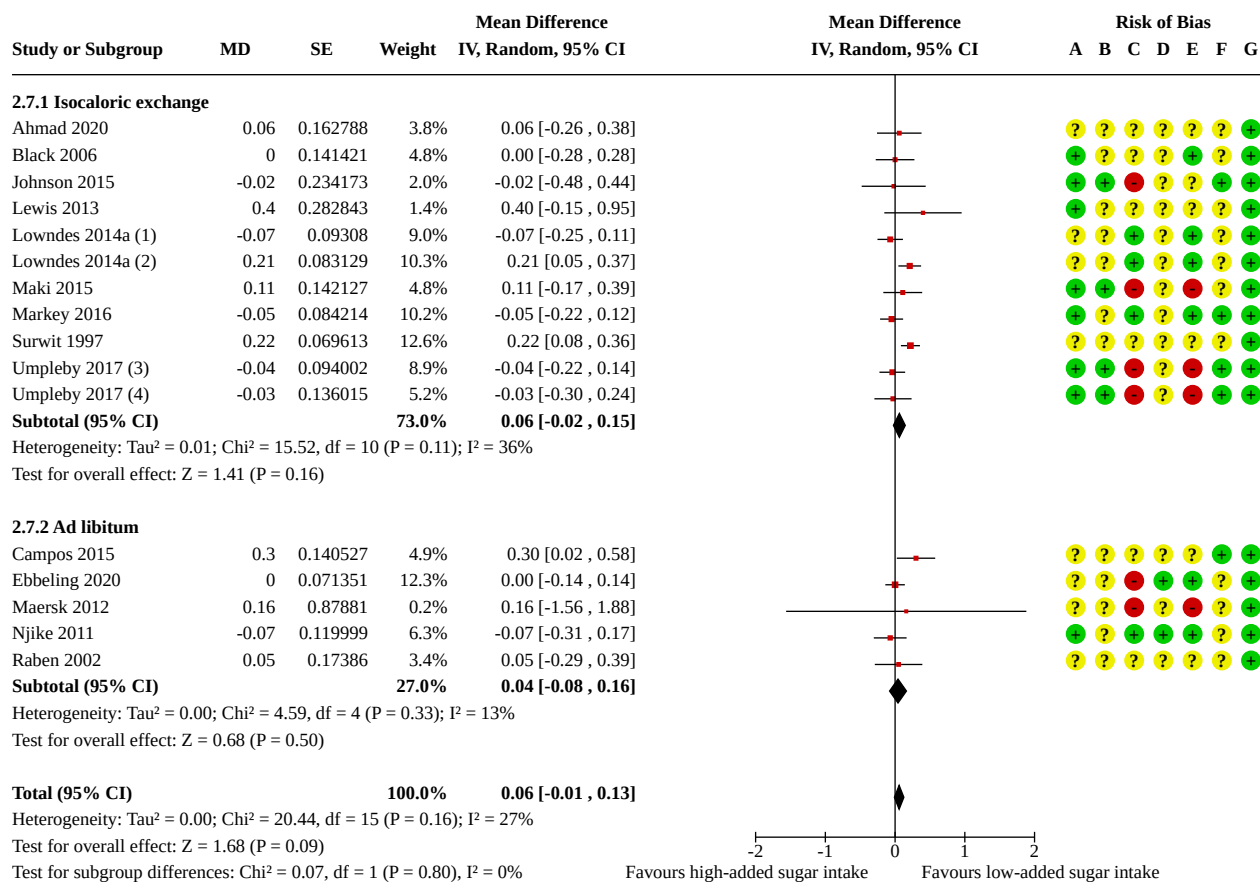


Footnotes

- (1) Shows the two arms of the intervention that compares high and low HFCS intake.
(2) Shows the two arms of the intervention that compares high and low sucrose intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Analysis 2.7. Comparison 2: Subgroup analysis - Isocaloric intake or ad libitum, Outcome 7: Fasting plasma glucose (mmol/L)**Footnotes**

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
 (2) Shows the two arms of the intervention that compares high and low HFCS intake.
 (3) Includes the individuals diagnosed with NAFLD.
 (4) Includes the individuals not diagnosed with NAFLD, the controls.

Risk of bias legend

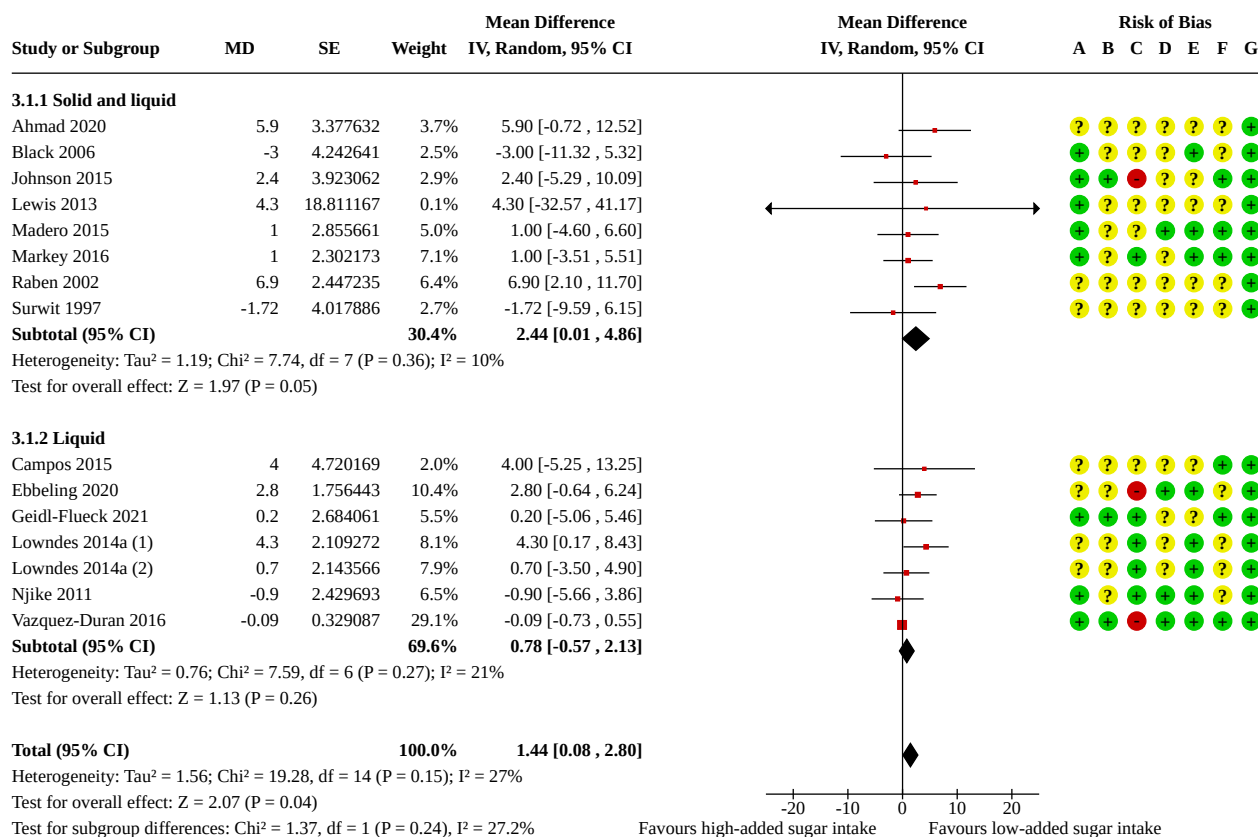
- (A) Random sequence generation (selection bias)
 (B) Allocation concealment (selection bias)
 (C) Blinding of participants and personnel (performance bias)
 (D) Blinding of outcome assessment (detection bias)
 (E) Incomplete outcome data (attrition bias)
 (F) Selective reporting (reporting bias)
 (G) Other bias

Comparison 3. Subgroup analysis - Solid and liquid or liquid state

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
3.1 Systolic blood pressure (mmHg)	14		Mean Difference (IV, Random, 95% CI)	1.44 [0.08, 2.80]
3.1.1 Solid and liquid	8		Mean Difference (IV, Random, 95% CI)	2.44 [0.01, 4.86]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
3.1.2 Liquid	6		Mean Difference (IV, Random, 95% CI)	0.78 [-0.57, 2.13]
3.2 Diastolic blood pressure (mmHg)	14		Mean Difference (IV, Random, 95% CI)	1.52 [0.67, 2.37]
3.2.1 Solid and liquid	8		Mean Difference (IV, Random, 95% CI)	1.11 [-0.60, 2.81]
3.2.2 Liquid	6		Mean Difference (IV, Random, 95% CI)	1.54 [0.49, 2.59]
3.3 Total cholesterol (mmol/L)	16		Mean Difference (IV, Random, 95% CI)	0.11 [0.01, 0.21]
3.3.1 Solid and liquid	12		Mean Difference (IV, Random, 95% CI)	0.13 [-0.03, 0.30]
3.3.2 Liquid	4		Mean Difference (IV, Random, 95% CI)	0.09 [-0.04, 0.23]
3.4 LDL-cholesterol (mmol/L)	12		Mean Difference (IV, Random, 95% CI)	0.01 [-0.08, 0.10]
3.4.1 Solid and liquid	8		Mean Difference (IV, Random, 95% CI)	0.03 [-0.13, 0.18]
3.4.2 Liquid	4		Mean Difference (IV, Random, 95% CI)	0.00 [-0.11, 0.11]
3.5 HDL-cholesterol (mmol/L)	16		Mean Difference (IV, Random, 95% CI)	0.01 [-0.02, 0.04]
3.5.1 Solid and liquid	11		Mean Difference (IV, Random, 95% CI)	0.02 [-0.02, 0.06]
3.5.2 Liquid	5		Mean Difference (IV, Random, 95% CI)	-0.00 [-0.05, 0.05]
3.6 Triglycerides (mmol/L)	14		Mean Difference (IV, Random, 95% CI)	0.10 [0.03, 0.17]
3.6.1 Solid and liquid	10		Mean Difference (IV, Random, 95% CI)	0.11 [-0.02, 0.25]
3.6.2 Liquid	4		Mean Difference (IV, Random, 95% CI)	0.11 [0.02, 0.19]
3.7 Fasting plasma glucose (mmol/L)	14		Mean Difference (IV, Random, 95% CI)	0.06 [-0.01, 0.13]
3.7.1 Solid and liquid	10		Mean Difference (IV, Random, 95% CI)	0.06 [-0.01, 0.14]
3.7.2 Liquid	4		Mean Difference (IV, Random, 95% CI)	0.06 [-0.07, 0.20]

Analysis 3.1. Comparison 3: Subgroup analysis - Solid and liquid or liquid state, Outcome 1: Systolic blood pressure (mmHg)



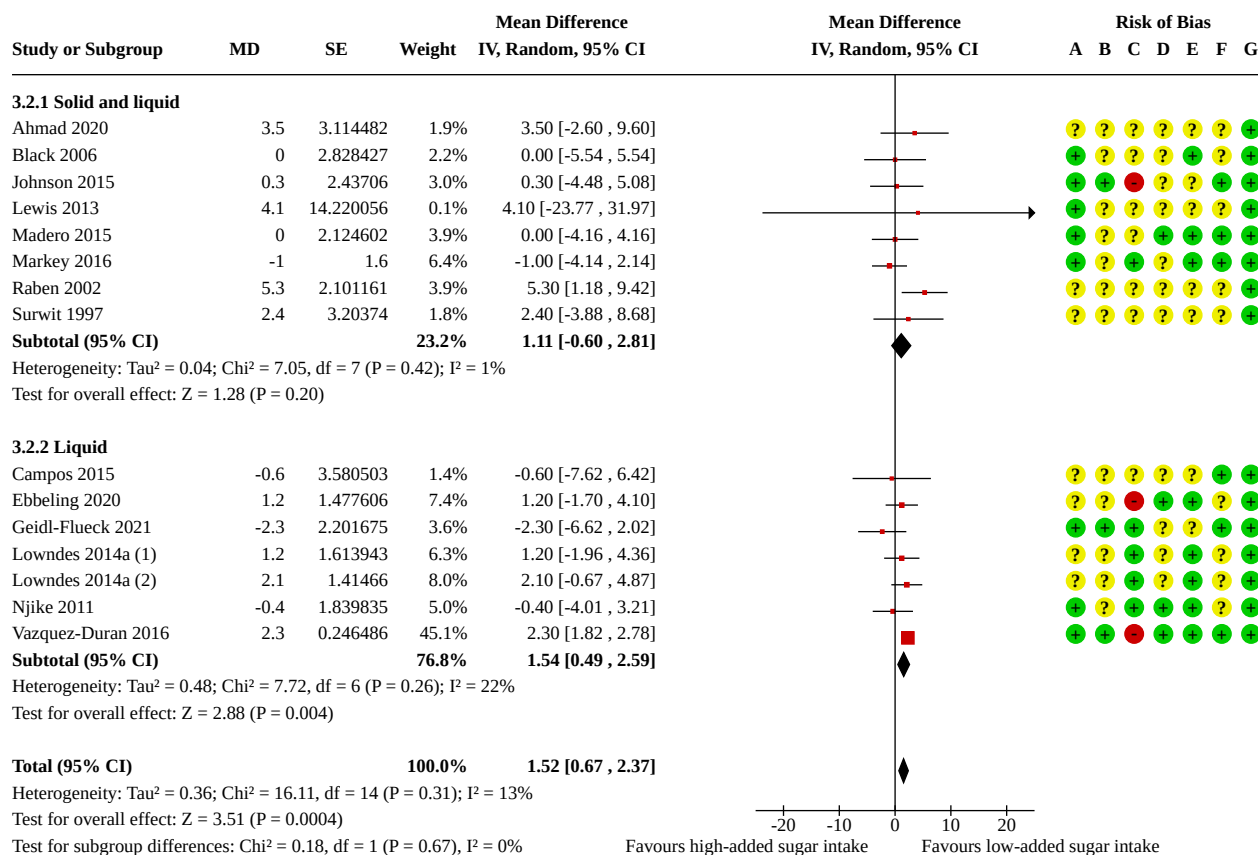
Footnotes

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
(2) Shows the two arms of the intervention that compares high and low HFCS intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Analysis 3.2. Comparison 3: Subgroup analysis - Solid and liquid or liquid state, Outcome 2: Diastolic blood pressure (mmHg)



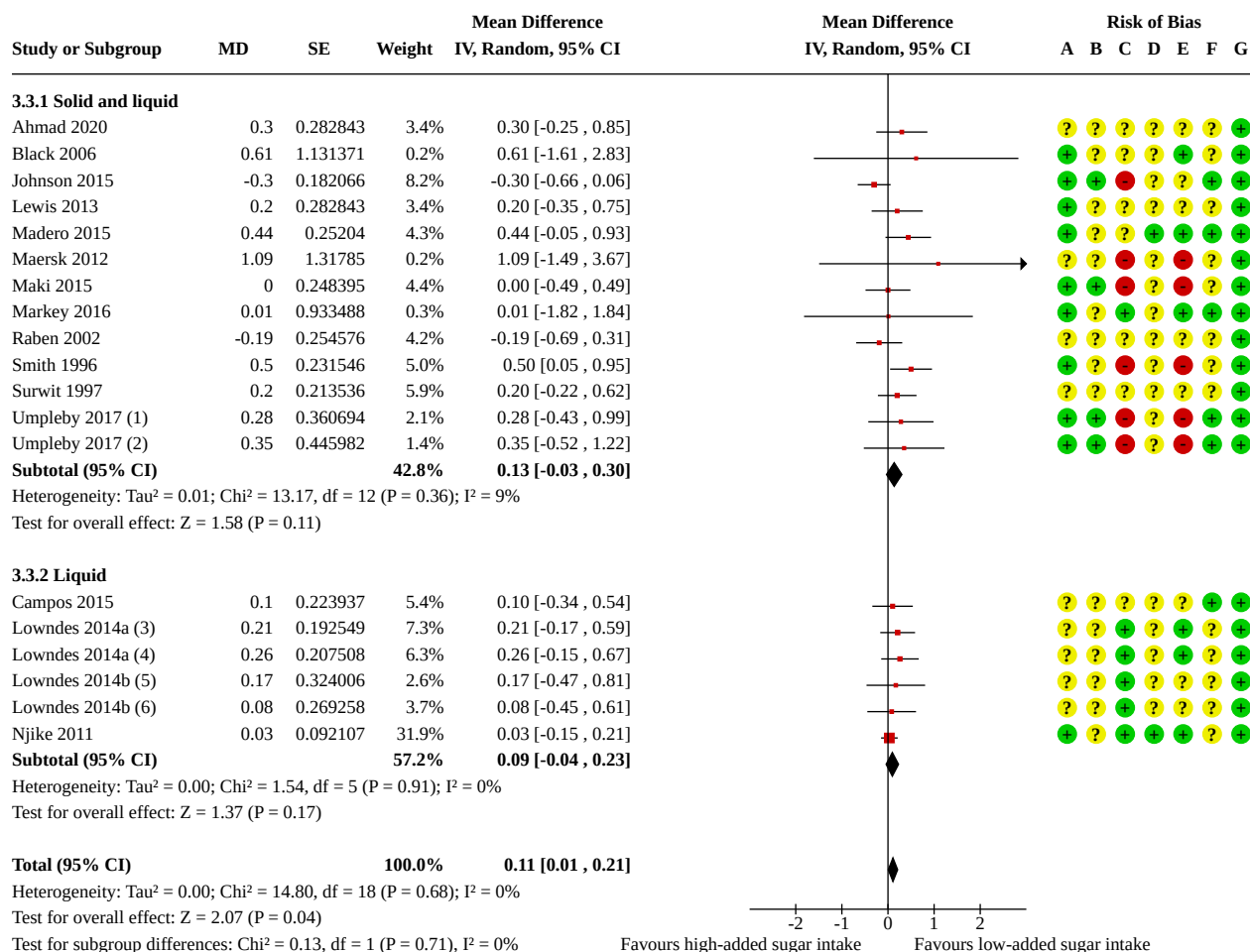
Footnotes

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
(2) Shows the two arms of the intervention that compares high and low HFCS intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Analysis 3.3. Comparison 3: Subgroup analysis - Solid and liquid or liquid state, Outcome 3: Total cholesterol (mmol/L)



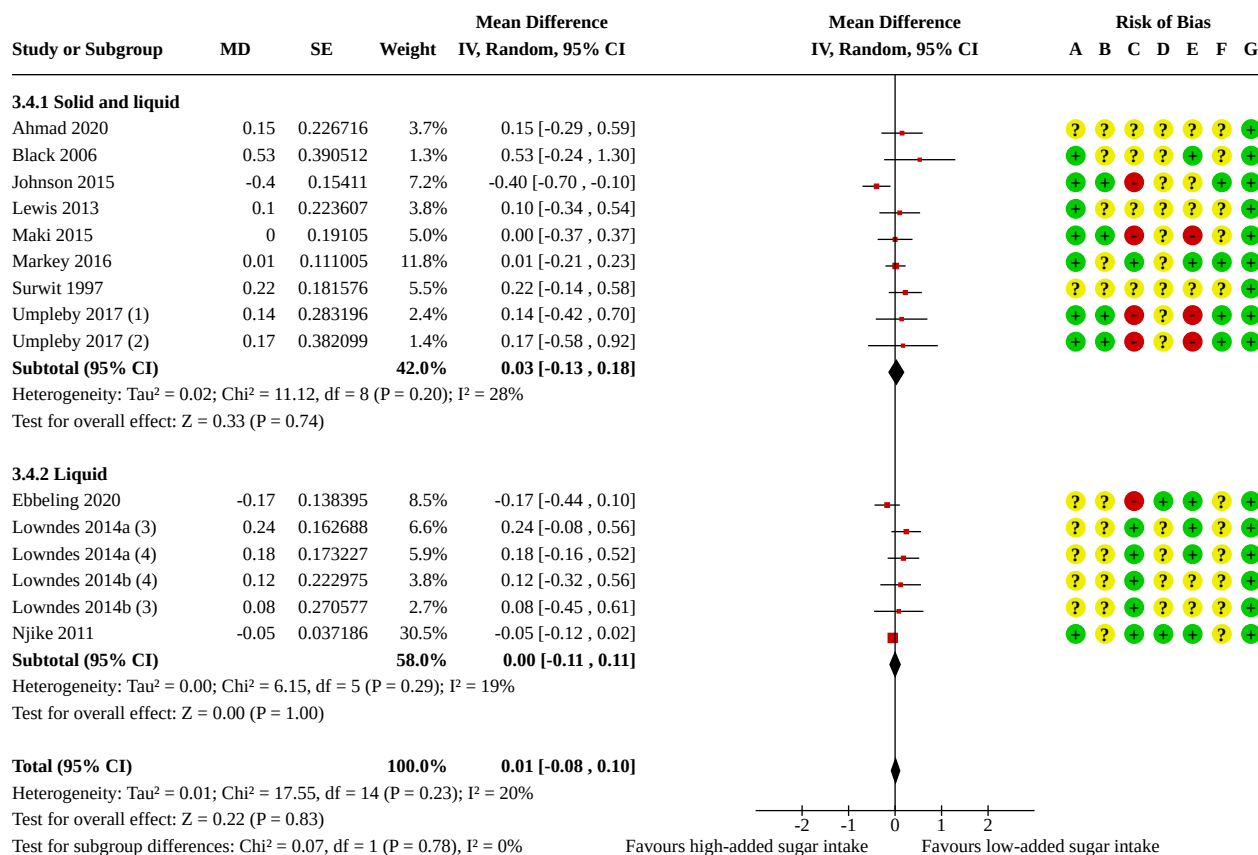
Footnotes

- (1) Includes the individuals not diagnosed with NAFLD, the controls.
- (2) Includes the individuals diagnosed with NAFLD.
- (3) Shows the two arms of the intervention that compares high and low HFCS intake
- (4) Shows the two arms of the intervention that compares high and low sucrose intake
- (5) Shows the two arms of the intervention that compares high and low HFCS intake.
- (6) Shows the two arms of the intervention that compares high and low sucrose intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 3.4. Comparison 3: Subgroup analysis - Solid and liquid or liquid state, Outcome 4: LDL-cholesterol (mmol/L)



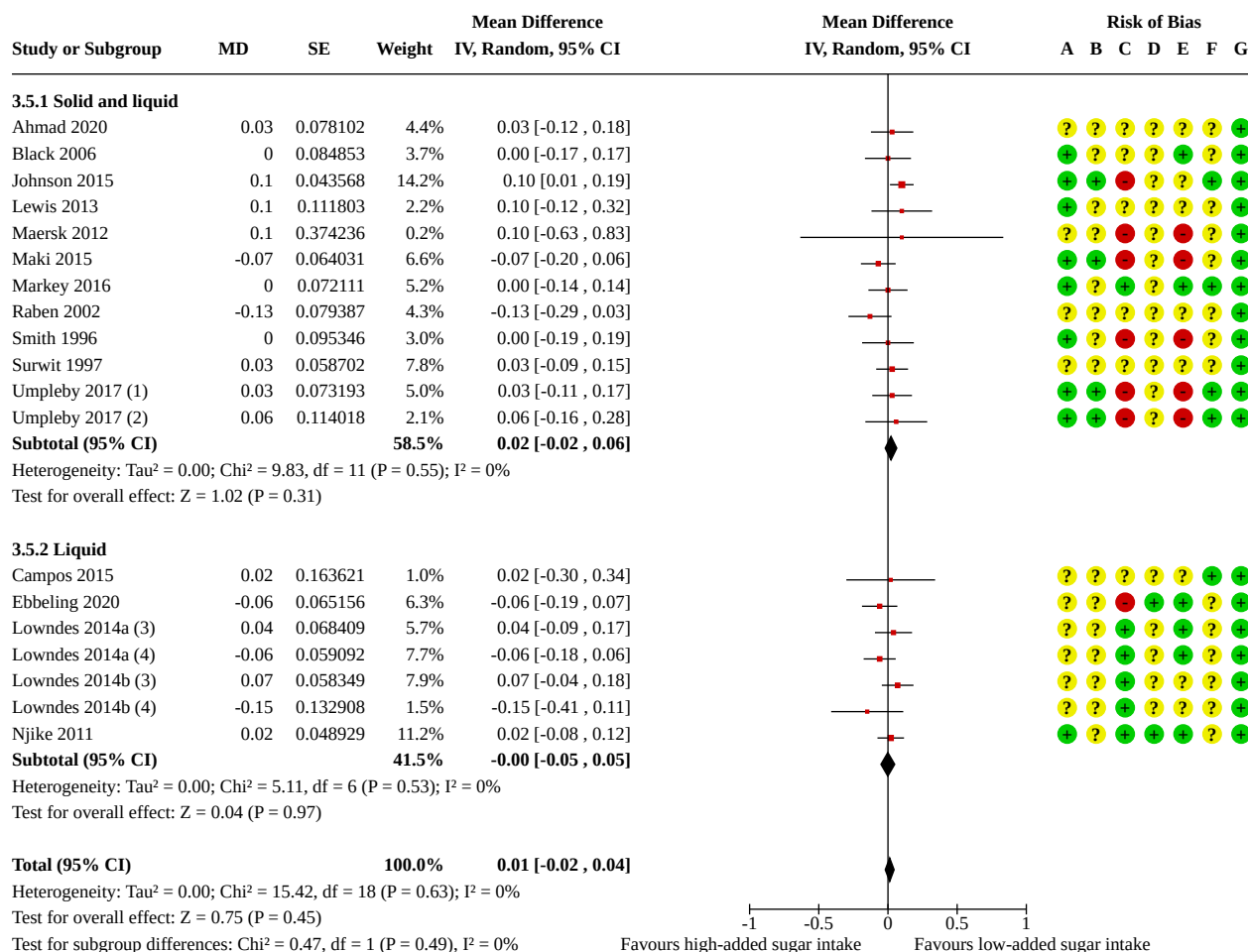
Footnotes

- (1) Includes the individuals not diagnosed with NAFLD, the controls.
- (2) Includes the individuals diagnosed with NAFLD.
- (3) Shows the two arms of the intervention that compares high and low HFCS intake.
- (4) Shows the two arms of the intervention that compares high and low sucrose intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 3.5. Comparison 3: Subgroup analysis - Solid and liquid or liquid state, Outcome 5: HDL-cholesterol (mmol/L)



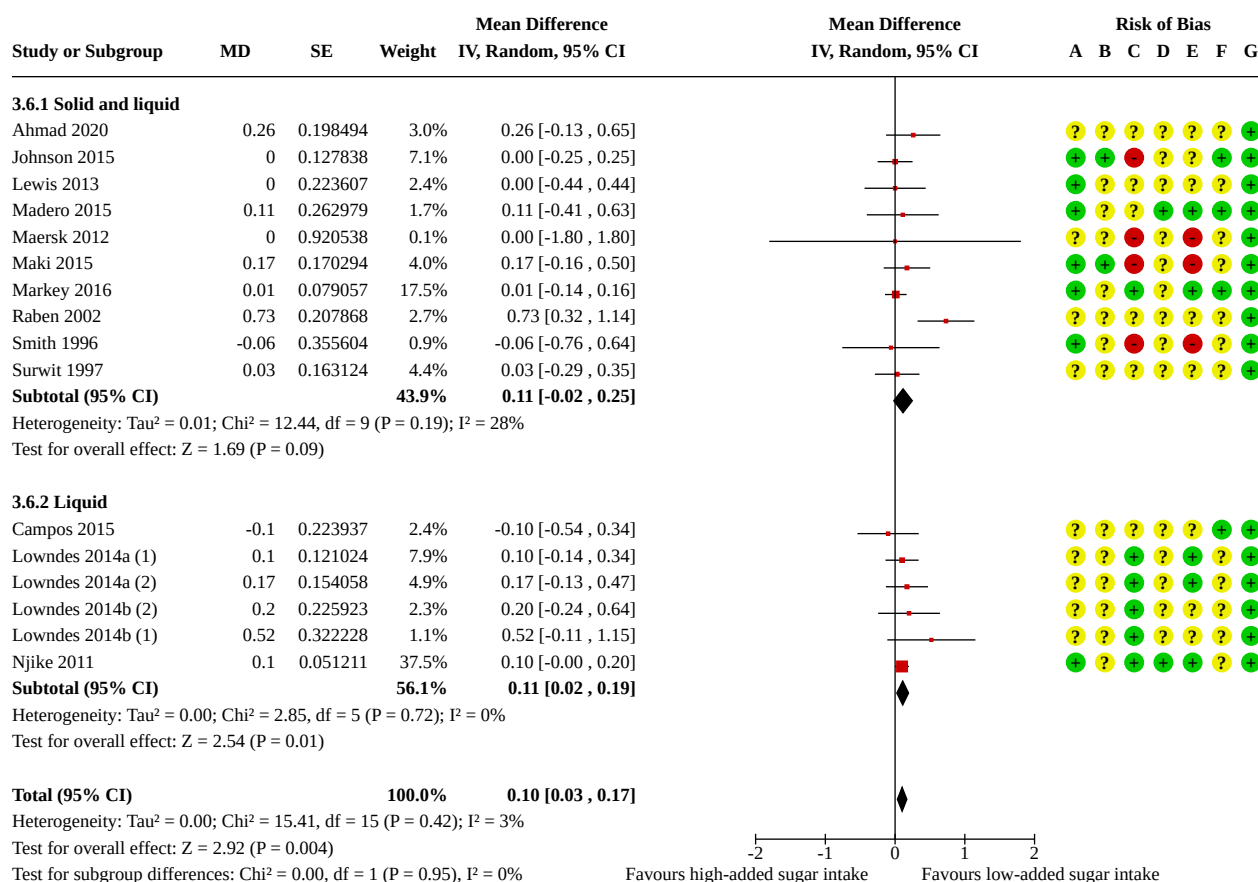
Footnotes

- (1) Includes the individuals not diagnosed with NAFLD, the controls.
- (2) Includes the individuals diagnosed with NAFLD.
- (3) Shows the two arms of the intervention that compares high and low sucrose intake.
- (4) Shows the two arms of the intervention that compares high and low HFCS intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 3.6. Comparison 3: Subgroup analysis - Solid and liquid or liquid state, Outcome 6: Triglycerides (mmol/L)

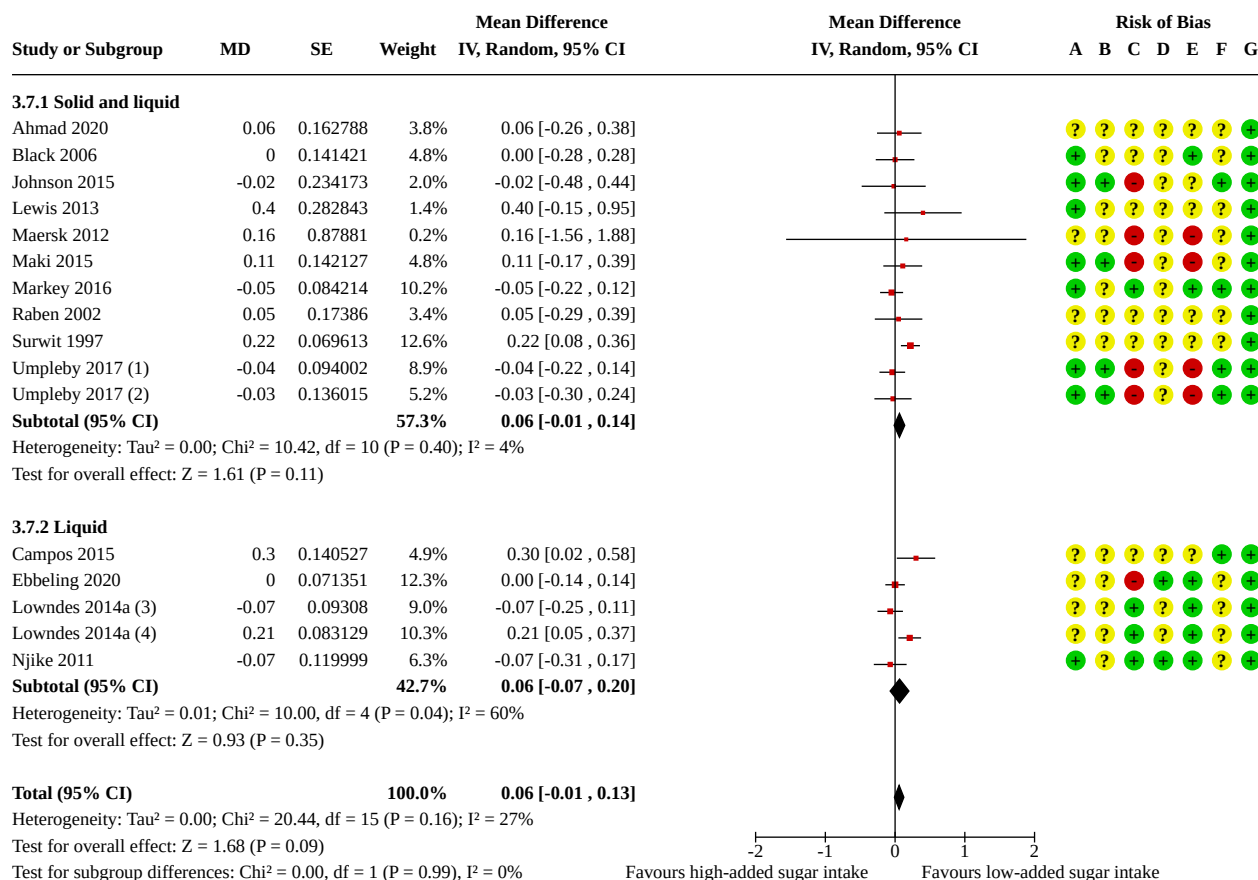


Footnotes

- (1) Shows the two arms of the intervention that compares high and low HFCS intake.
(2) Shows the two arms of the intervention that compares high and low sucrose intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Analysis 3.7. Comparison 3: Subgroup analysis - Solid and liquid or liquid state, Outcome 7: Fasting plasma glucose (mmol/L)**Footnotes**

- (1) Includes the individuals diagnosed with NAFLD.
 (2) Includes the individuals not diagnosed with NAFLD, the controls.
 (3) Shows the two arms of the intervention that compares high and low sucrose intake.
 (4) Shows the two arms of the intervention that compares high and low HFCS intake.

Risk of bias legend

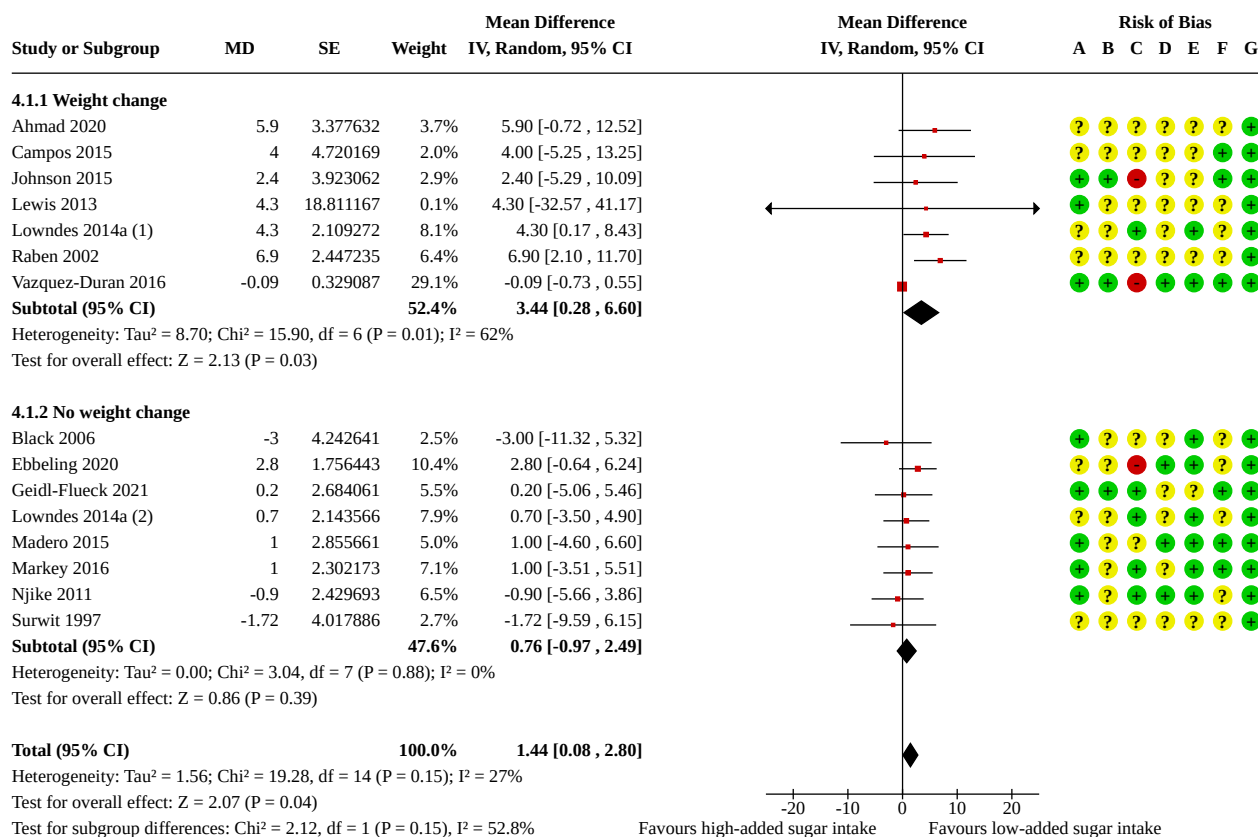
- (A) Random sequence generation (selection bias)
 (B) Allocation concealment (selection bias)
 (C) Blinding of participants and personnel (performance bias)
 (D) Blinding of outcome assessment (detection bias)
 (E) Incomplete outcome data (attrition bias)
 (F) Selective reporting (reporting bias)
 (G) Other bias

Comparison 4. Subgroup analysis - Difference in weight change

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
4.1 Systolic blood pressure (mmHg)	14		Mean Difference (IV, Random, 95% CI)	1.44 [0.08, 2.80]
4.1.1 Weight change	7		Mean Difference (IV, Random, 95% CI)	3.44 [0.28, 6.60]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
4.1.2 No weight change	8		Mean Difference (IV, Random, 95% CI)	0.76 [-0.97, 2.49]
4.2 Diastolic blood pressure (mmHg)	14		Mean Difference (IV, Random, 95% CI)	1.52 [0.67, 2.37]
4.2.1 Weight change	7		Mean Difference (IV, Random, 95% CI)	2.29 [1.82, 2.76]
4.2.2 No weight change	8		Mean Difference (IV, Random, 95% CI)	0.37 [-0.92, 1.66]
4.3 Total cholesterol (mmol/L)	16		Mean Difference (IV, Random, 95% CI)	0.11 [0.01, 0.21]
4.3.1 Weight change	8		Mean Difference (IV, Random, 95% CI)	0.12 [-0.07, 0.32]
4.3.2 No weight change	9		Mean Difference (IV, Random, 95% CI)	0.11 [-0.02, 0.24]
4.4 LDL-cholesterol (mmol/L)	12		Mean Difference (IV, Random, 95% CI)	0.01 [-0.08, 0.10]
4.4.1 Weight change	5		Mean Difference (IV, Random, 95% CI)	0.02 [-0.22, 0.25]
4.4.2 No weight change	8		Mean Difference (IV, Random, 95% CI)	-0.00 [-0.09, 0.08]
4.5 HDL-cholesterol (mmol/L)	16		Mean Difference (IV, Random, 95% CI)	0.01 [-0.02, 0.04]
4.5.1 Weight change	7		Mean Difference (IV, Random, 95% CI)	0.02 [-0.04, 0.08]
4.5.2 No weight change	10		Mean Difference (IV, Random, 95% CI)	0.00 [-0.04, 0.05]
4.6 Triglycerides (mmol/L)	14		Mean Difference (IV, Random, 95% CI)	0.10 [0.03, 0.17]
4.6.1 Weight change	6		Mean Difference (IV, Random, 95% CI)	0.17 [-0.05, 0.39]
4.6.2 No weight change	9		Mean Difference (IV, Random, 95% CI)	0.09 [0.01, 0.16]
4.7 Fasting plasma glucose (mmol/L)	14		Mean Difference (IV, Random, 95% CI)	0.06 [-0.01, 0.13]
4.7.1 Weight change	7		Mean Difference (IV, Random, 95% CI)	0.02 [-0.07, 0.12]
4.7.2 No weight change	8		Mean Difference (IV, Random, 95% CI)	0.07 [-0.02, 0.17]

Analysis 4.1. Comparison 4: Subgroup analysis - Difference in weight change, Outcome 1: Systolic blood pressure (mmHg)



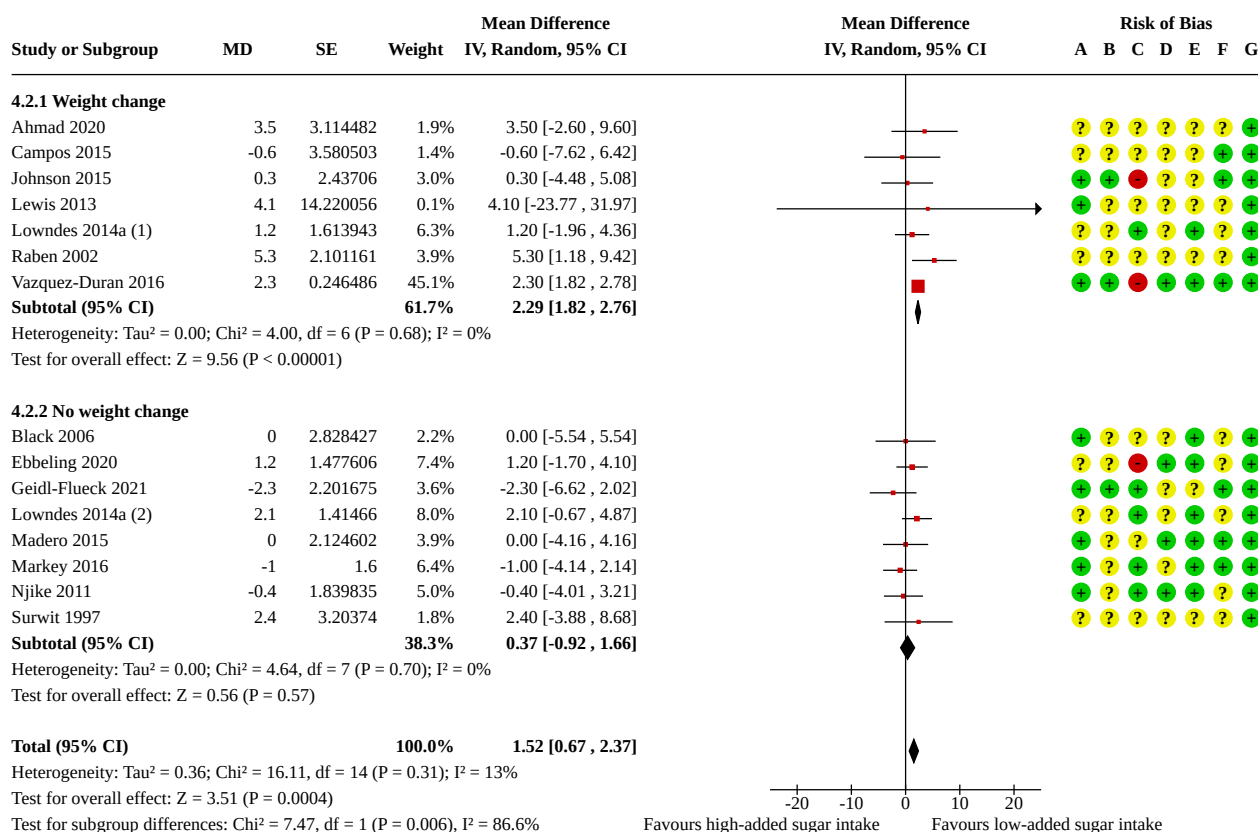
Footnotes

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
(2) Shows the two arms of the intervention that compares high and low HFCS intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Analysis 4.2. Comparison 4: Subgroup analysis - Difference in weight change, Outcome 2: Diastolic blood pressure (mmHg)



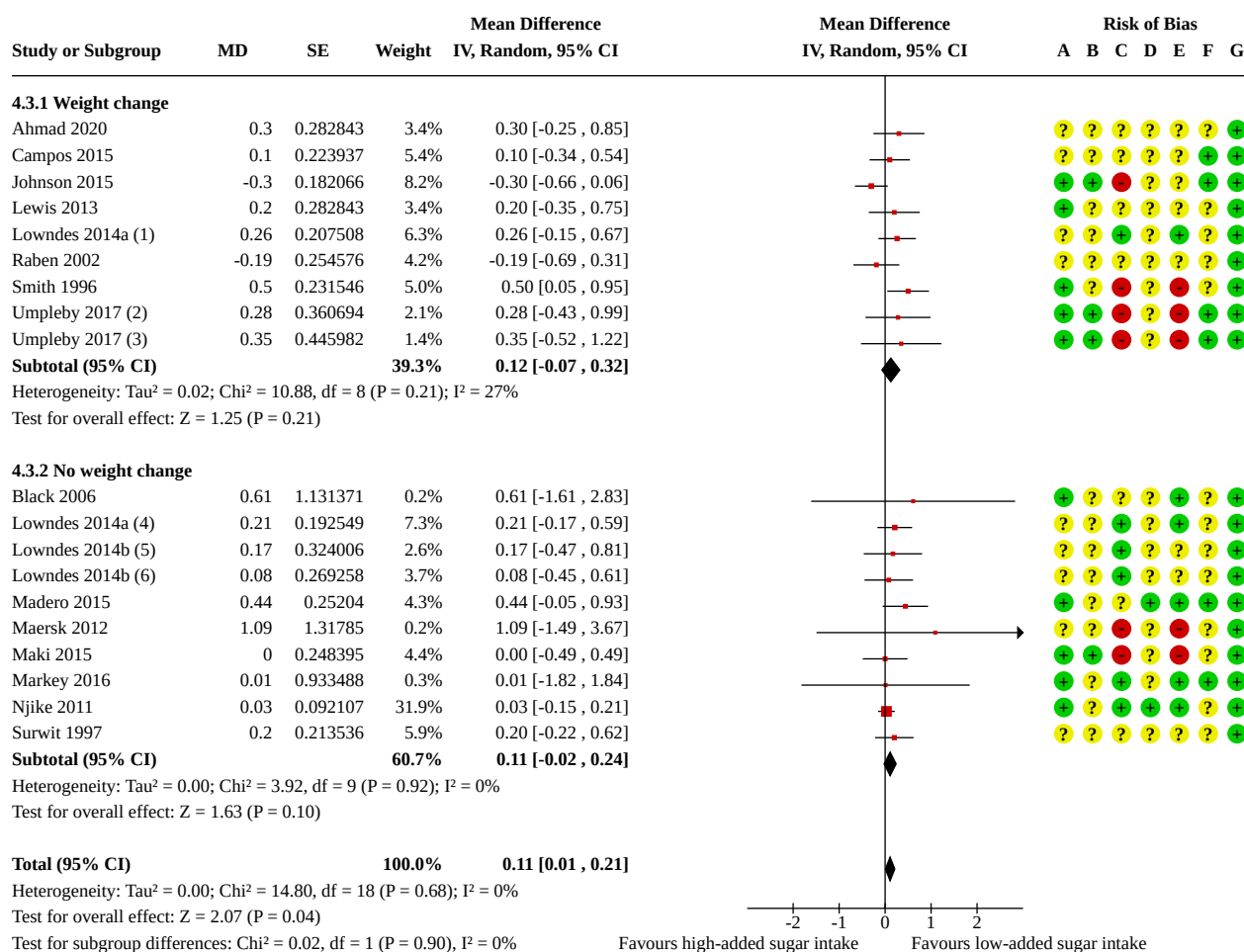
Footnotes

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
(2) Shows the two arms of the intervention that compares high and low HFCS intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Analysis 4.3. Comparison 4: Subgroup analysis - Difference in weight change, Outcome 3: Total cholesterol (mmol/L)



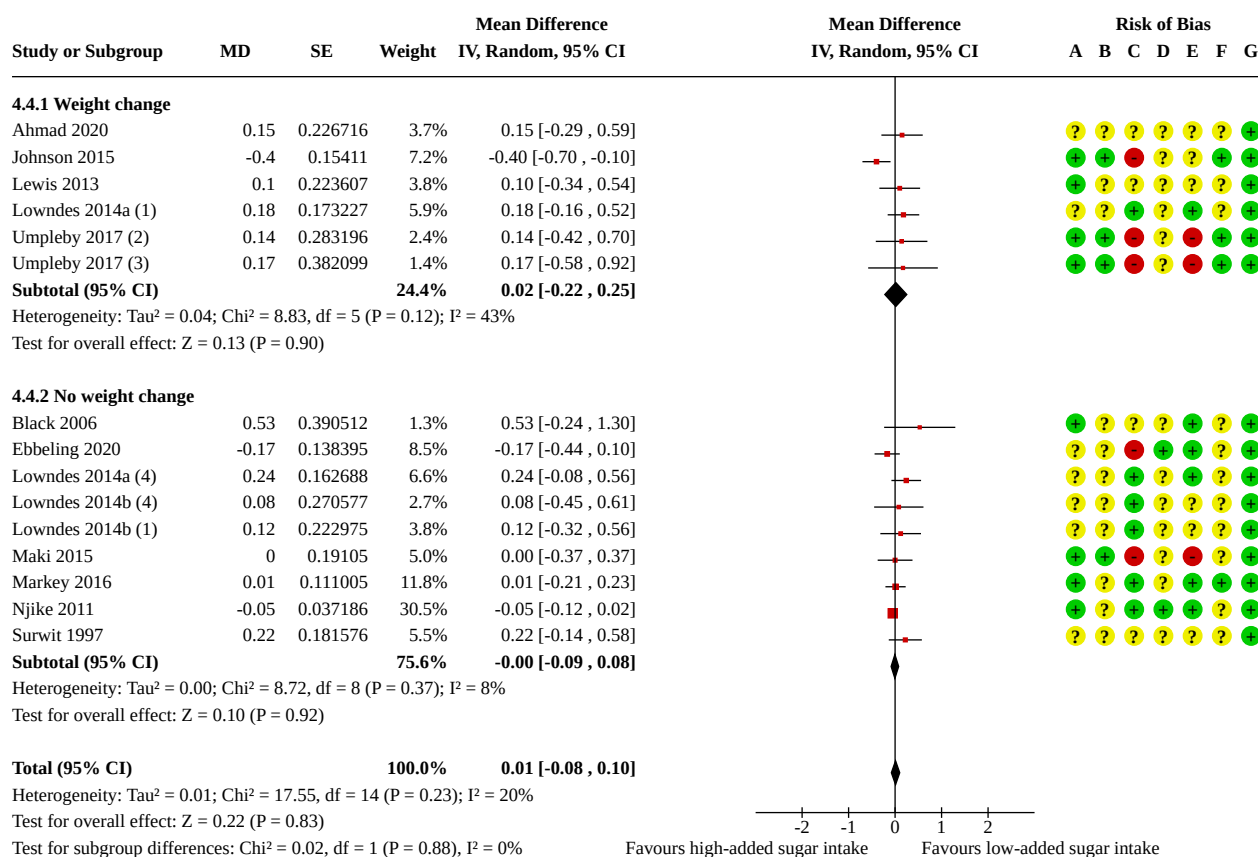
Footnotes

- (1) Shows the two arms of the intervention that compares high and low sucrose intake
- (2) Includes the individuals not diagnosed with NAFLD, the controls.
- (3) Includes the individuals diagnosed with NAFLD.
- (4) Shows the two arms of the intervention that compares high and low HFCS intake
- (5) Shows the two arms of the intervention that compares high and low HFCS intake.
- (6) Shows the two arms of the intervention that compares high and low sucrose intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 4.4. Comparison 4: Subgroup analysis - Difference in weight change, Outcome 4: LDL-cholesterol (mmol/L)



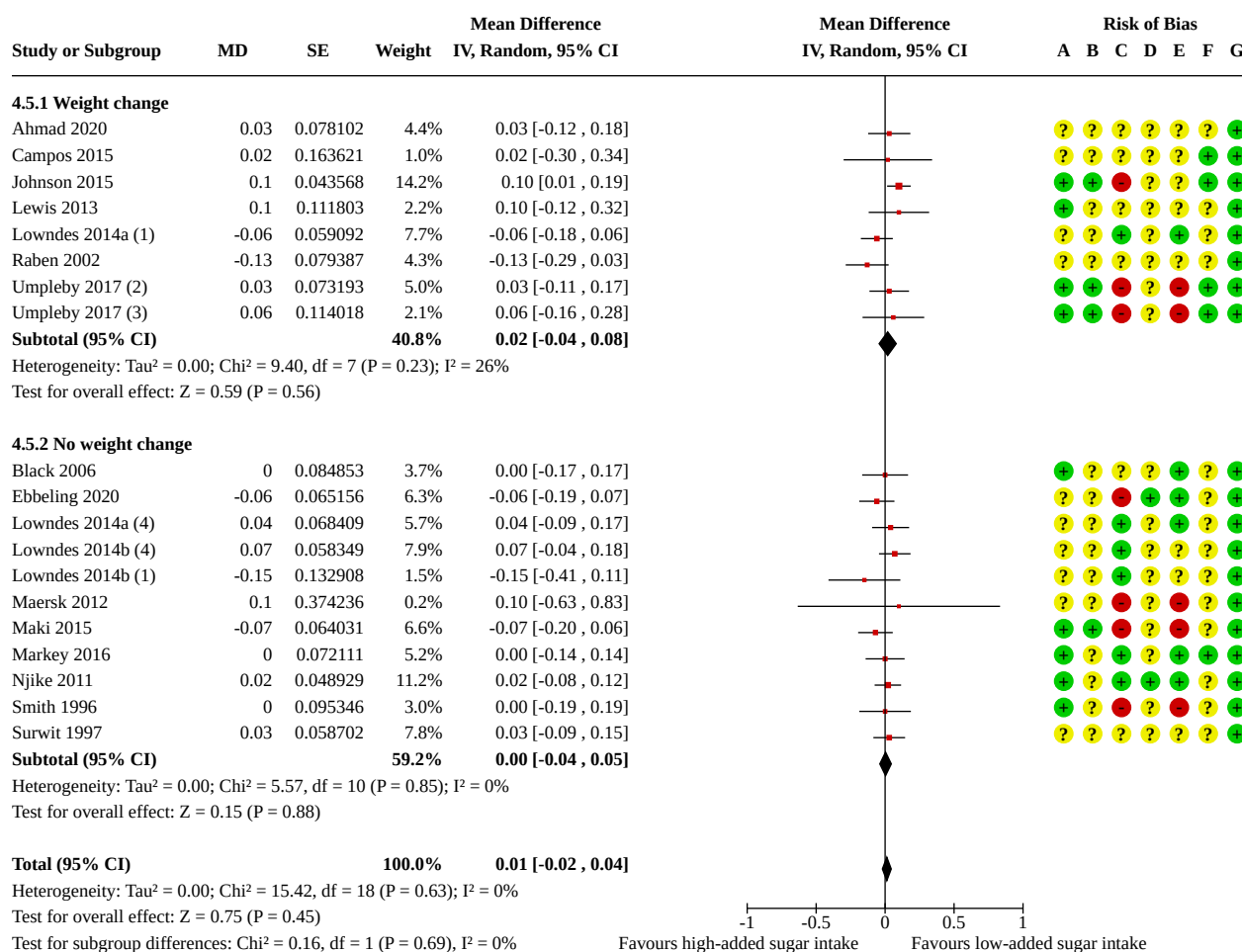
Footnotes

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
- (2) Includes the individuals not diagnosed with NAFLD, the controls.
- (3) Includes the individuals diagnosed with NAFLD.
- (4) Shows the two arms of the intervention that compares high and low HFCS intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 4.5. Comparison 4: Subgroup analysis - Difference in weight change, Outcome 5: HDL-cholesterol (mmol/L)



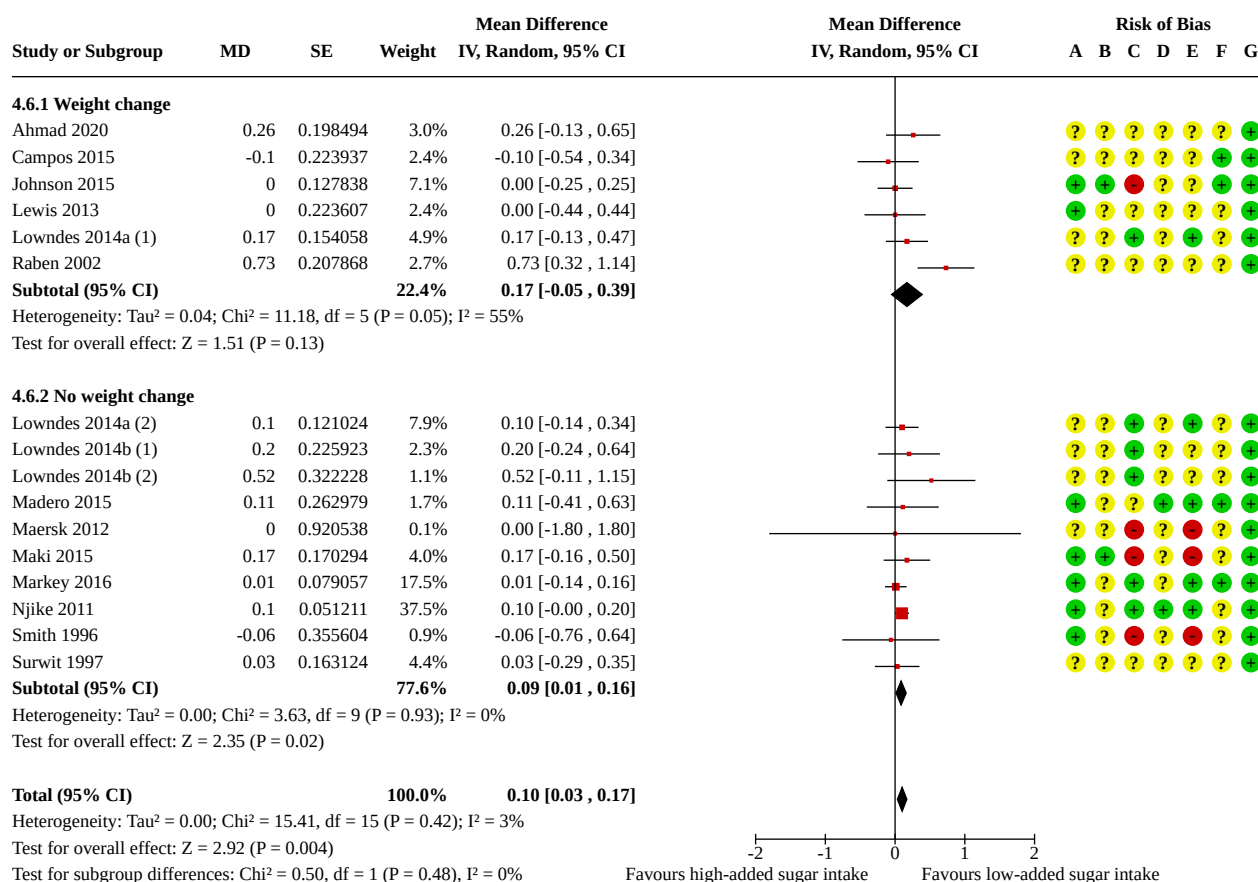
Footnotes

- (1) Shows the two arms of the intervention that compares high and low HFCS intake.
- (2) Includes the individuals not diagnosed with NAFLD, the controls.
- (3) Includes the individuals diagnosed with NAFLD.
- (4) Shows the two arms of the intervention that compares high and low sucrose intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 4.6. Comparison 4: Subgroup analysis - Difference in weight change, Outcome 6: Triglycerides (mmol/L)

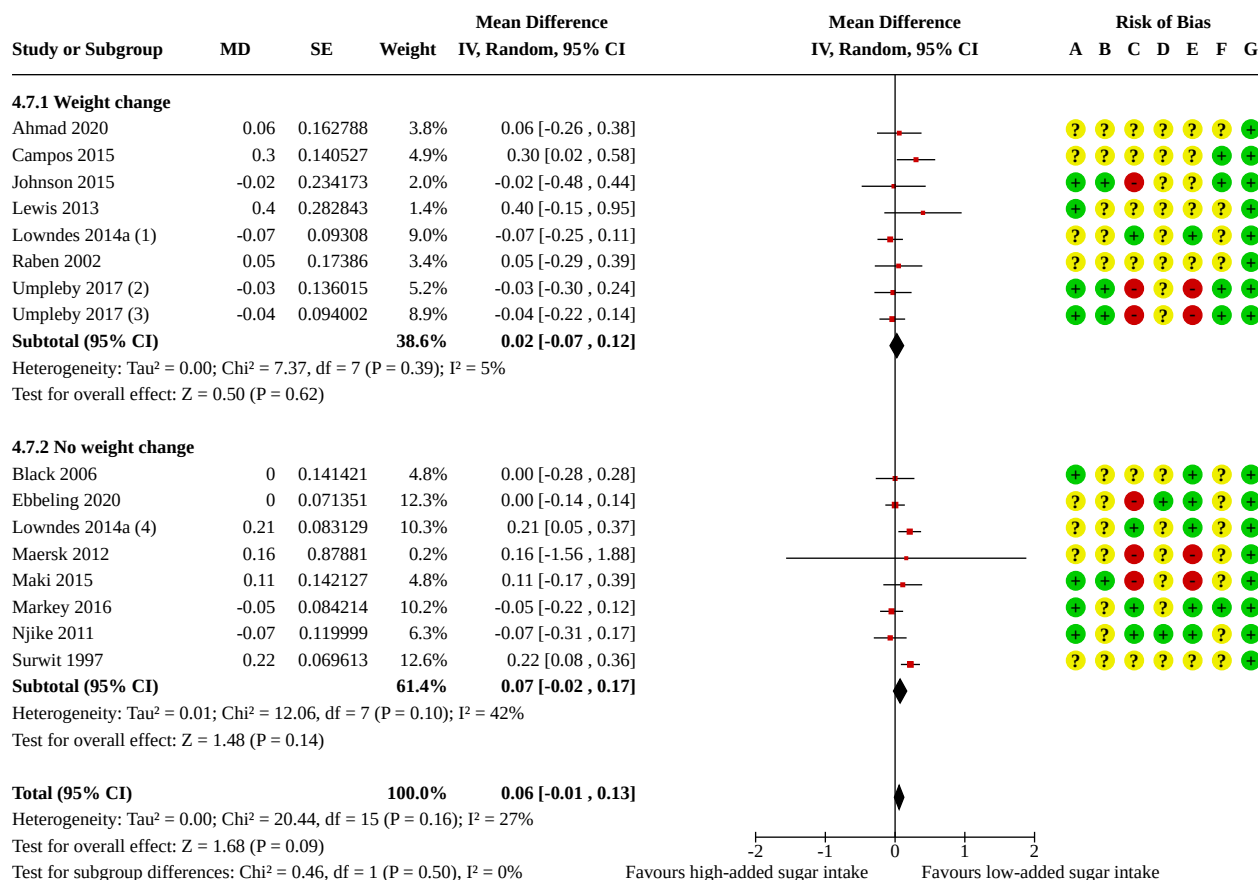


Footnotes

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
(2) Shows the two arms of the intervention that compares high and low HFCS intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Analysis 4.7. Comparison 4: Subgroup analysis - Difference in weight change, Outcome 7: Fasting plasma glucose (mmol/L)**Footnotes**

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
 (2) Includes the individuals not diagnosed with NAFLD, the controls.
 (3) Includes the individuals diagnosed with NAFLD.
 (4) Shows the two arms of the intervention that compares high and low HFCS intake.

Risk of bias legend

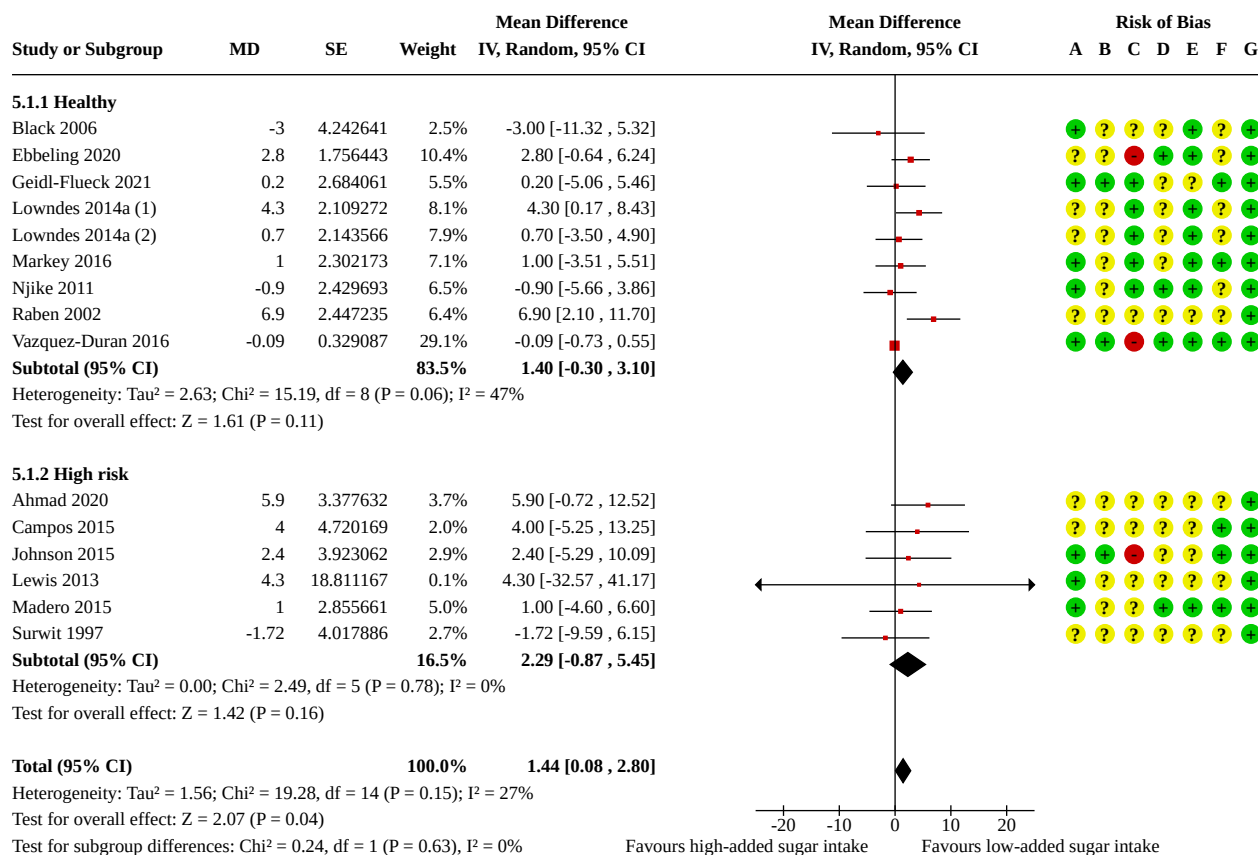
- (A) Random sequence generation (selection bias)
 (B) Allocation concealment (selection bias)
 (C) Blinding of participants and personnel (performance bias)
 (D) Blinding of outcome assessment (detection bias)
 (E) Incomplete outcome data (attrition bias)
 (F) Selective reporting (reporting bias)
 (G) Other bias

Comparison 5. Subgroup analysis - Healthy or high risk population

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
5.1 Systolic blood pressure (mmHg)	14		Mean Difference (IV, Random, 95% CI)	1.44 [0.08, 2.80]
5.1.1 Healthy	8		Mean Difference (IV, Random, 95% CI)	1.40 [-0.30, 3.10]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
5.1.2 High risk	6		Mean Difference (IV, Random, 95% CI)	2.29 [-0.87, 5.45]
5.2 Diastolic blood pressure (mmHg)	14		Mean Difference (IV, Random, 95% CI)	1.52 [0.67, 2.37]
5.2.1 Healthy	8		Mean Difference (IV, Random, 95% CI)	1.29 [0.07, 2.51]
5.2.2 High risk	6		Mean Difference (IV, Random, 95% CI)	0.92 [-1.47, 3.31]
5.3 Total cholesterol (mmol/L)	16		Mean Difference (IV, Random, 95% CI)	0.11 [0.01, 0.21]
5.3.1 Healthy	6		Mean Difference (IV, Random, 95% CI)	0.07 [-0.06, 0.21]
5.3.2 High risk	10		Mean Difference (IV, Random, 95% CI)	0.16 [-0.01, 0.33]
5.4 LDL-cholesterol (mmol/L)	12		Mean Difference (IV, Random, 95% CI)	0.01 [-0.08, 0.10]
5.4.1 Healthy	6		Mean Difference (IV, Random, 95% CI)	0.00 [-0.09, 0.10]
5.4.2 High risk	6		Mean Difference (IV, Random, 95% CI)	0.01 [-0.19, 0.21]
5.5 HDL-cholesterol (mmol/L)	16		Mean Difference (IV, Random, 95% CI)	0.01 [-0.02, 0.04]
5.5.1 Healthy	7		Mean Difference (IV, Random, 95% CI)	-0.01 [-0.06, 0.03]
5.5.2 High risk	9		Mean Difference (IV, Random, 95% CI)	0.04 [-0.01, 0.09]
5.6 Triglycerides (mmol/L)	14		Mean Difference (IV, Random, 95% CI)	0.10 [0.03, 0.17]
5.6.1 Healthy	6		Mean Difference (IV, Random, 95% CI)	0.15 [0.02, 0.27]
5.6.2 High risk	8		Mean Difference (IV, Random, 95% CI)	0.06 [-0.07, 0.20]
5.7 Fasting plasma glucose (mmol/L)	14		Mean Difference (IV, Random, 95% CI)	0.06 [-0.01, 0.13]
5.7.1 Healthy	6		Mean Difference (IV, Random, 95% CI)	0.01 [-0.07, 0.10]
5.7.2 High risk	8		Mean Difference (IV, Random, 95% CI)	0.12 [0.02, 0.21]

Analysis 5.1. Comparison 5: Subgroup analysis - Healthy or high risk population, Outcome 1: Systolic blood pressure (mmHg)



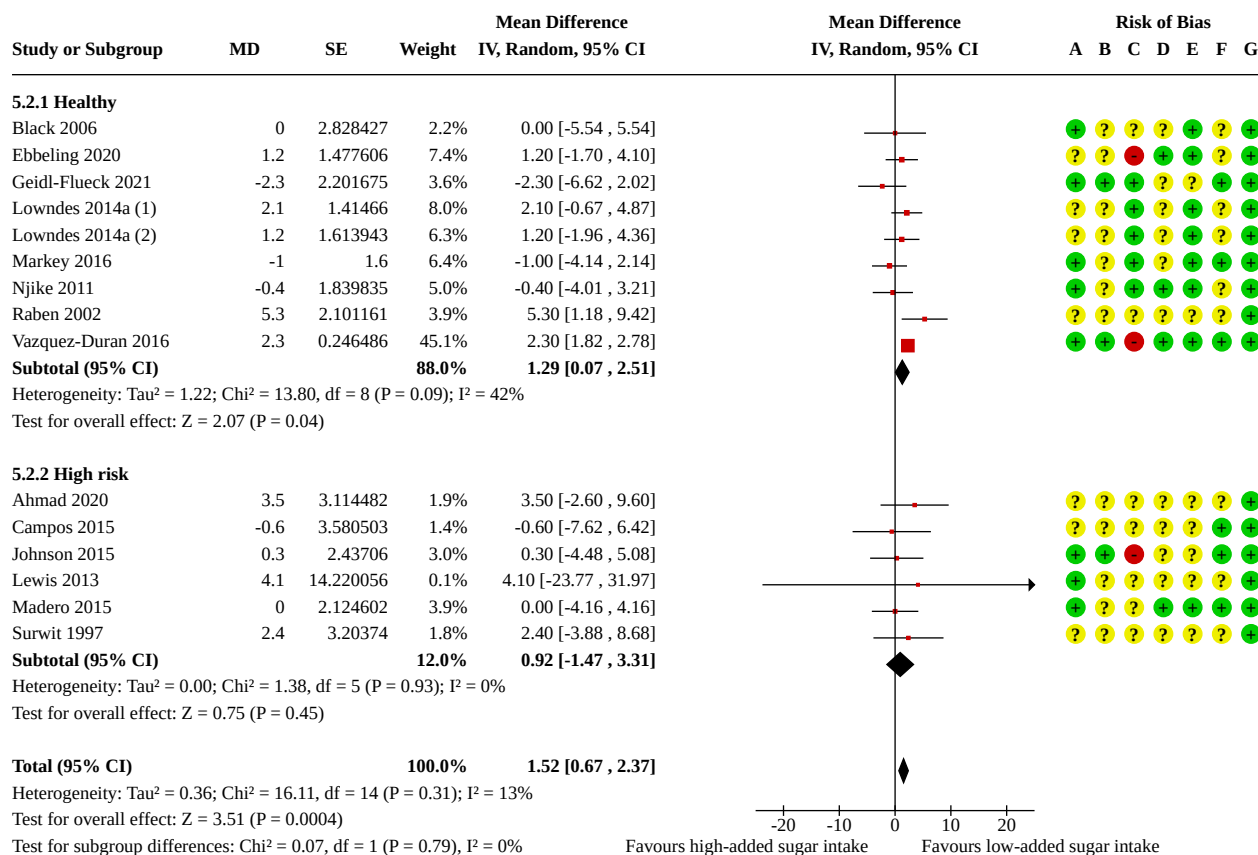
Footnotes

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
- (2) Shows the two arms of the intervention that compares high and low HFCS intake.

Risk of bias legend

- Random sequence generation (selection bias)
- Allocation concealment (selection bias)
- Blinding of participants and personnel (performance bias)
- Blinding of outcome assessment (detection bias)
- Incomplete outcome data (attrition bias)
- Selective reporting (reporting bias)
- Other bias

Analysis 5.2. Comparison 5: Subgroup analysis - Healthy or high risk population, Outcome 2: Diastolic blood pressure (mmHg)



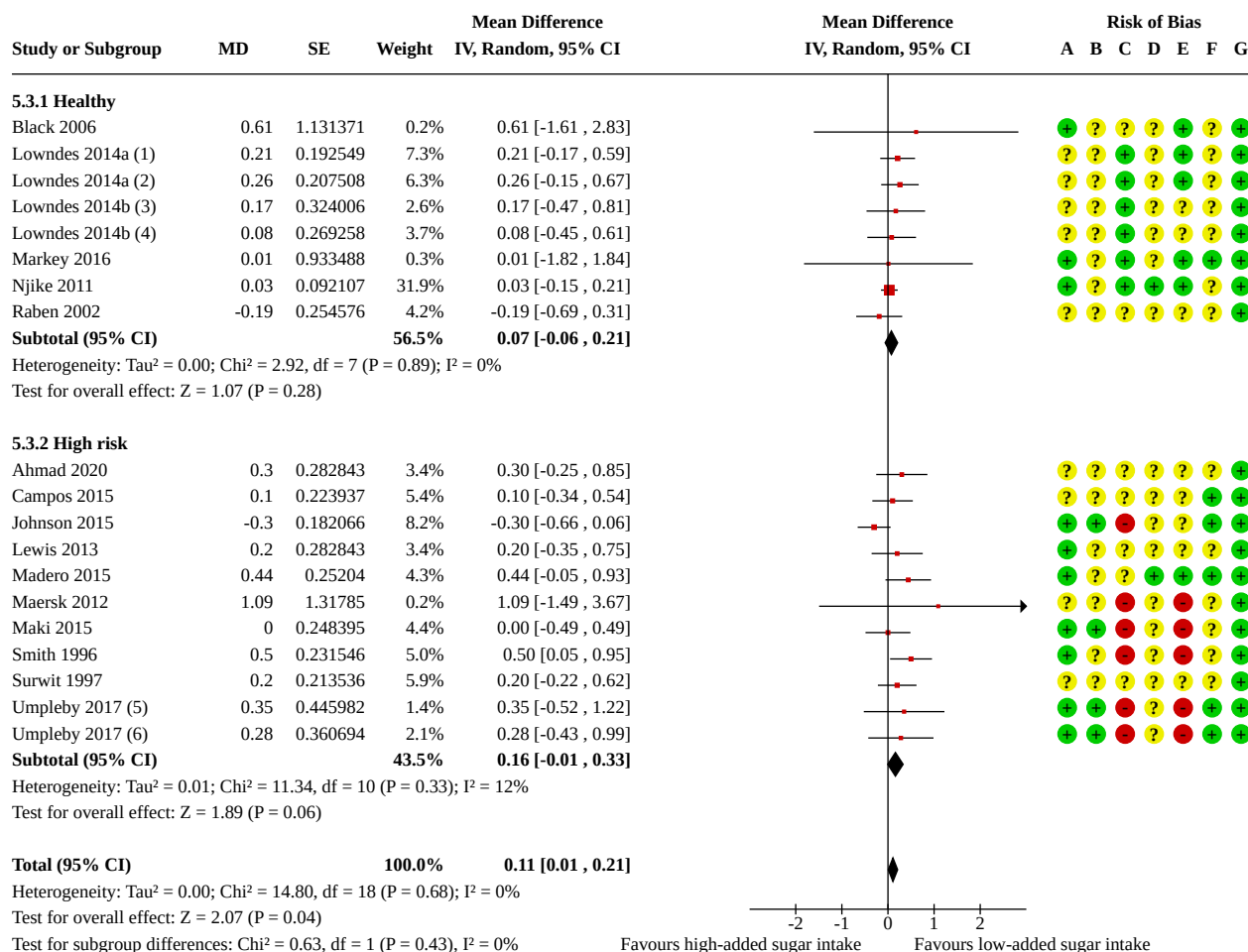
Footnotes

- (1) Shows the two arms of the intervention that compares high and low HFCS intake.
- (2) Shows the two arms of the intervention that compares high and low sucrose intake.

Risk of bias legend

- Random sequence generation (selection bias)
- Allocation concealment (selection bias)
- Blinding of participants and personnel (performance bias)
- Blinding of outcome assessment (detection bias)
- Incomplete outcome data (attrition bias)
- Selective reporting (reporting bias)
- Other bias

Analysis 5.3. Comparison 5: Subgroup analysis - Healthy or high risk population, Outcome 3: Total cholesterol (mmol/L)



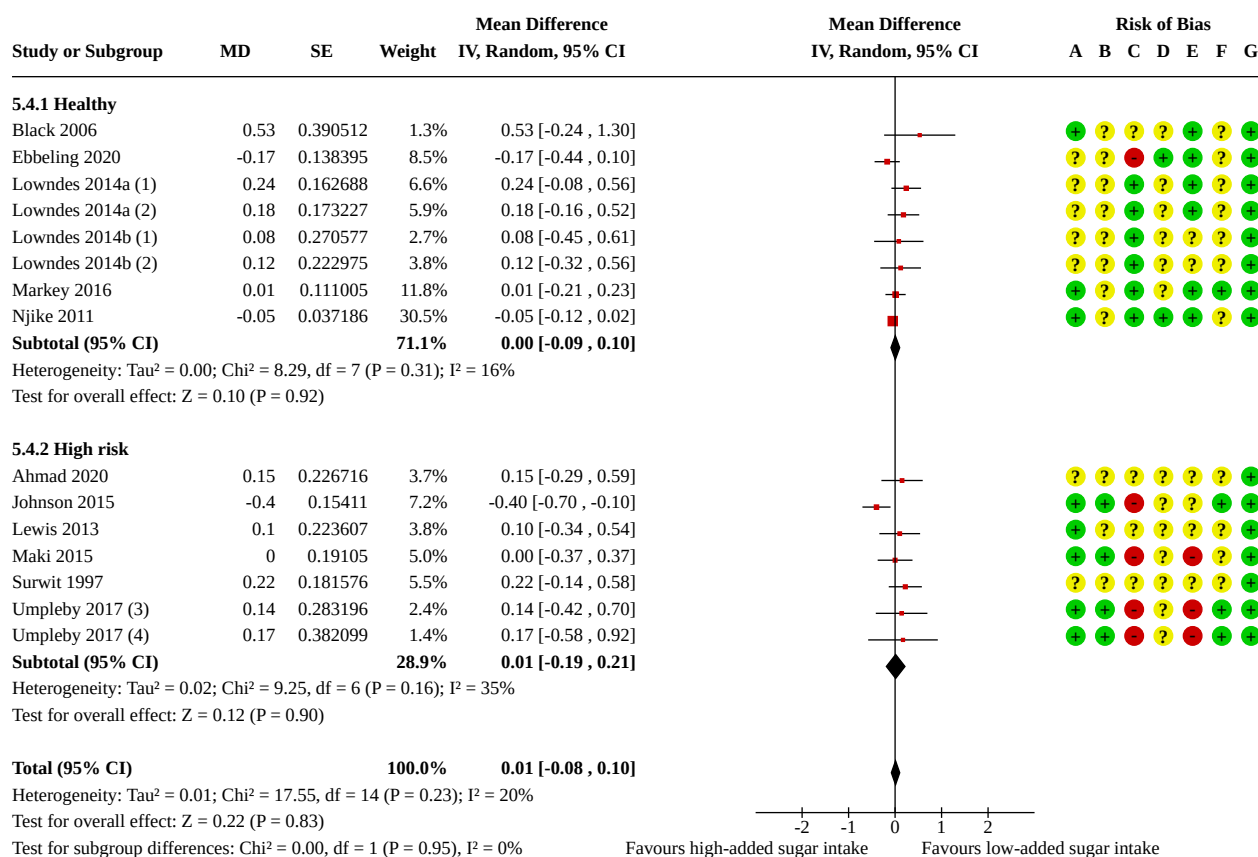
Footnotes

- (1) Shows the two arms of the intervention that compares high and low HFCS intake
- (2) Shows the two arms of the intervention that compares high and low sucrose intake
- (3) Shows the two arms of the intervention that compares high and low HFCS intake.
- (4) Shows the two arms of the intervention that compares high and low sucrose intake.
- (5) Includes the individuals diagnosed with NAFLD.
- (6) Includes the individuals not diagnosed with NAFLD, the controls.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 5.4. Comparison 5: Subgroup analysis - Healthy or high risk population, Outcome 4: LDL-cholesterol (mmol/L)



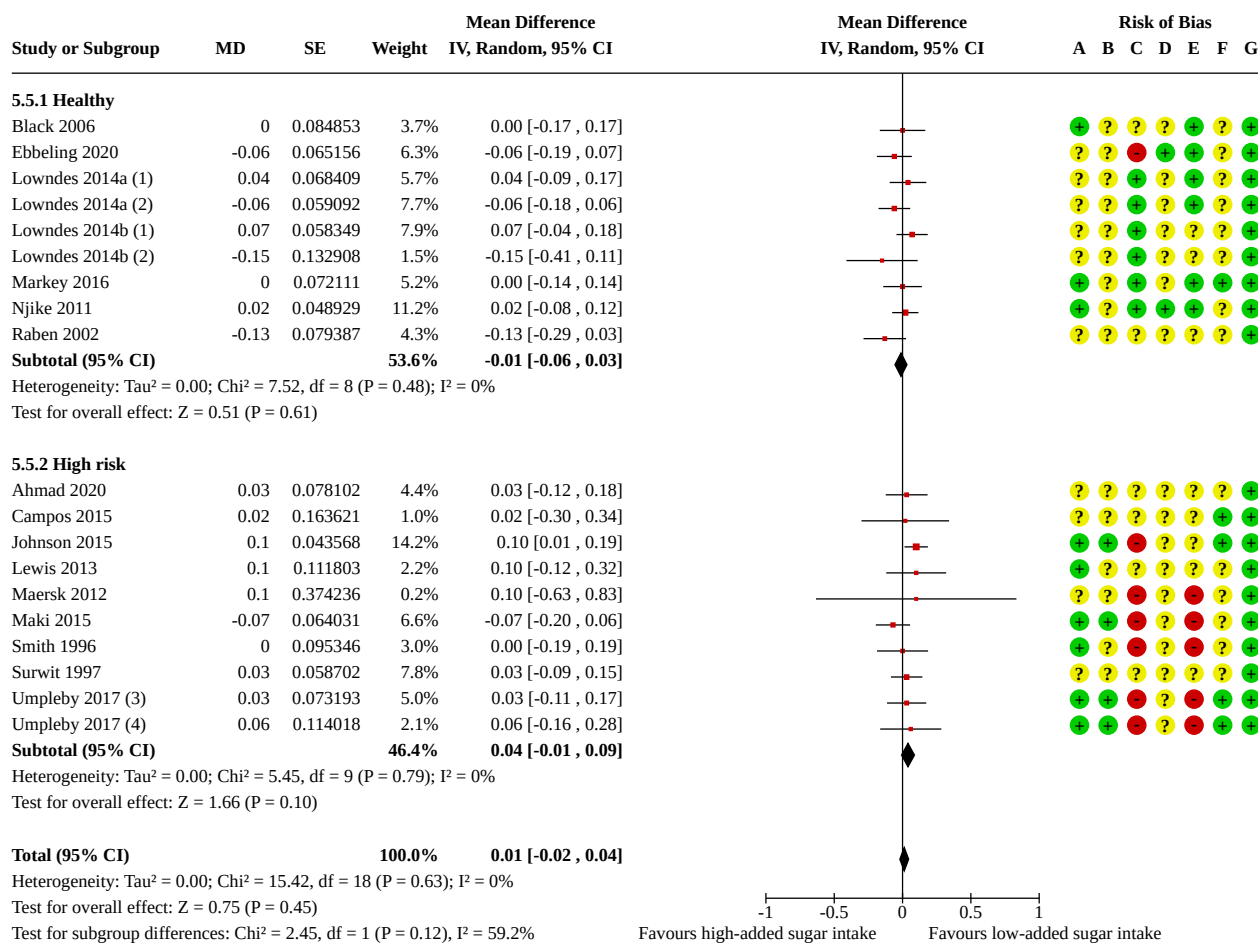
Footnotes

- (1) Shows the two arms of the intervention that compares high and low HFCS intake.
- (2) Shows the two arms of the intervention that compares high and low sucrose intake.
- (3) Includes the individuals not diagnosed with NAFLD, the controls.
- (4) Includes the individuals diagnosed with NAFLD.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 5.5. Comparison 5: Subgroup analysis - Healthy or high risk population, Outcome 5: HDL-cholesterol (mmol/L)



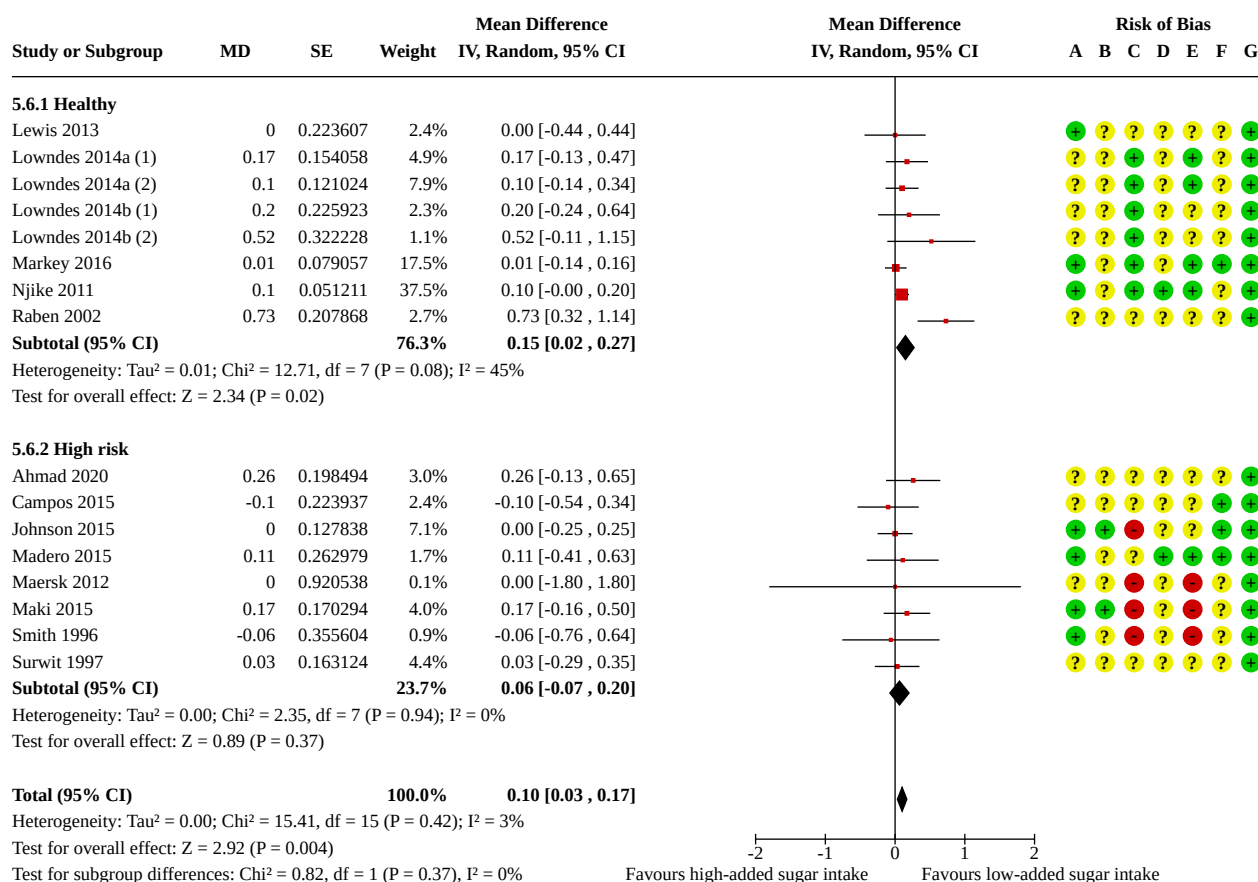
Footnotes

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
- (2) Shows the two arms of the intervention that compares high and low HFCS intake.
- (3) Includes the individuals not diagnosed with NAFLD, the controls.
- (4) Includes the individuals diagnosed with NAFLD.

Risk of bias legend

- Random sequence generation (selection bias)
- Allocation concealment (selection bias)
- Blinding of participants and personnel (performance bias)
- Blinding of outcome assessment (detection bias)
- Incomplete outcome data (attrition bias)
- Selective reporting (reporting bias)
- Other bias

Analysis 5.6. Comparison 5: Subgroup analysis - Healthy or high risk population, Outcome 6: Triglycerides (mmol/L)

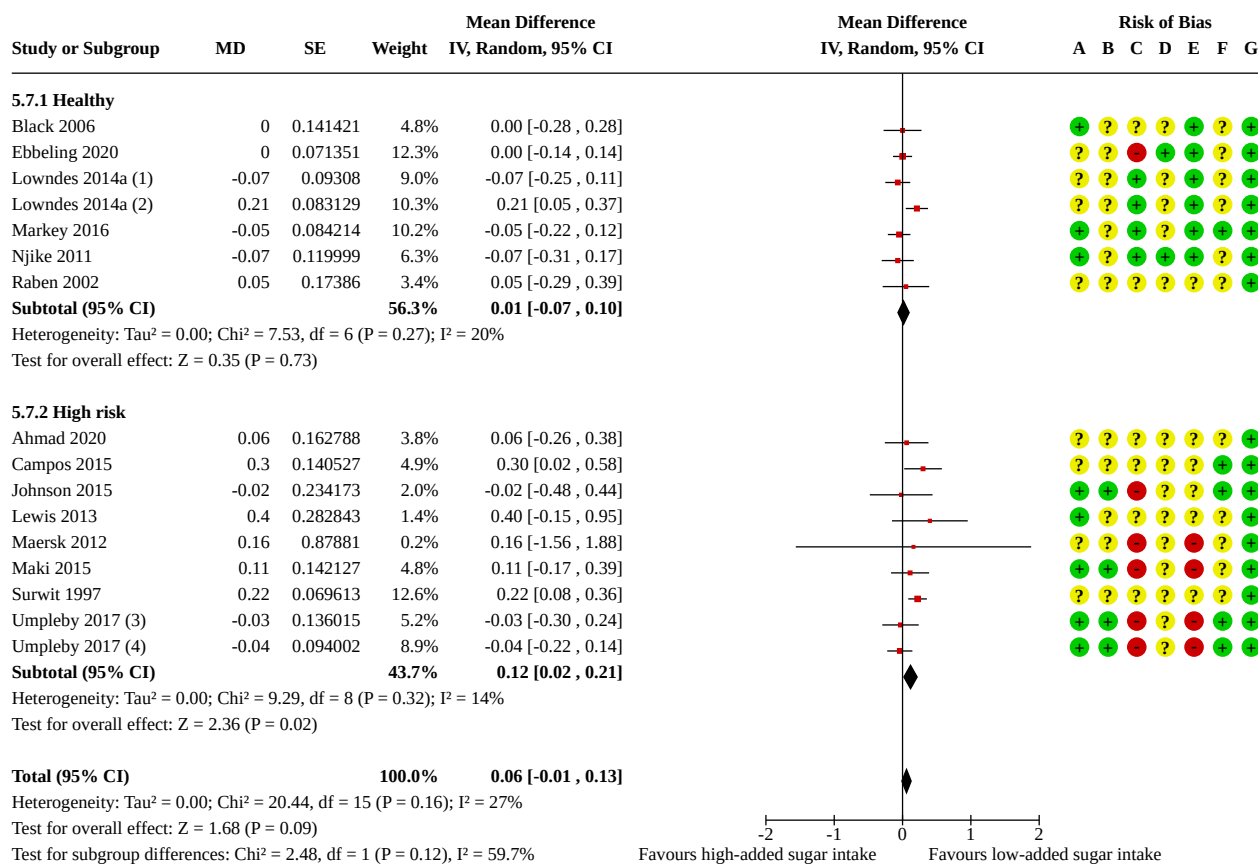


Footnotes

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
(2) Shows the two arms of the intervention that compares high and low HFCS intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Analysis 5.7. Comparison 5: Subgroup analysis - Healthy or high risk population, Outcome 7: Fasting plasma glucose (mmol/L)**Footnotes**

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
 (2) Shows the two arms of the intervention that compares high and low HFCS intake.
 (3) Includes the individuals not diagnosed with NAFLD, the controls.
 (4) Includes the individuals diagnosed with NAFLD.

Risk of bias legend

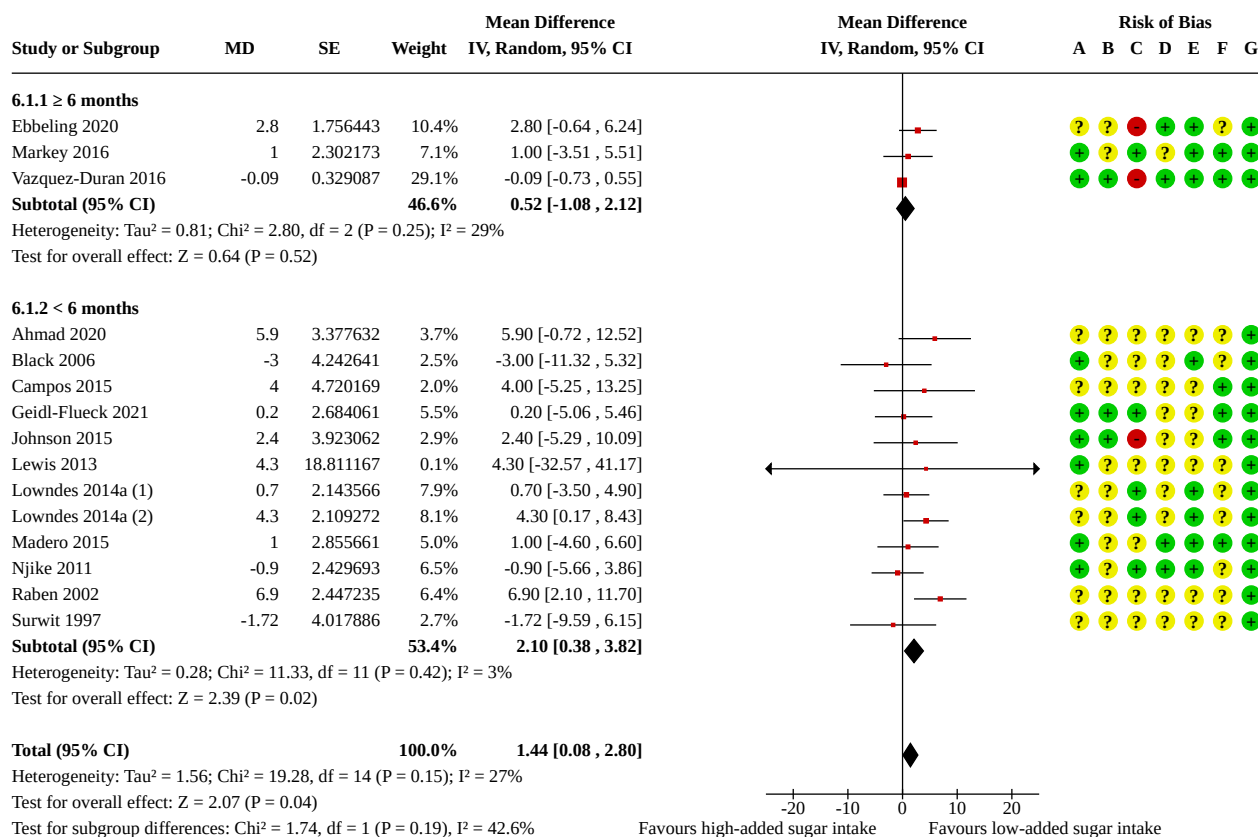
- (A) Random sequence generation (selection bias)
 (B) Allocation concealment (selection bias)
 (C) Blinding of participants and personnel (performance bias)
 (D) Blinding of outcome assessment (detection bias)
 (E) Incomplete outcome data (attrition bias)
 (F) Selective reporting (reporting bias)
 (G) Other bias

Comparison 6. Subgroup analysis - Intervention <6 months or ≥6 months

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
6.1 Systolic blood pressure (mmHg)	14		Mean Difference (IV, Random, 95% CI)	1.44 [0.08, 2.80]
6.1.1 ≥ 6 months	3		Mean Difference (IV, Random, 95% CI)	0.52 [-1.08, 2.12]
6.1.2 < 6 months	11		Mean Difference (IV, Random, 95% CI)	2.10 [0.38, 3.82]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
6.2 Diastolic blood pressure (mmHg)	14		Mean Difference (IV, Random, 95% CI)	1.52 [0.67, 2.37]
6.2.1 ≥ 6 months	2		Mean Difference (IV, Random, 95% CI)	2.27 [1.79, 2.75]
6.2.2 < 6 months	12		Mean Difference (IV, Random, 95% CI)	0.84 [-0.33, 2.01]
6.3 Total cholesterol (mmol/L)	16		Mean Difference (IV, Random, 95% CI)	0.11 [0.01, 0.21]
6.3.1 ≥ 6 months	2		Mean Difference (IV, Random, 95% CI)	0.52 [0.07, 0.96]
6.3.2 6 months	14		Mean Difference (IV, Random, 95% CI)	0.09 [-0.02, 0.19]
6.4 LDL-cholesterol (mmol/L)	12		Mean Difference (IV, Random, 95% CI)	0.01 [-0.08, 0.10]
6.4.1 ≥ 6 months	1		Mean Difference (IV, Random, 95% CI)	-0.17 [-0.44, 0.10]
6.4.2 < 6 months	11		Mean Difference (IV, Random, 95% CI)	0.03 [-0.07, 0.12]
6.5 HDL-cholesterol (mmol/L)	16		Mean Difference (IV, Random, 95% CI)	0.01 [-0.02, 0.04]
6.5.1 ≥ 6 months	3		Mean Difference (IV, Random, 95% CI)	-0.04 [-0.14, 0.07]
6.5.2 < 6 months	13		Mean Difference (IV, Random, 95% CI)	0.02 [-0.02, 0.05]
6.6 Triglycerides (mmol/L)	14		Mean Difference (IV, Random, 95% CI)	0.10 [0.03, 0.17]
6.6.1 ≥ 6 months	2		Mean Difference (IV, Random, 95% CI)	-0.05 [-0.70, 0.60]
6.6.2 < 6 months	12		Mean Difference (IV, Random, 95% CI)	0.11 [0.03, 0.19]
6.7 Fasting plasma glucose (mmol/L)	14		Mean Difference (IV, Random, 95% CI)	0.06 [-0.01, 0.13]
6.7.1 ≥ 6 months	2		Mean Difference (IV, Random, 95% CI)	0.00 [-0.14, 0.14]
6.7.2 < 6 months	12		Mean Difference (IV, Random, 95% CI)	0.07 [-0.01, 0.14]

Analysis 6.1. Comparison 6: Subgroup analysis - Intervention <6 months or ≥6 months, Outcome 1: Systolic blood pressure (mmHg)



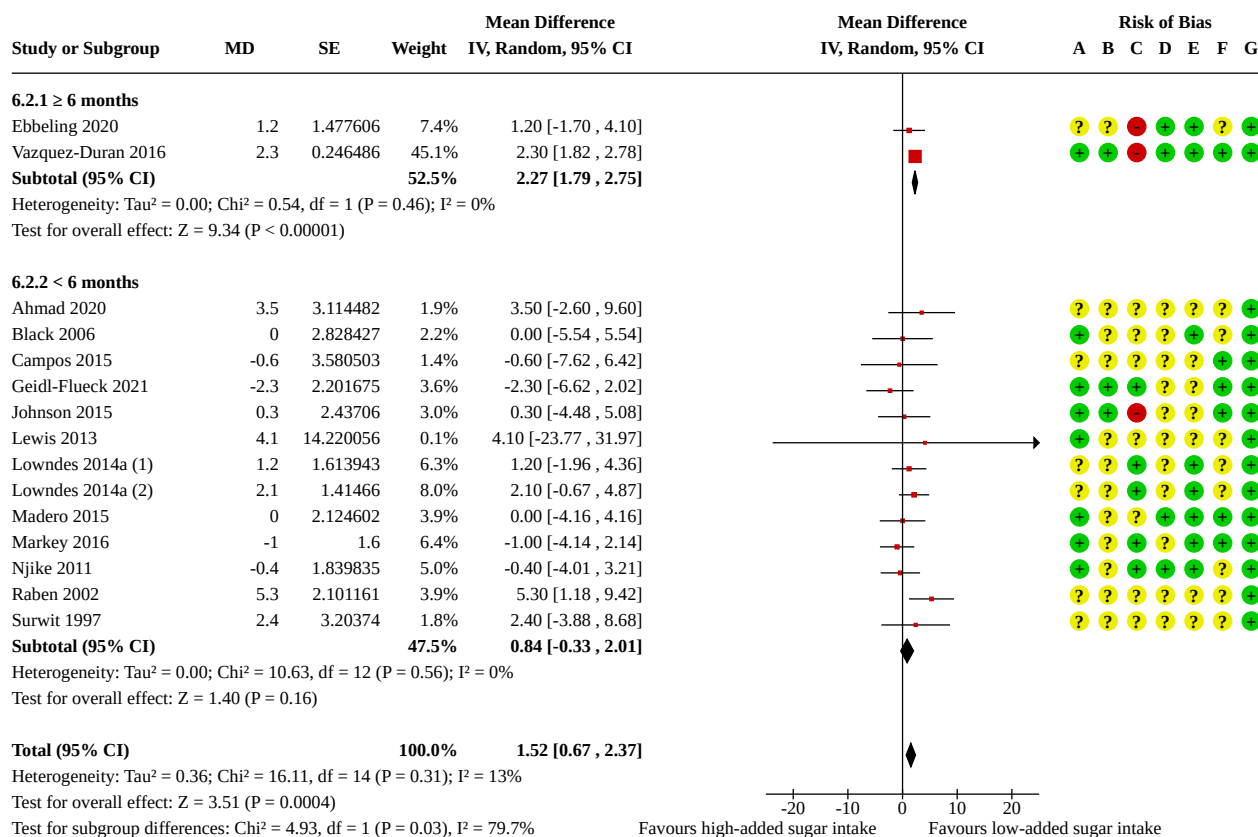
Footnotes

- (1) Shows the two arms of the intervention that compares high and low HFCS intake.
- (2) Shows the two arms of the intervention that compares high and low sucrose intake.

Risk of bias legend

- Random sequence generation (selection bias)
- Allocation concealment (selection bias)
- Blinding of participants and personnel (performance bias)
- Blinding of outcome assessment (detection bias)
- Incomplete outcome data (attrition bias)
- Selective reporting (reporting bias)
- Other bias

Analysis 6.2. Comparison 6: Subgroup analysis - Intervention <6 months or ≥6 months, Outcome 2: Diastolic blood pressure (mmHg)



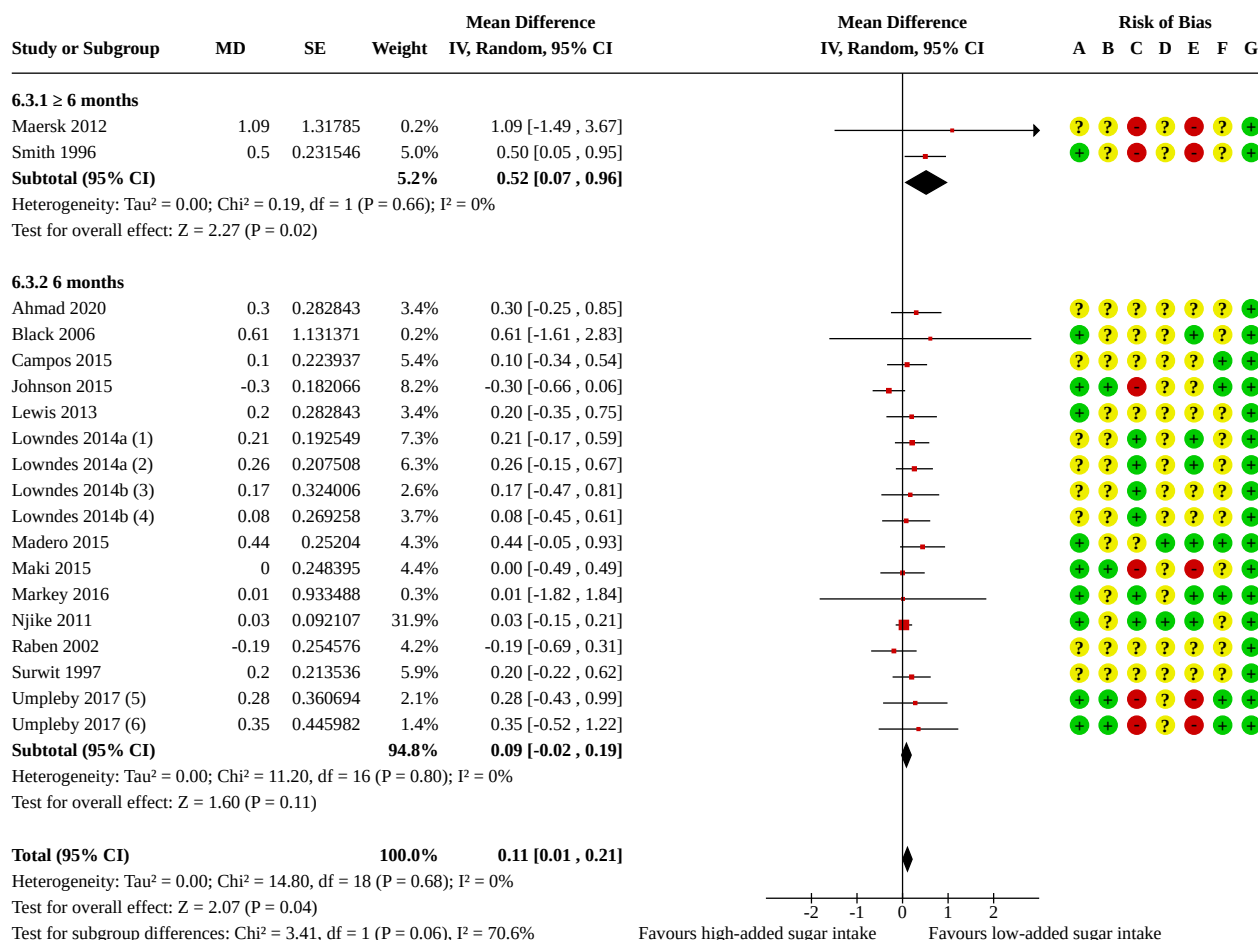
Footnotes

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
(2) Shows the two arms of the intervention that compares high and low HFCS intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Analysis 6.3. Comparison 6: Subgroup analysis - Intervention <6 months or ≥6 months, Outcome 3: Total cholesterol (mmol/L)



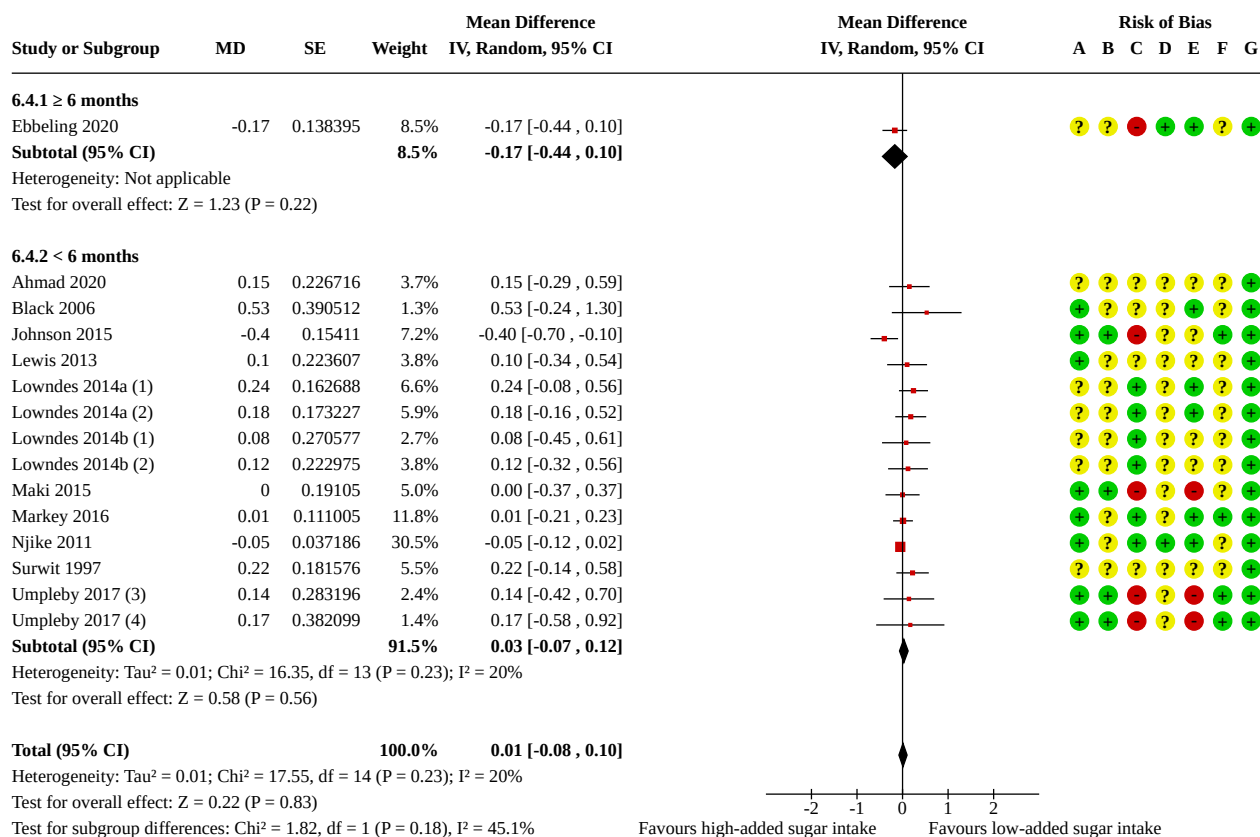
Footnotes

- (1) Shows the two arms of the intervention that compares high and low HFCS intake
- (2) Shows the two arms of the intervention that compares high and low sucrose intake
- (3) Shows the two arms of the intervention that compares high and low HFCS intake.
- (4) Shows the two arms of the intervention that compares high and low sucrose intake.
- (5) Includes the individuals not diagnosed with NAFLD, the controls.
- (6) Includes the individuals diagnosed with NAFLD.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 6.4. Comparison 6: Subgroup analysis - Intervention <6 months or ≥6 months, Outcome 4: LDL-cholesterol (mmol/L)



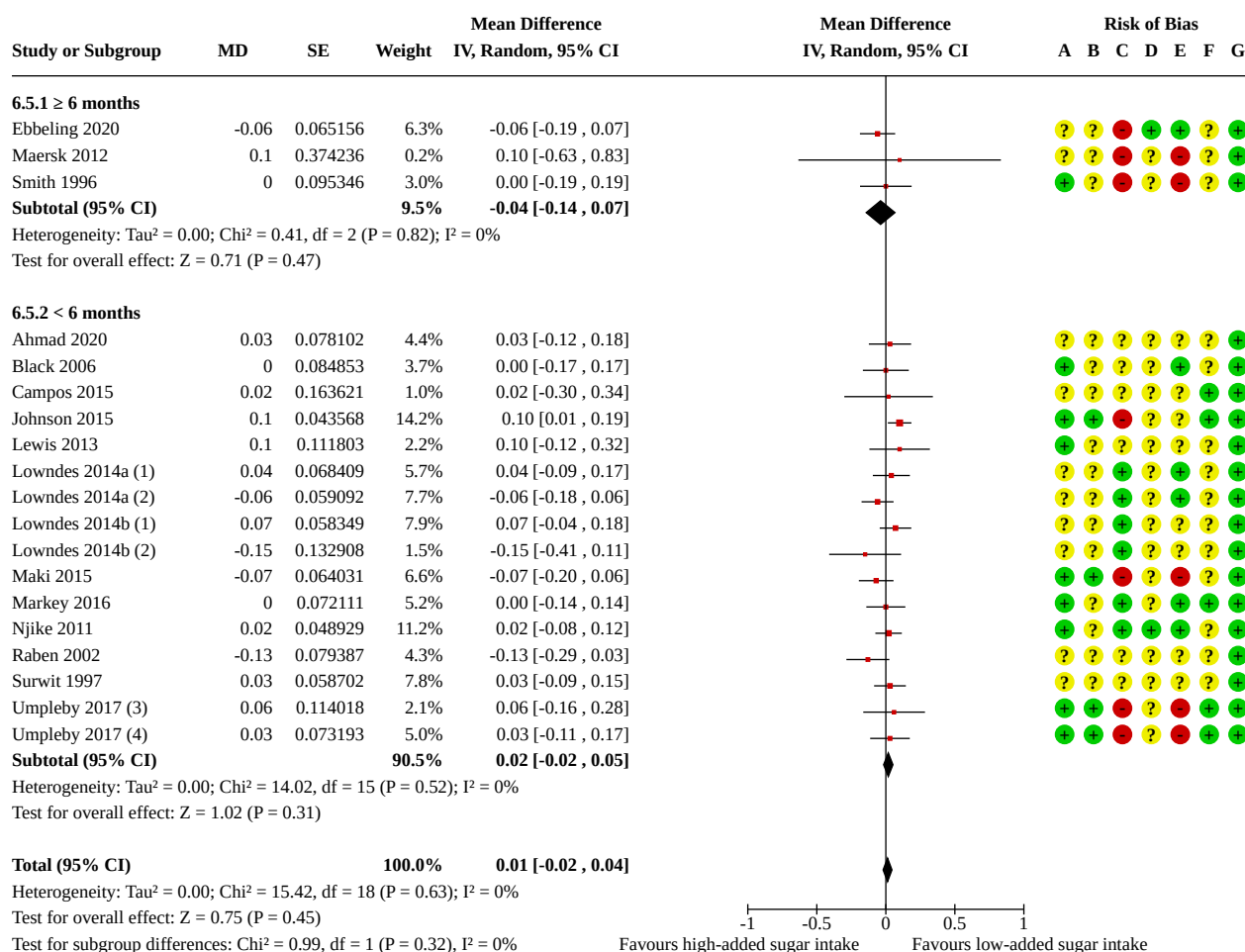
Footnotes

- (1) Shows the two arms of the intervention that compares high and low HFCS intake.
- (2) Shows the two arms of the intervention that compares high and low sucrose intake.
- (3) Includes the individuals not diagnosed with NAFLD, the controls.
- (4) Includes the individuals diagnosed with NAFLD.

Risk of bias legend

- Random sequence generation (selection bias)
- Allocation concealment (selection bias)
- Blinding of participants and personnel (performance bias)
- Blinding of outcome assessment (detection bias)
- Incomplete outcome data (attrition bias)
- Selective reporting (reporting bias)
- Other bias

Analysis 6.5. Comparison 6: Subgroup analysis - Intervention <6 months or ≥6 months, Outcome 5: HDL-cholesterol (mmol/L)



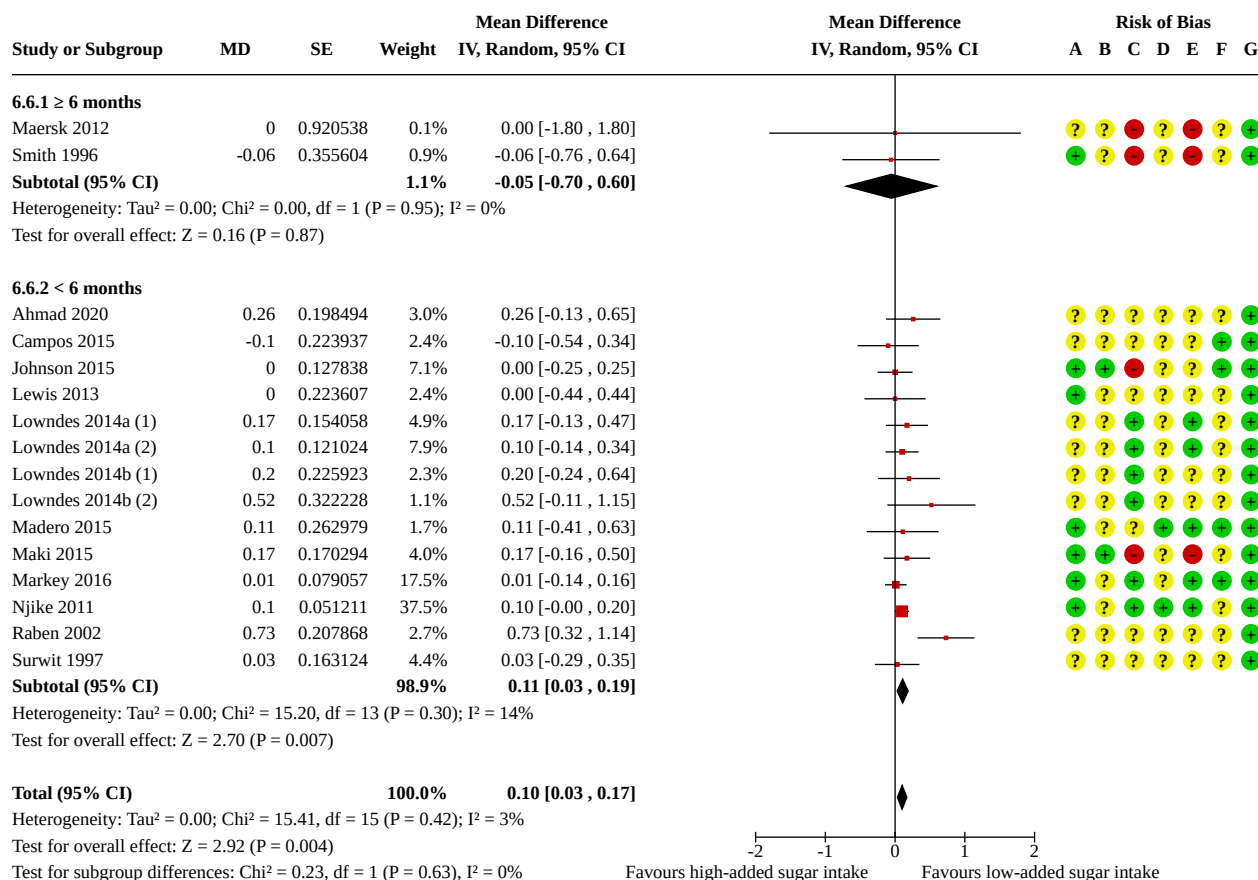
Footnotes

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
- (2) Shows the two arms of the intervention that compares high and low HFCS intake.
- (3) Includes the individuals diagnosed with NAFLD.
- (4) Includes the individuals not diagnosed with NAFLD, the controls.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 6.6. Comparison 6: Subgroup analysis - Intervention <6 months or ≥6 months, Outcome 6: Triglycerides (mmol/L)

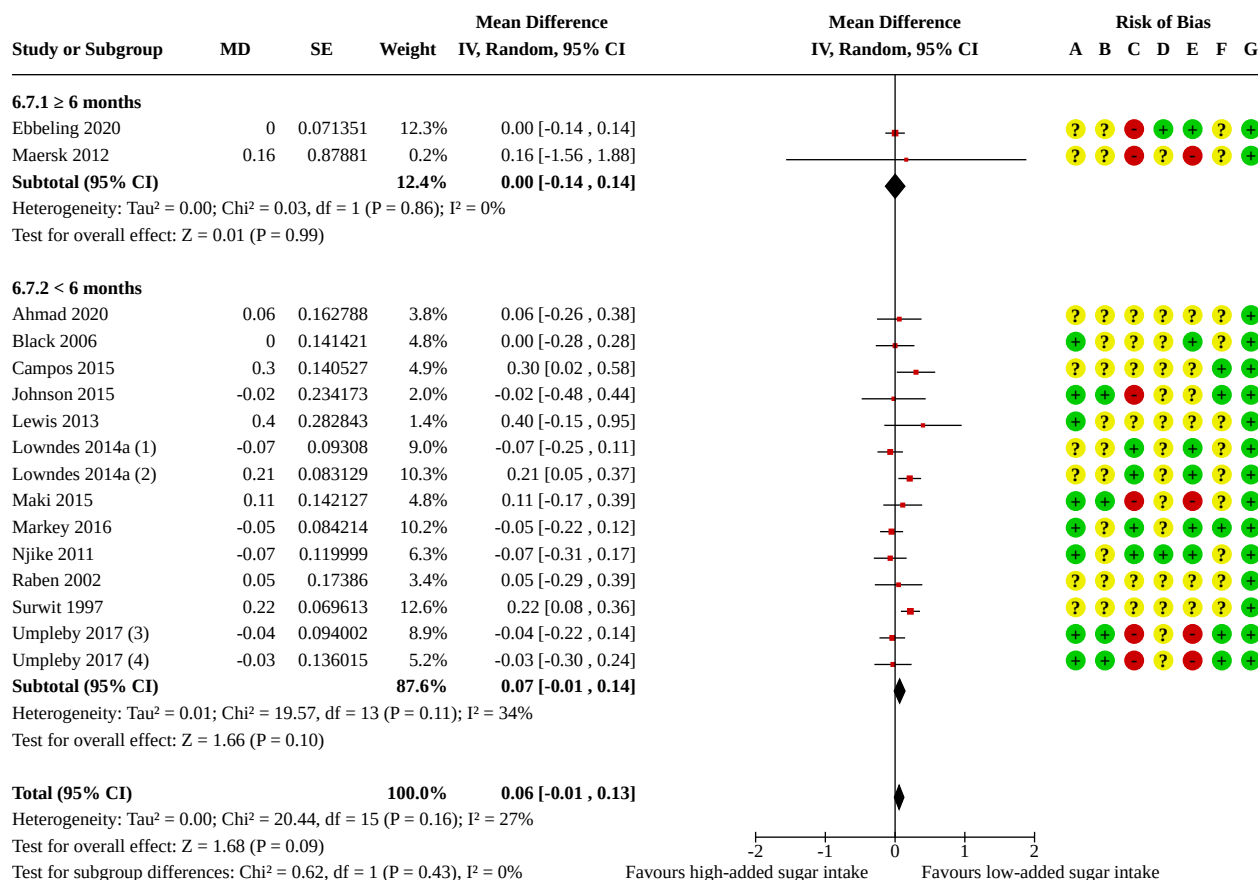


Footnotes

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
(2) Shows the two arms of the intervention that compares high and low HFCS intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Analysis 6.7. Comparison 6: Subgroup analysis - Intervention <6 months or ≥6 months, Outcome 7: Fasting plasma glucose (mmol/L)**Footnotes**

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
- (2) Shows the two arms of the intervention that compares high and low HFCS intake.
- (3) Includes the individuals diagnosed with NAFLD.
- (4) Includes the individuals not diagnosed with NAFLD, the controls.

Risk of bias legend

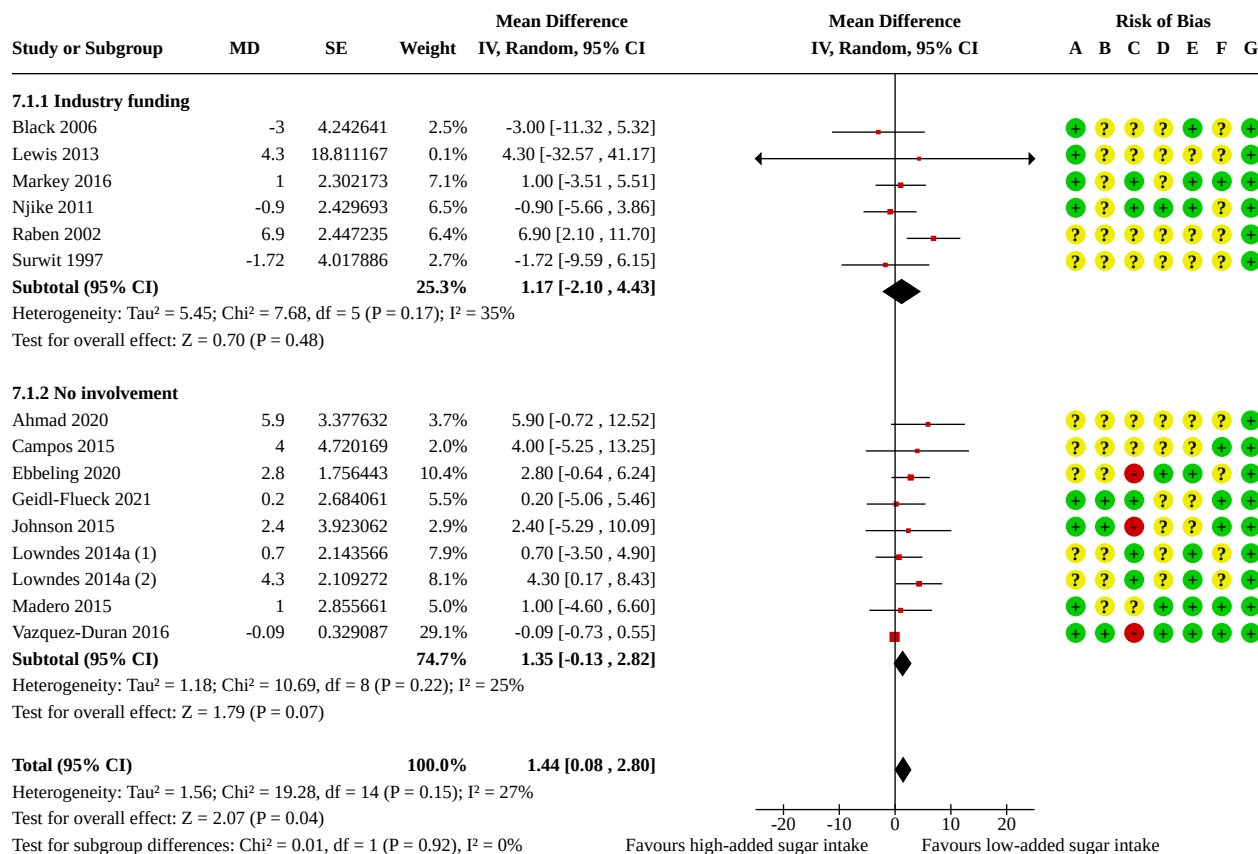
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Comparison 7. Subgroup analysis - Industry funding

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
7.1 Systolic blood pressure (mmHg)	14		Mean Difference (IV, Random, 95% CI)	1.44 [0.08, 2.80]
7.1.1 Industry funding	6		Mean Difference (IV, Random, 95% CI)	1.17 [-2.10, 4.43]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
7.1.2 No involvement	8		Mean Difference (IV, Random, 95% CI)	1.35 [-0.13, 2.82]
7.2 Diastolic blood pressure (mmHg)	14		Mean Difference (IV, Random, 95% CI)	1.52 [0.67, 2.37]
7.2.1 Industry funding	6		Mean Difference (IV, Random, 95% CI)	1.00 [-1.26, 3.25]
7.2.2 No involvement	8		Mean Difference (IV, Random, 95% CI)	2.14 [1.69, 2.60]
7.3 Total cholesterol (mmol/L)	16		Mean Difference (IV, Random, 95% CI)	0.11 [0.01, 0.21]
7.3.1 Industry funding	8		Mean Difference (IV, Random, 95% CI)	0.05 [-0.10, 0.19]
7.3.2 No involvement	8		Mean Difference (IV, Random, 95% CI)	0.17 [0.02, 0.33]
7.4 LDL-cholesterol (mmol/L)	12		Mean Difference (IV, Random, 95% CI)	0.01 [-0.08, 0.10]
7.4.1 Industry funding	6		Mean Difference (IV, Random, 95% CI)	-0.03 [-0.09, 0.04]
7.4.2 No involvement	6		Mean Difference (IV, Random, 95% CI)	0.02 [-0.15, 0.19]
7.5 HDL-cholesterol (mmol/L)	16		Mean Difference (IV, Random, 95% CI)	0.01 [-0.02, 0.04]
7.5.1 Industry funding	8		Mean Difference (IV, Random, 95% CI)	-0.01 [-0.06, 0.04]
7.5.2 No involvement	8		Mean Difference (IV, Random, 95% CI)	0.03 [-0.02, 0.07]
7.6 Triglycerides (mmol/L)	14		Mean Difference (IV, Random, 95% CI)	0.10 [0.03, 0.17]
7.6.1 Industry funding	7		Mean Difference (IV, Random, 95% CI)	0.12 [-0.01, 0.26]
7.6.2 No involvement	7		Mean Difference (IV, Random, 95% CI)	0.11 [-0.01, 0.23]
7.7 Fasting plasma glucose (mmol/L)	14		Mean Difference (IV, Random, 95% CI)	0.06 [-0.01, 0.13]
7.7.1 Industry funding	8		Mean Difference (IV, Random, 95% CI)	0.07 [-0.04, 0.18]
7.7.2 No involvement	6		Mean Difference (IV, Random, 95% CI)	0.05 [-0.05, 0.14]

Analysis 7.1. Comparison 7: Subgroup analysis - Industry funding, Outcome 1: Systolic blood pressure (mmHg)



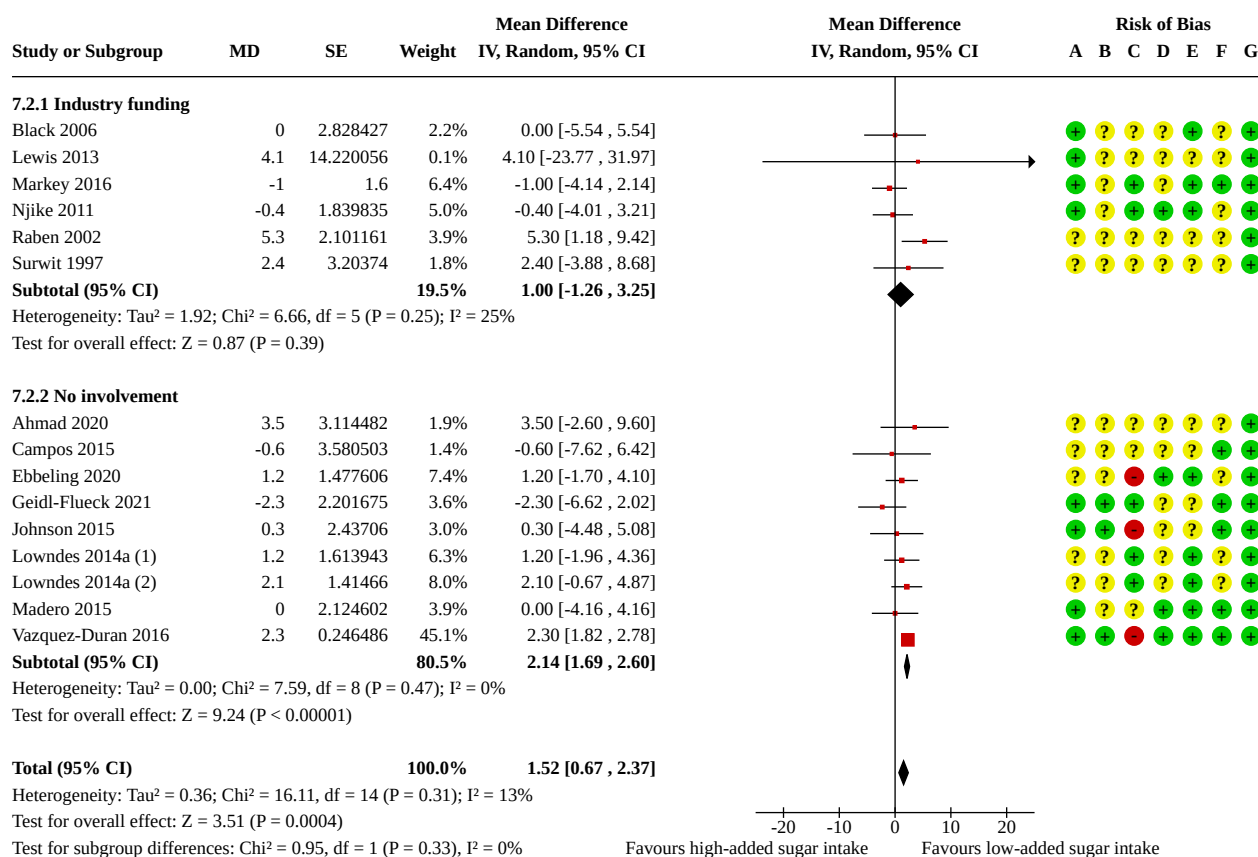
Footnotes

- (1) Shows the two arms of the intervention that compares high and low HFCS intake.
- (2) Shows the two arms of the intervention that compares high and low sucrose intake.

Risk of bias legend

- Random sequence generation (selection bias)
- Allocation concealment (selection bias)
- Blinding of participants and personnel (performance bias)
- Blinding of outcome assessment (detection bias)
- Incomplete outcome data (attrition bias)
- Selective reporting (reporting bias)
- Other bias

Analysis 7.2. Comparison 7: Subgroup analysis - Industry funding, Outcome 2: Diastolic blood pressure (mmHg)



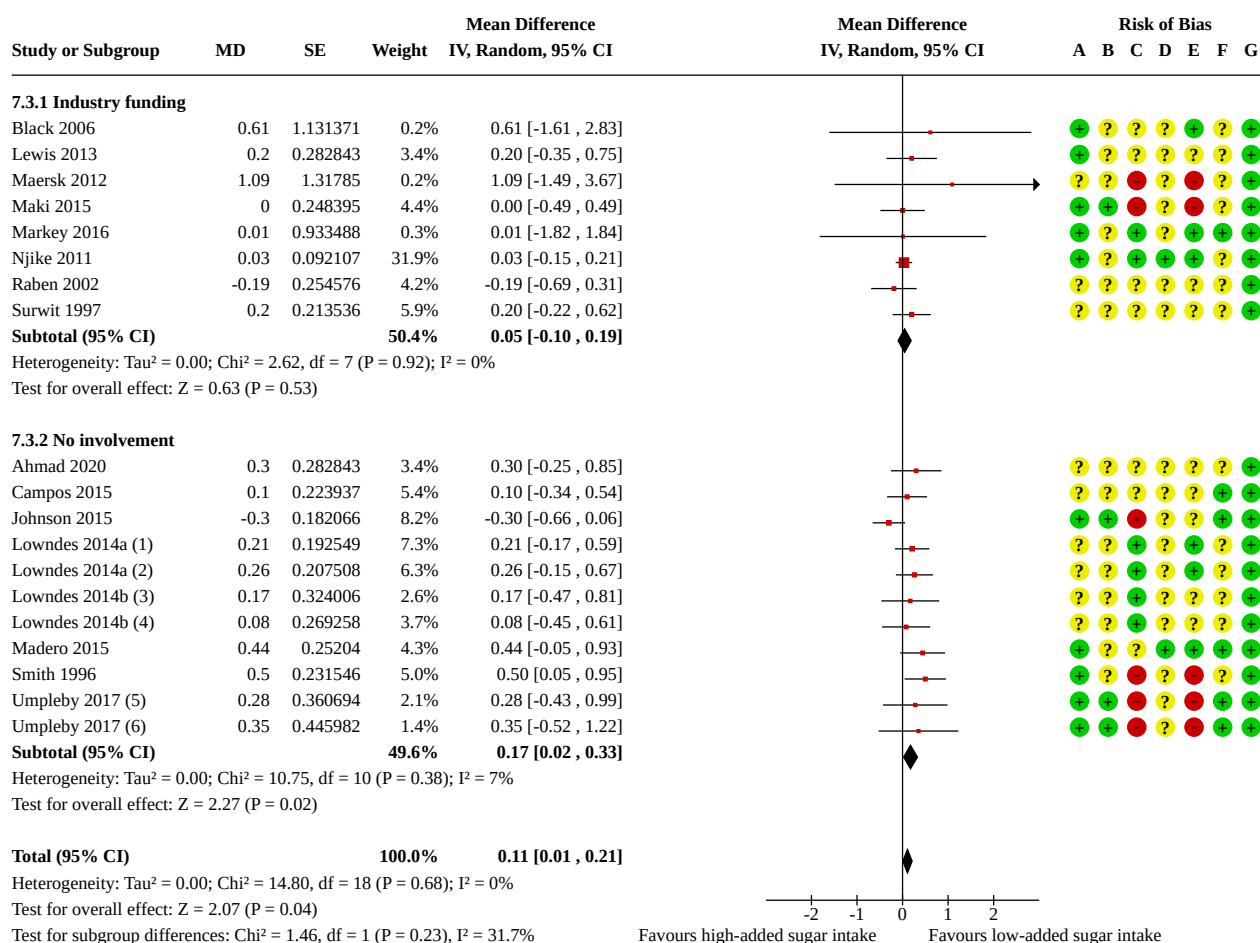
Footnotes

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
(2) Shows the two arms of the intervention that compares high and low HFCS intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Analysis 7.3. Comparison 7: Subgroup analysis - Industry funding, Outcome 3: Total cholesterol (mmol/L)



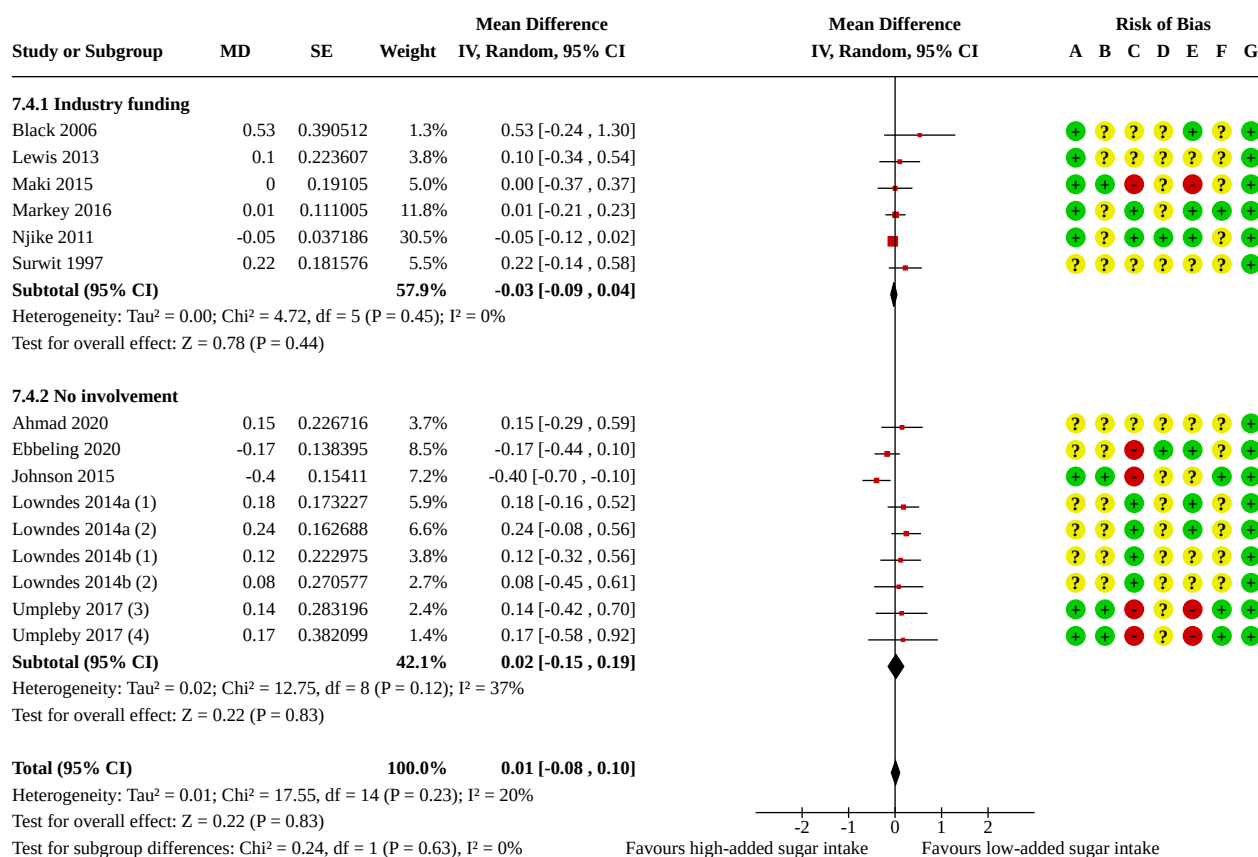
Footnotes

- (1) Shows the two arms of the intervention that compares high and low HFCS intake
- (2) Shows the two arms of the intervention that compares high and low sucrose intake
- (3) Shows the two arms of the intervention that compares high and low HFCS intake.
- (4) Shows the two arms of the intervention that compares high and low sucrose intake.
- (5) Includes the individuals not diagnosed with NAFLD, the controls.
- (6) Includes the individuals diagnosed with NAFLD.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 7.4. Comparison 7: Subgroup analysis - Industry funding, Outcome 4: LDL-cholesterol (mmol/L)



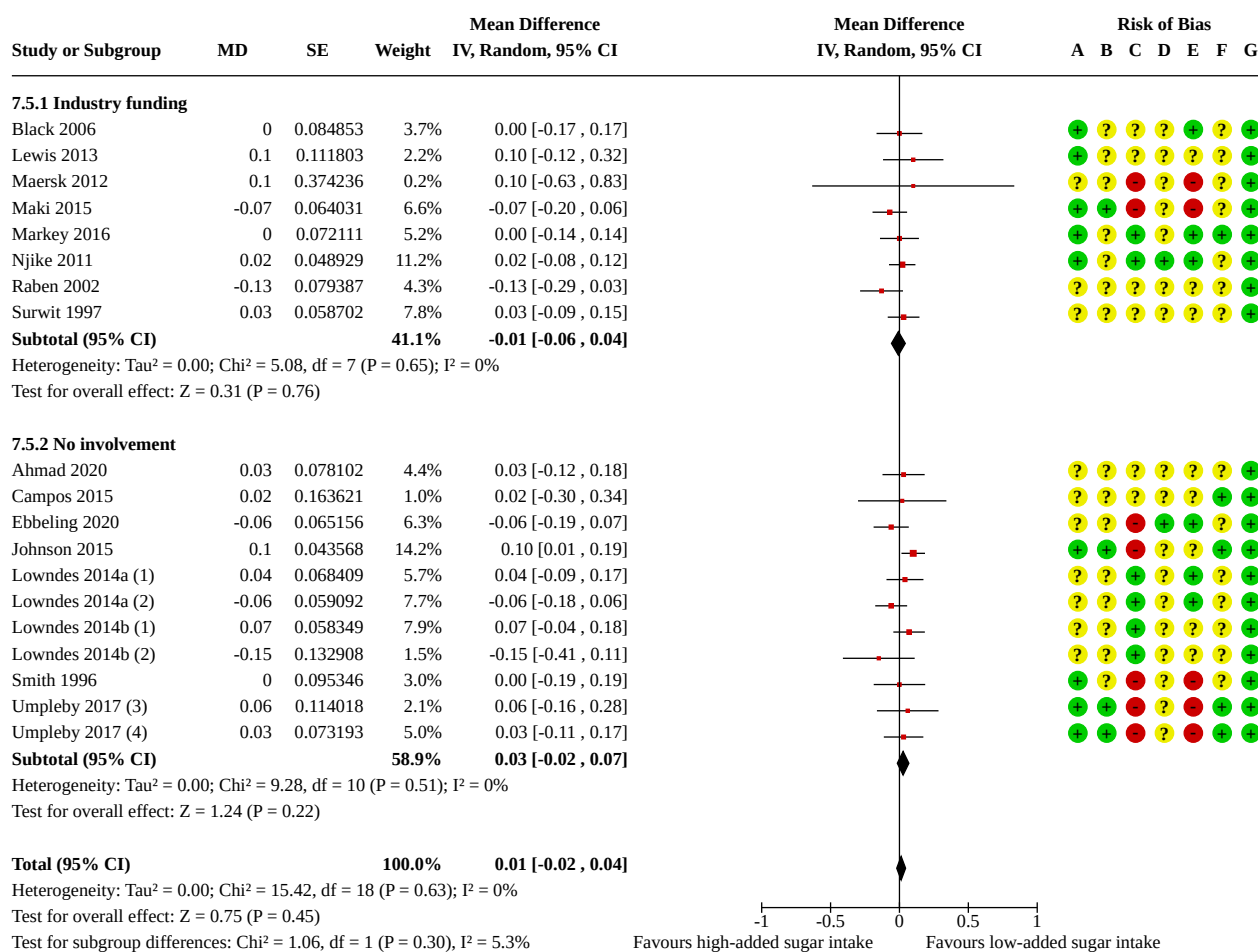
Footnotes

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
- (2) Shows the two arms of the intervention that compares high and low HFCS intake.
- (3) Includes the individuals not diagnosed with NAFLD, the controls.
- (4) Includes the individuals diagnosed with NAFLD.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 7.5. Comparison 7: Subgroup analysis - Industry funding, Outcome 5: HDL-cholesterol (mmol/L)



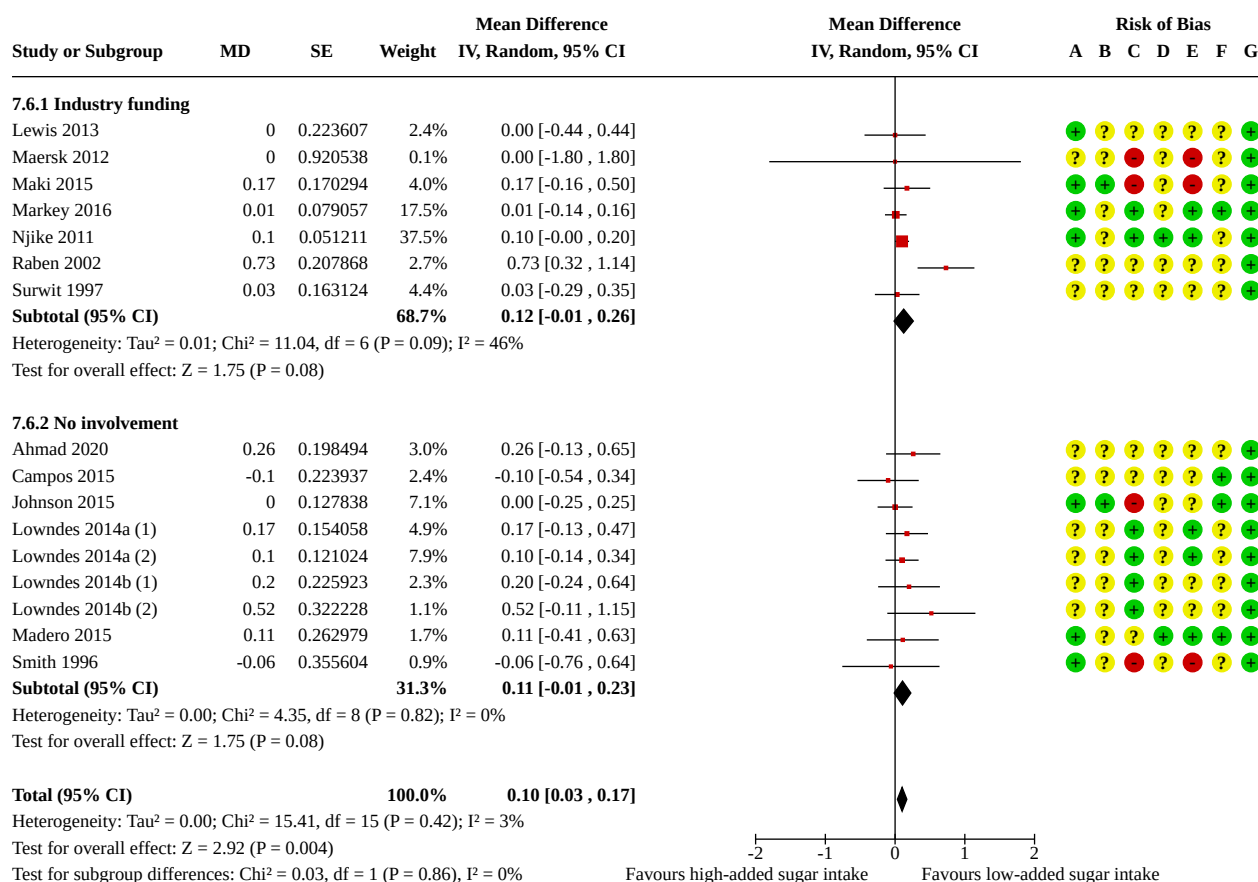
Footnotes

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
- (2) Shows the two arms of the intervention that compares high and low HFCS intake.
- (3) Includes the individuals diagnosed with NAFLD.
- (4) Includes the individuals not diagnosed with NAFLD, the controls.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 7.6. Comparison 7: Subgroup analysis - Industry funding, Outcome 6: Triglycerides (mmol/L)

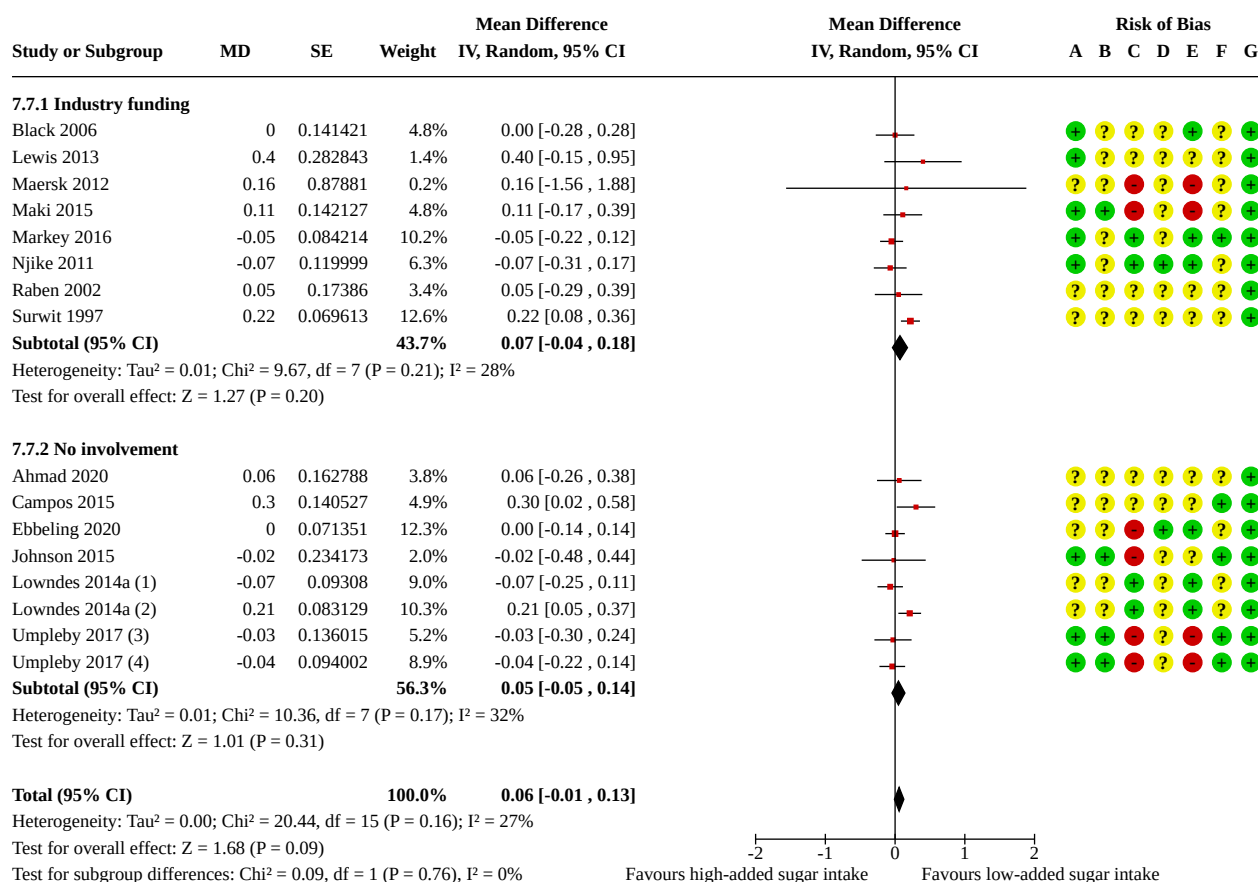


Footnotes

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
(2) Shows the two arms of the intervention that compares high and low HFCS intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Analysis 7.7. Comparison 7: Subgroup analysis - Industry funding, Outcome 7: Fasting plasma glucose (mmol/L)**Footnotes**

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
 (2) Shows the two arms of the intervention that compares high and low HFCS intake.
 (3) Includes the individuals not diagnosed with NAFLD, the controls.
 (4) Includes the individuals diagnosed with NAFLD.

Risk of bias legend

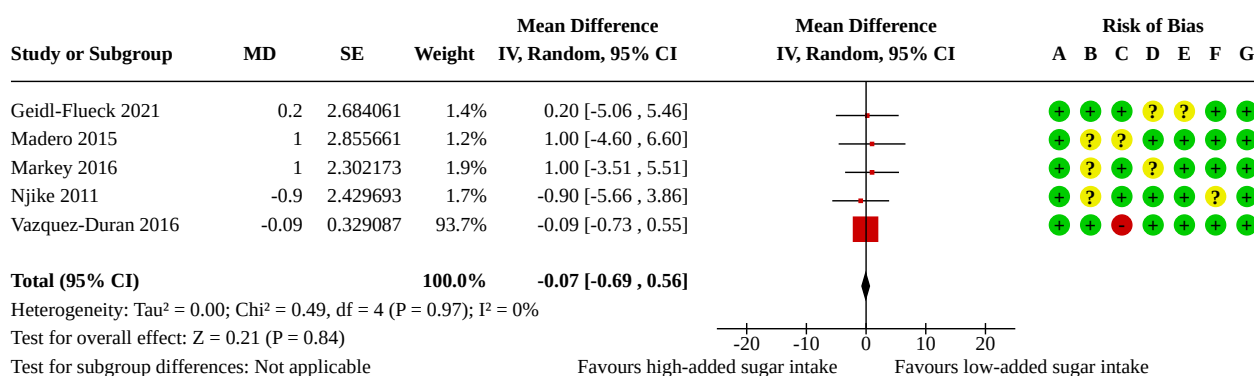
- (A) Random sequence generation (selection bias)
 (B) Allocation concealment (selection bias)
 (C) Blinding of participants and personnel (performance bias)
 (D) Blinding of outcome assessment (detection bias)
 (E) Incomplete outcome data (attrition bias)
 (F) Selective reporting (reporting bias)
 (G) Other bias

Comparison 8. Sensitivity analysis - Low risk of bias

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
8.1 Systolic blood pressure (mmHg)	5		Mean Difference (IV, Random, 95% CI)	-0.07 [-0.69, 0.56]
8.2 Diastolic blood pressure (mmHg)	5		Mean Difference (IV, Random, 95% CI)	0.19 [-1.93, 2.31]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
8.3 Total cholesterol (mmol/L)	3		Mean Difference (IV, Random, 95% CI)	0.11 [-0.13, 0.36]
8.4 LDL-cholesterol (mmol/L)	2		Mean Difference (IV, Random, 95% CI)	-0.04 [-0.11, 0.03]
8.5 HDL-cholesterol (mmol/L)	2		Mean Difference (IV, Random, 95% CI)	0.01 [-0.07, 0.09]
8.6 Triglycerides (mmol/L)	3		Mean Difference (IV, Random, 95% CI)	0.07 [-0.01, 0.16]
8.7 Fasting plasma glucose (mmol/L)	2		Mean Difference (IV, Random, 95% CI)	-0.06 [-0.19, 0.08]

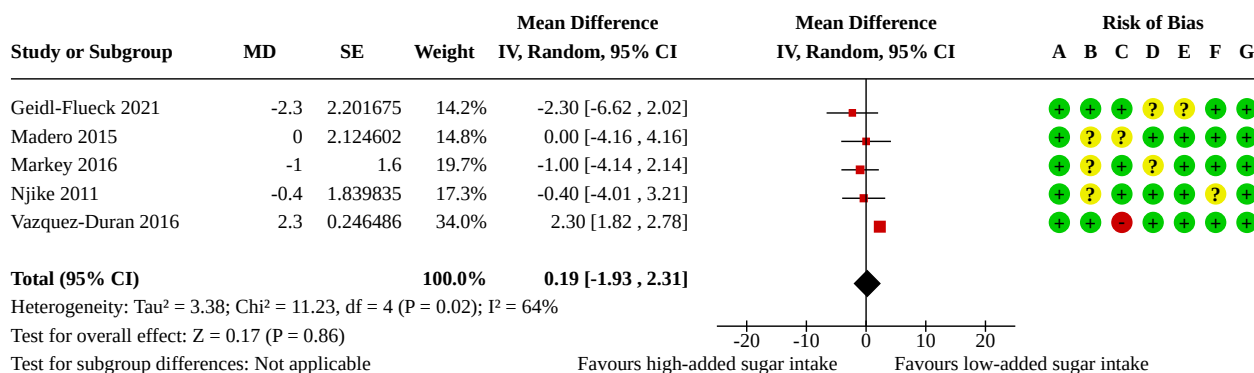
Analysis 8.1. Comparison 8: Sensitivity analysis - Low risk of bias, Outcome 1: Systolic blood pressure (mmHg)



Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

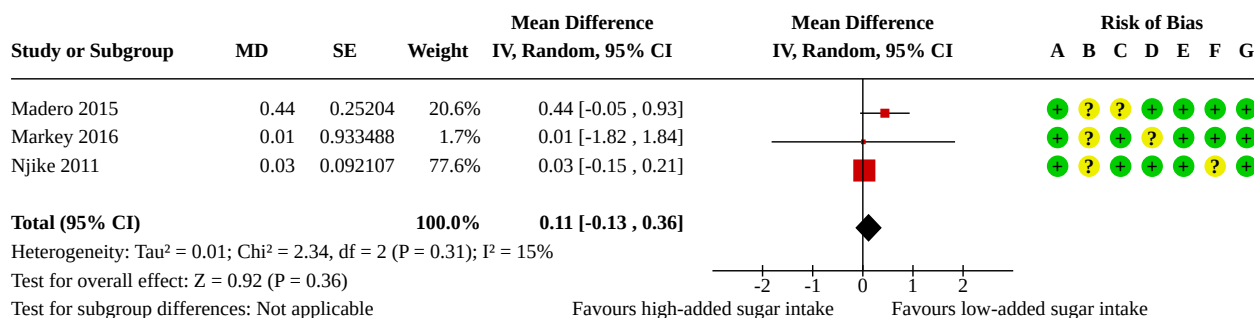
Analysis 8.2. Comparison 8: Sensitivity analysis - Low risk of bias, Outcome 2: Diastolic blood pressure (mmHg)



Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

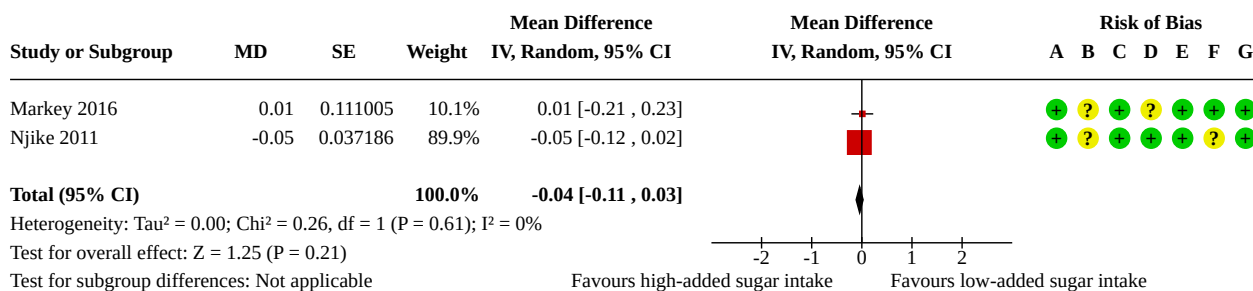
Analysis 8.3. Comparison 8: Sensitivity analysis - Low risk of bias, Outcome 3: Total cholesterol (mmol/L)



Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

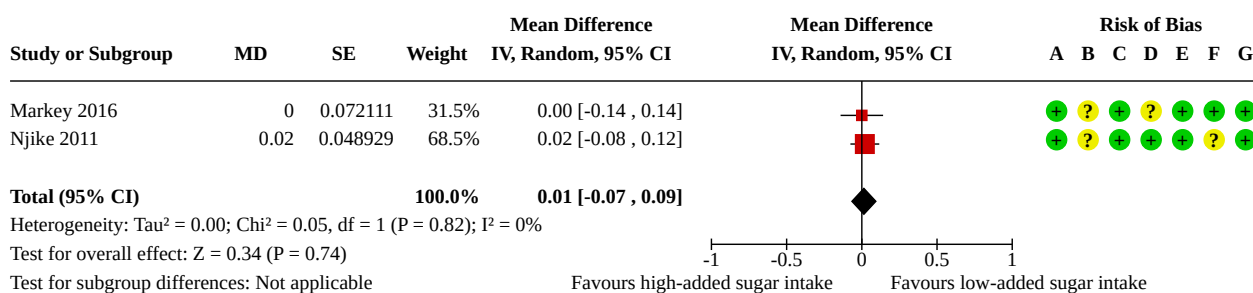
Analysis 8.4. Comparison 8: Sensitivity analysis - Low risk of bias, Outcome 4: LDL-cholesterol (mmol/L)



Risk of bias legend

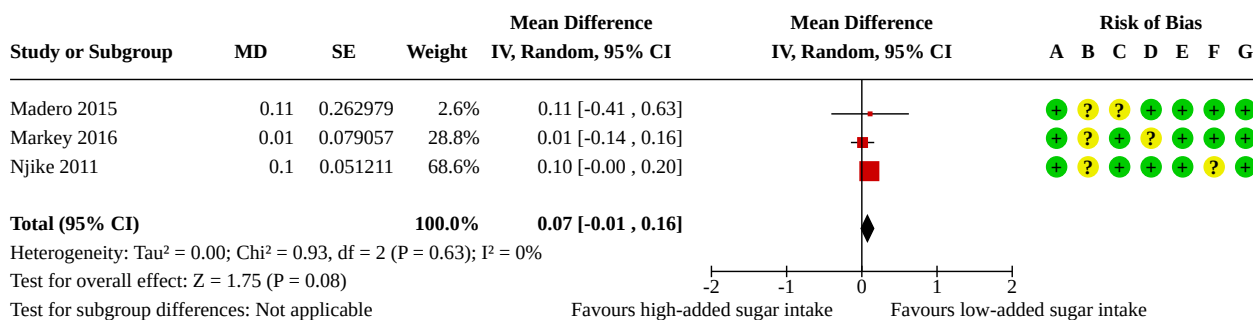
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 8.5. Comparison 8: Sensitivity analysis - Low risk of bias, Outcome 5: HDL-cholesterol (mmol/L)

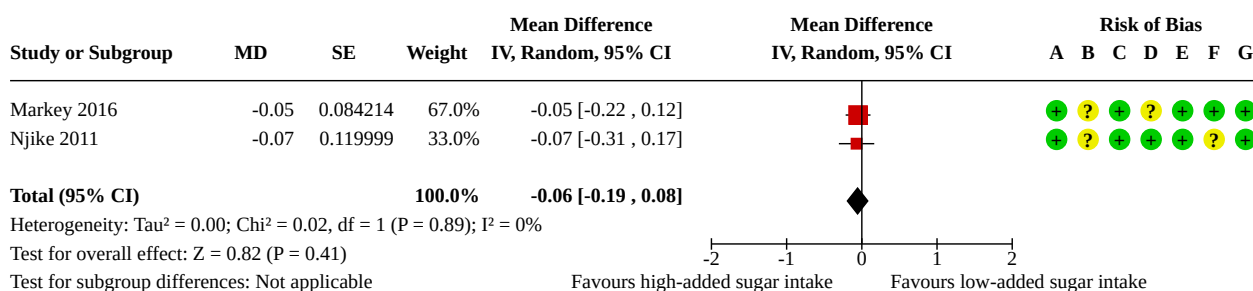


Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 8.6. Comparison 8: Sensitivity analysis - Low risk of bias, Outcome 6: Triglycerides (mmol/L)**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 8.7. Comparison 8: Sensitivity analysis - Low risk of bias, Outcome 7: Fasting plasma glucose (mmol/L)**Risk of bias legend**

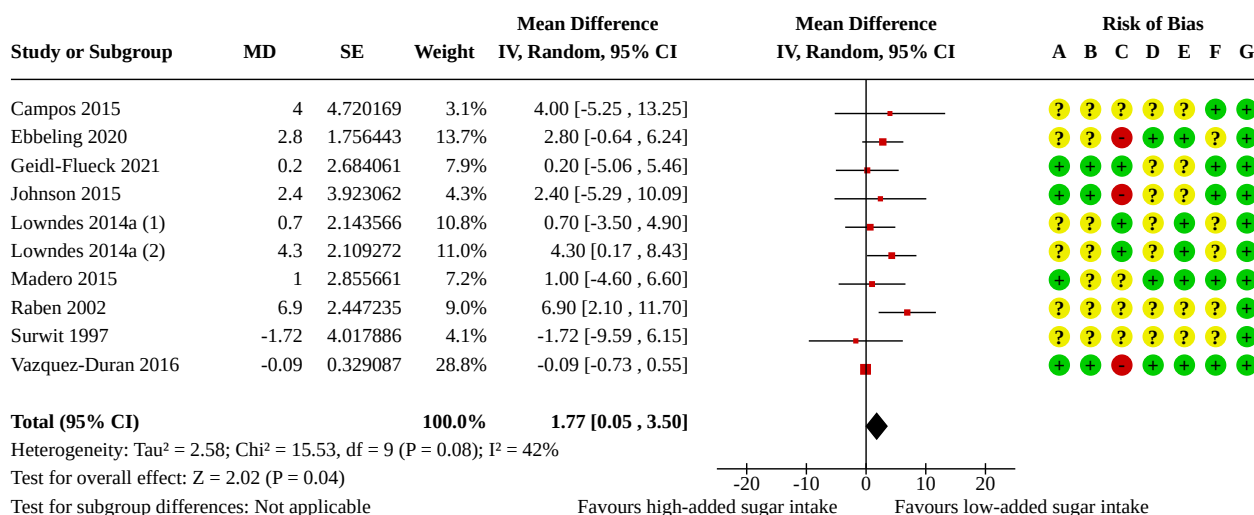
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Comparison 9. Sensitivity analysis - Cross-overs excluded

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
9.1 Systolic blood pressure (mmHg)	9		Mean Difference (IV, Random, 95% CI)	1.77 [0.05, 3.50]
9.2 Diastolic blood pressure (mmHg)	9		Mean Difference (IV, Random, 95% CI)	1.98 [1.23, 2.72]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
9.3 Total cholesterol (mmol/L)	10		Mean Difference (IV, Random, 95% CI)	0.13 [-0.02, 0.27]
9.4 LDL-cholesterol (mmol/L)	5		Mean Difference (IV, Random, 95% CI)	0.02 [-0.18, 0.22]
9.5 HDL-cholesterol (mmol/L)	9		Mean Difference (IV, Random, 95% CI)	0.01 [-0.04, 0.06]
9.6 Triglycerides (mmol/L)	9		Mean Difference (IV, Random, 95% CI)	0.14 [0.01, 0.28]
9.7 Fasting plasma glucose (mmol/L)	7		Mean Difference (IV, Random, 95% CI)	0.11 [0.00, 0.21]

Analysis 9.1. Comparison 9: Sensitivity analysis - Cross-overs excluded, Outcome 1: Systolic blood pressure (mmHg)



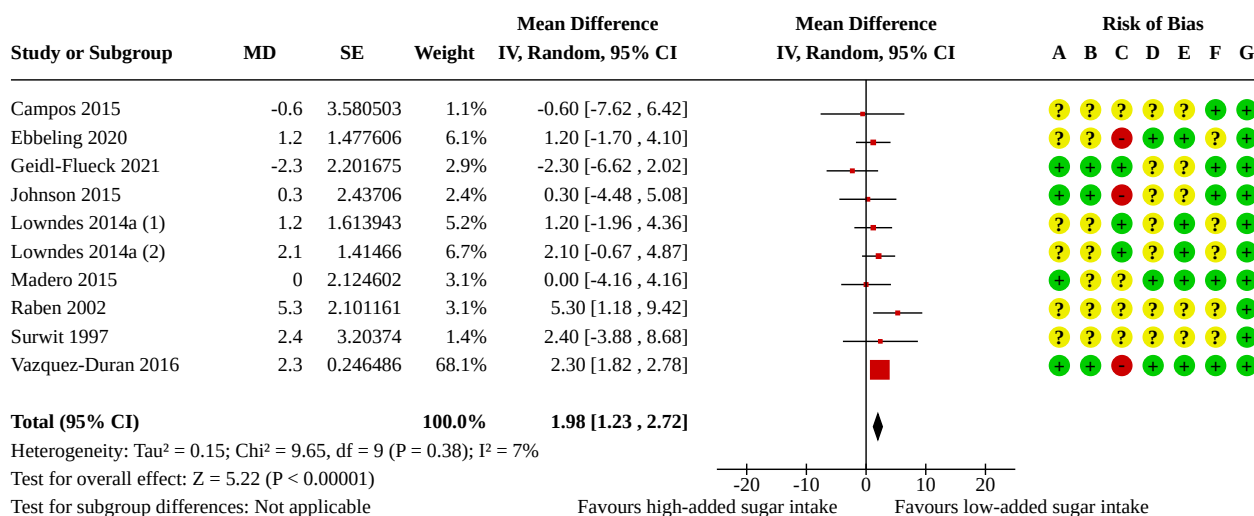
Footnotes

- (1) Shows the two arms of the intervention that compares high and low HFCS intake.
(2) Shows the two arms of the intervention that compares high and low sucrose intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
(B) Allocation concealment (selection bias)
(C) Blinding of participants and personnel (performance bias)
(D) Blinding of outcome assessment (detection bias)
(E) Incomplete outcome data (attrition bias)
(F) Selective reporting (reporting bias)
(G) Other bias

Analysis 9.2. Comparison 9: Sensitivity analysis - Cross-overs excluded, Outcome 2: Diastolic blood pressure (mmHg)



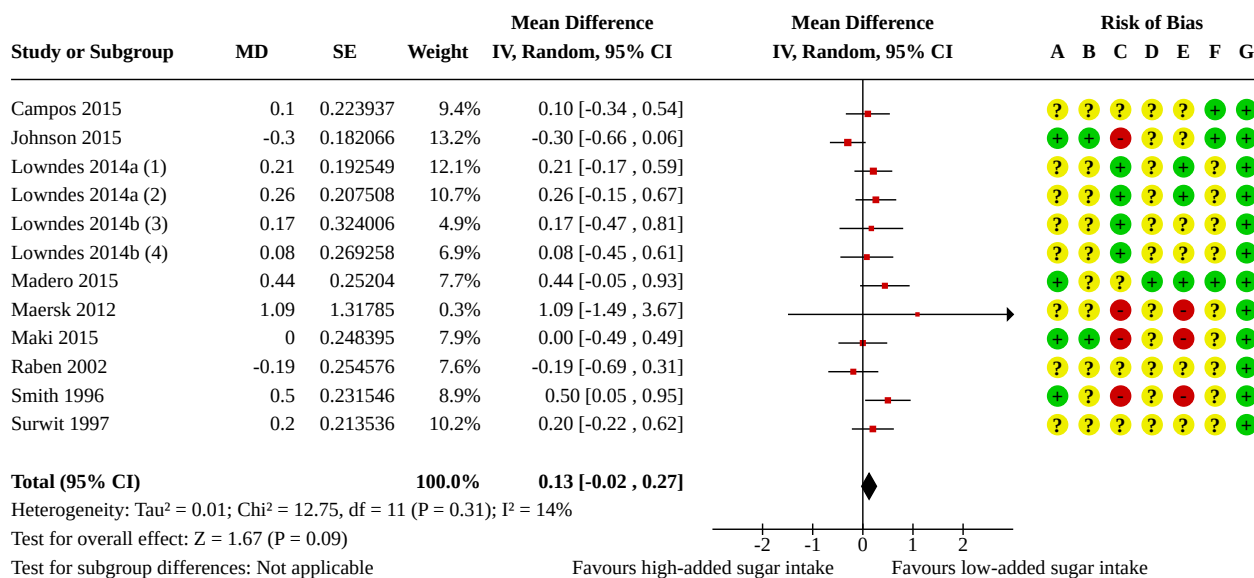
Footnotes

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
 (2) Shows the two arms of the intervention that compares high and low HFCS intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
 (B) Allocation concealment (selection bias)
 (C) Blinding of participants and personnel (performance bias)
 (D) Blinding of outcome assessment (detection bias)
 (E) Incomplete outcome data (attrition bias)
 (F) Selective reporting (reporting bias)
 (G) Other bias

Analysis 9.3. Comparison 9: Sensitivity analysis - Cross-overs excluded, Outcome 3: Total cholesterol (mmol/L)



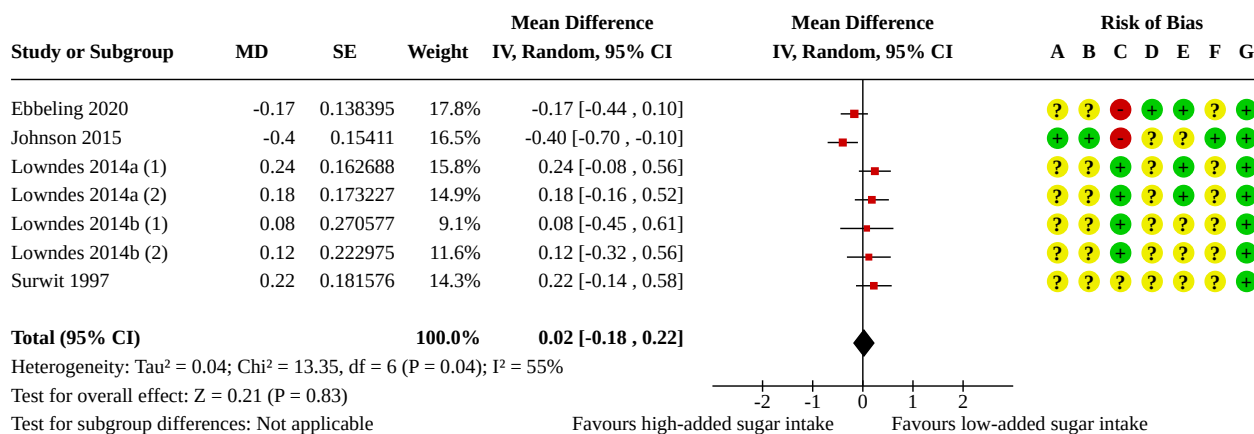
Footnotes

- (1) Shows the two arms of the intervention that compares high and low HFCS intake
- (2) Shows the two arms of the intervention that compares high and low sucrose intake
- (3) Shows the two arms of the intervention that compares high and low HFCS intake.
- (4) Shows the two arms of the intervention that compares high and low sucrose intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Analysis 9.4. Comparison 9: Sensitivity analysis - Cross-overs excluded, Outcome 4: LDL-cholesterol (mmol/L)



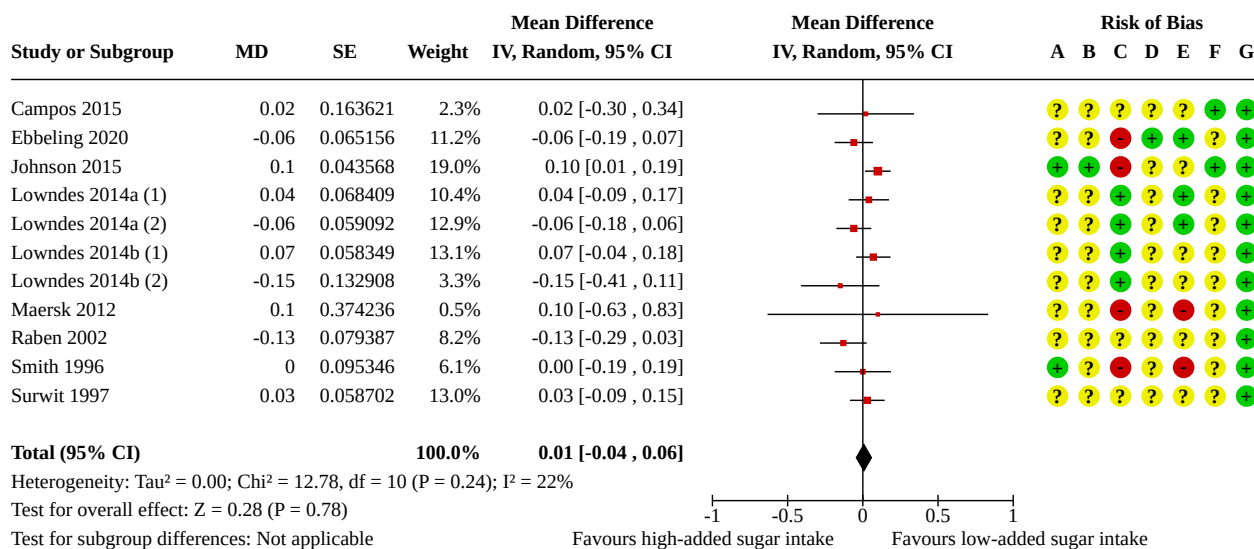
Footnotes

- (1) Shows the two arms of the intervention that compares high and low HFCS intake.
 (2) Shows the two arms of the intervention that compares high and low sucrose intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
 (B) Allocation concealment (selection bias)
 (C) Blinding of participants and personnel (performance bias)
 (D) Blinding of outcome assessment (detection bias)
 (E) Incomplete outcome data (attrition bias)
 (F) Selective reporting (reporting bias)
 (G) Other bias

Analysis 9.5. Comparison 9: Sensitivity analysis - Cross-overs excluded, Outcome 5: HDL-cholesterol (mmol/L)



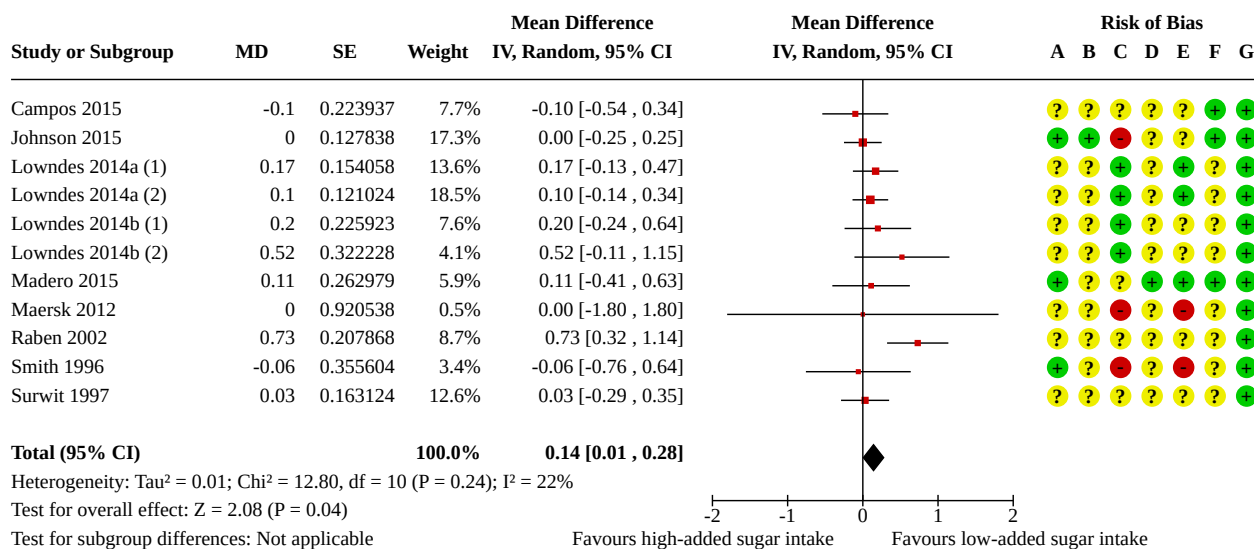
Footnotes

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
 (2) Shows the two arms of the intervention that compares high and low HFCS intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
 (B) Allocation concealment (selection bias)
 (C) Blinding of participants and personnel (performance bias)
 (D) Blinding of outcome assessment (detection bias)
 (E) Incomplete outcome data (attrition bias)
 (F) Selective reporting (reporting bias)
 (G) Other bias

Analysis 9.6. Comparison 9: Sensitivity analysis - Cross-overs excluded, Outcome 6: Triglycerides (mmol/L)



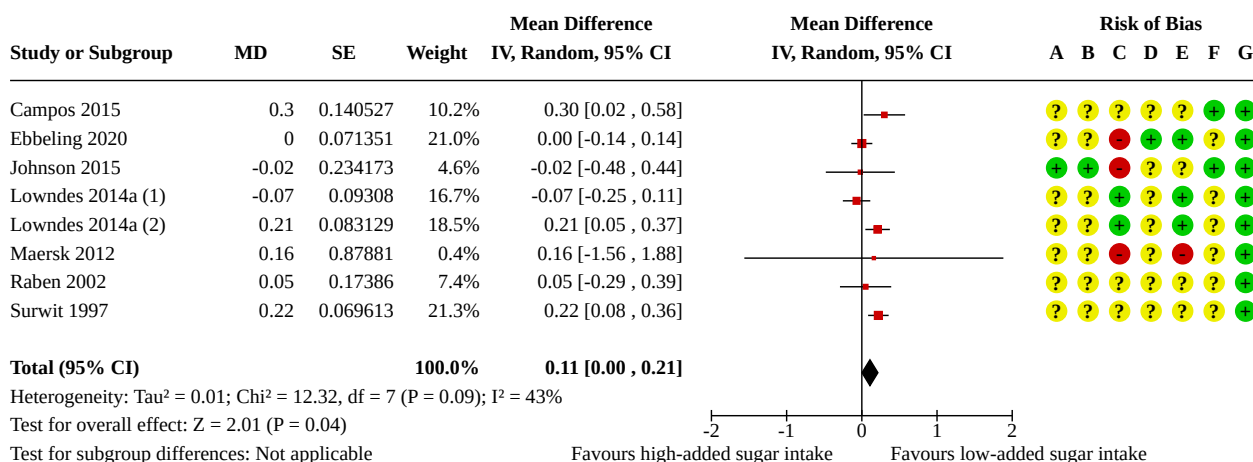
Footnotes

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
 (2) Shows the two arms of the intervention that compares high and low HFCS intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
 (B) Allocation concealment (selection bias)
 (C) Blinding of participants and personnel (performance bias)
 (D) Blinding of outcome assessment (detection bias)
 (E) Incomplete outcome data (attrition bias)
 (F) Selective reporting (reporting bias)
 (G) Other bias

Analysis 9.7. Comparison 9: Sensitivity analysis - Cross-overs excluded, Outcome 7: Fasting plasma glucose (mmol/L)



Footnotes

- (1) Shows the two arms of the intervention that compares high and low sucrose intake.
 (2) Shows the two arms of the intervention that compares high and low HFCS intake.

Risk of bias legend

- (A) Random sequence generation (selection bias)
 (B) Allocation concealment (selection bias)
 (C) Blinding of participants and personnel (performance bias)
 (D) Blinding of outcome assessment (detection bias)
 (E) Incomplete outcome data (attrition bias)
 (F) Selective reporting (reporting bias)
 (G) Other bias

APPENDICES

Appendix 1. Search strategies

CENTRAL

#1 sugar*:ti,ab

#2 (monosaccharide* or disaccharide* or glucose or galactose or fructose or dextrose or sucrose or lactose or maltose or "corn syrup" or honey):ti,ab

#3 (sweetener* or sweetening or (maize next syrup*) or ("glucose fructose" next syrup*) or isoglucose):ti,ab

#4 ((soft or fizzy or carbonated) NEAR/3 drink*):ti,ab

#5 {or #1-#4}

#6 (cardio* or cardia* or heart* or coronary* or angina* or ventric* or myocard* or pericard* or ischem* or ischaem* or emboli* or arrhythmi* or thrombo* or "atrial fibrillat*" or tachycardi* or endocardi* or "sick sinus"):ti,ab

#7 (stroke or strokes or cerebrovasc* or "cerebral vascular" or apoplexy or (brain next accident*) or (brain next infarct*) or (cerebral next infarct*) or (lacunar next infarct*)):ti,ab

#8 (hypertensi* or (peripheral next arter* next disease) or "high blood pressure" or (increas* next "blood pressure") or "elevated blood pressure"):ti,ab

#9 (hyperlipid* or hyperlipemia* or hyperlipaemia* or hypercholesterol* or hypercholesteremia* or hypercholesteremia* or hyperlipoproteinemia* or hyperlipoproteinaemia* or hypertriglyceridemia* or hypertriglyceridaemia*):ti,ab

#10 (diet* NEAR/2 cholesterol*):ti,ab
 #11 (HDL* NEAR/3 cholesterol*):ti,ab
 #12 ("high density" NEAR/2 cholesterol*):ti,ab
 #13 "alpha lipoprotein cholesterol":ti,ab
 #14 (LDL NEAR/2 cholesterol*):ti,ab
 #15 ("low density" NEAR/2 cholesterol*):ti,ab
 #16 "beta lipoprotein cholesterol":ti,ab
 #17 ("coronary risk" next factor*):ti,ab
 #18 "blood pressure":ti,ab
 #19 (triglyceride* or triacylglycerol*):ti,ab
 #20 (lipoprote* or "fasting blood sugar" or "fasting plasma glucose"):ti,ab
 #21 {or #6-#20}
 #22 (diet* or consum* or eat* or food* or nutri* or intake):ti,ab
 #23 #5 and #21 and #22

MEDLINE Ovid

1. exp Dietary Sugars/
2. sugar*.tw.
3. exp Monosaccharides/
4. exp Disaccharides/
5. monosaccharide*.tw.
6. disaccharide*.tw.
7. glucose.tw.
8. galactose.tw.
9. fructose.tw.
10. dextrose.tw.
11. sucrose.tw.
12. lactose.tw.
13. maltose.tw.
14. corn syrup.tw.
15. honey.tw.
16. exp Sweetening Agents/
17. sweetener*.tw.
18. sweetening.tw.
19. maize syrup*.tw.
20. glucose-fructose syrup*.tw.

21. isoglucose.tw.
22. Carbonated Beverages/
23. ((soft or fizzy or carbonated) adj3 drink*).tw.
24. or/1-23
25. exp Cardiovascular Diseases/
26. cardio*.tw.
27. cardia*.tw.
28. heart*.tw.
29. coronary*.tw.
30. angina*.tw.
31. ventric*.tw.
32. myocard*.tw.
33. pericard*.tw.
34. isch?em*.tw.
35. emboli*.tw.
36. arrhythmi*.tw.
37. thrombo*.tw.
38. atrial fibrillat*.tw.
39. tachycardi*.tw.
40. endocardi*.tw.
41. (sick adj sinus).tw.
42. exp Stroke/
43. (stroke or strokes).tw.
44. cerebrovasc*.tw.
45. cerebral vascular.tw.
46. apoplexy.tw.
47. (brain adj2 accident*).tw.
48. ((brain* or cerebral or lacunar) adj2 infarct*).tw.
49. exp Hypertension/
50. hypertensi*.tw.
51. peripheral arter* disease*.tw.
52. ((high or increased or elevated) adj2 blood pressure).tw.
53. exp Hyperlipidemias/
54. hyperlipid*.tw.
55. hyperlip?emia*.tw.

56. hypercholesterol*.tw.
57. hypercholester?emia*.tw.
58. hyperlipoprotein?emia*.tw.
59. hypertriglycerid?emia*.tw.
60. exp Arteriosclerosis/
61. Cholesterol, Dietary/
62. (diet* adj2 cholesterol*).tw.
63. Cholesterol, HDL/
64. (HDL* adj3 cholesterol*).tw.
65. (high density adj2 cholesterol*).tw.
66. alpha lipoprotein cholesterol.tw.
67. Cholesterol, LDL/
68. (LDL adj2 cholester*).tw.
69. (low density adj2 cholesterol*).tw.
70. beta lipoprotein cholesterol.tw.
71. "coronary risk factor"*.tw.
72. Blood Pressure/
73. blood pressure.tw.
74. exp Triglycerides/
75. triglyceride*.tw.
76. triacylglycerol*.tw.
77. exp Lipoproteins/
78. lipoprote*.tw.
79. "fasting blood sugar".tw.
80. "fasting plasma glucose".tw.
81. or/25-80
82. diet/
83. diet*.tw.
84. consum*.tw.
85. eat*.tw.
86. food*.tw.
87. nutri*.tw.
88. intake.tw.
89. or/82-88
90. 24 and 81 and 89

91. randomized controlled trial.pt.
92. controlled clinical trial.pt.
93. randomized.ab.
94. placebo.ab.
95. clinical trials as topic.sh.
96. randomly.ab.
97. trial.ti.
98. 91 or 92 or 93 or 94 or 95 or 96 or 97
99. exp animals/ not humans.sh.
100. 98 not 99
101. 90 and 100

Embase Ovid

1. sugar intake/
2. sugar*.tw.
3. exp monosaccharide/
4. *disaccharide/
5. monosaccharide*.tw.
6. disaccharide*.tw.
7. glucose.tw.
8. galactose.tw.
9. fructose.tw.
10. dextrose.tw.
11. sucrose.tw.
12. lactose.tw.
13. maltose.tw.
14. corn syrup.tw.
15. honey.tw.
16. *sweetening agent/
17. sweetener*.tw.
18. sweetening.tw.
19. maize syrup*.tw.
20. glucose-fructose syrup*.tw.
21. isoglucose.tw.
22. carbonated beverage/
23. ((soft or fizzy or carbonated) adj3 drink*).tw.

24. or/1-23
25. exp cardiovascular disease/
26. cardio*.tw.
27. cardia*.tw.
28. heart*.tw.
29. coronary*.tw.
30. angina*.tw.
31. ventric*.tw.
32. myocard*.tw.
33. pericard*.tw.
34. isch?em*.tw.
35. emboli*.tw.
36. arrhythmi*.tw.
37. thrombo*.tw.
38. atrial fibrillat*.tw.
39. tachycardi*.tw.
40. endocardi*.tw.
41. (sick adj sinus).tw.
42. exp cerebrovascular accident/
43. (stroke or strokes).tw.
44. cerebrovasc*.tw.
45. cerebral vascular.tw.
46. apoplexy.tw.
47. (brain adj2 accident*).tw.
48. ((brain* or cerebral or lacunar) adj2 infarct*).tw.
49. exp hypertension/
50. hypertensi*.tw.
51. peripheral arter* disease*.tw.
52. ((high or increased or elevated) adj2 blood pressure).tw.
53. exp hyperlipidemia/
54. hyperlipid*.tw.
55. hyperlip?emia*.tw.
56. hypercholesterol*.tw.
57. hypercholester?emia*.tw.
58. hyperlipoprotein?emia*.tw.

59. hypertriglycerid?emia*.tw.
60. exp arteriosclerosis/
61. cholesterol intake/
62. (diet* adj2 cholesterol*).tw.
63. high density lipoprotein cholesterol/
64. (HDL* adj3 cholesterol*).tw.
65. (high density adj2 cholesterol*).tw.
66. alpha lipoprotein cholesterol.tw.
67. low density lipoprotein cholesterol/
68. (LDL adj2 choleser*).tw.
69. (low density adj2 cholesterol*).tw.
70. beta lipoprotein cholesterol.tw.
71. "coronary risk factor* ".tw.
72. blood pressure/
73. blood pressure.tw.
74. exp triacylglycerol/
75. triglyceride*.tw.
76. triacylglycerol*.tw.
77. exp lipoprotein/
78. lipoprote*.tw.
79. "fasting blood sugar".tw.
80. "fasting plasma glucose".tw.
81. or/25-80
82. diet/
83. diet*.tw.
84. consum*.tw.
85. eat*.tw.
86. food*.tw.
87. nutri*.tw.
88. intake.tw.
89. or/82-88
90. 24 and 81 and 89
91. random\$.tw.
92. factorial\$.tw.
93. crossover\$.tw.

94. cross over\$.tw.
95. cross-over\$.tw.
96. placebo\$.tw.
97. (doubl\$ adj blind\$).tw.
98. (singl\$ adj blind\$).tw.
99. assign\$.tw.
100. allocat\$.tw.
101. volunteer\$.tw.
102. crossover procedure/
103. double blind procedure/
104. randomized controlled trial/
105. single blind procedure/
106. 91 or 92 or 93 or 94 or 95 or 96 or 97 or 98 or 99 or 100 or 101 or 102 or 103 or 104 or 105
107. (animal/ or nonhuman/) not human/
108. 106 not 107
109. 90 and 108
110. limit 109 to embase

CPCI-S

#16 #15 AND #14

#15 TS=(random* or blind* or allocat* or assign* or trial* or placebo* or crossover* or cross-over*)

#14 #13 AND #12 AND #4

#13 TS=(diet* or consum* or eat* or food* or nutri* or intake)

#12 #11 OR #10 OR #9 OR #8 OR #7 OR #6 OR #5

#11 TS=("blood pressure" or triglyceride* or triacylglycerol* or lipoprote* or "fasting blood sugar" or "fasting plasma glucose")

#10 TS=("coronary risk factor*")

#9 TS=(arterioscleros* or "diet* cholesterol*" or HDL or LDL or "high density cholesterol*" or "low density cholesterol*" or "alpha lipoprotein cholesterol" or "beta lipoprotein cholesterol")

#8 TS=(hyperlipid* or hyperlipemia* or hyperlipaemia* or hypercholesterol* or hypercholesteremia* or hypercholesteremia* or hyperlipoproteinemia* or hyperlipoproteinaemia* or hypertriglyceridemia* or hypertriglyceridaemia*)

#7 TS=(hypertensi* or "peripheral arter* disease*" or "high blood pressure" or "increas* blood pressure" or "elevated blood pressure")

#6 TS=(stroke or strokes or cerebrovasc* or "cerebral vascular" or apoplexy or "brain accident*" or "brain infarct*" or "cerebral infarct*" or "lacunar infarct*")

#5 TS=(cardio* or cardia* or heart* or coronary* or angina* or ventric* or myocard* or pericard* or ischem* or ischaem* or emboli* or arrhythmi* or thrombo* or "atrial fibrillat*" or tachycardi* or endocardi* or "sick sinus")

#4 #3 OR #2 OR #1

#3 TS=("soft drink*" or "fizzy drink*" or "carbonated drink*" or "carbonated beverage*")

#2 TS=(sweetener* or sweetening or "maize syrup*" or "glucose-fructose syrup*" or isoglucose)

High versus low-added sugar consumption for the primary prevention of cardiovascular disease (Review)

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#1 TS=(sugar* or monosaccharide* or disaccharide* or glucose or galactose or fructose or dextrose or sucrose or lactose or maltose or "corn syrup" or honey)

Clinicaltrials.gov

Advanced search

Interventional studies

Intervention: sugar

ICTRP

Recruitment: All

Intervention: sugar

HISTORY

Protocol first published: Issue 4, 2019

CONTRIBUTIONS OF AUTHORS

Sara Bergwall and Anna Johansson screened titles and abstracts and assessed studies for inclusion or exclusion, performed the analyses and wrote the review.

Emily Sonestedt and Stefan Acosta reviewed and contributed with expertise.

DECLARATIONS OF INTEREST

Sara Bergwall has declared that she has no conflicts of interest.

Anna Johansson reports to work as a health professional related to the topic area of this review.

Emily Sonestedt has declared that she has no conflicts of interest.

Stefan Acosta has declared that he has no conflicts of interest.

SOURCES OF SUPPORT

Internal sources

- No sources of support provided

External sources

- This project was supported by the NIHR via Cochrane Infrastructure funding to the Heart Group. The views and opinions expressed therein are those of the authors and do not necessarily reflect those of the Systematic Reviews Programme, NIHR, NHS or the Department of Health and Social Care, UK

NIHR

DIFFERENCES BETWEEN PROTOCOL AND REVIEW

Since it is no longer recommended that industry-funding should be taken into consideration in the risk of bias domain 'other potential sources of bias', only compliance with the intervention was assessed in this section and industry-funding was reported in the text only.

As no studies reported our primary outcomes, we instead generated a funnel plot for total cholesterol, the secondary outcome with the highest number of studies and with the most clinical relevance. For the same reason, none of the other planned analyses on the primary outcomes could be conducted.

We also changed a subgroup analysis from solid versus liquid to solid and liquid versus liquid, since no included trials only provided solid foods to the participants. We also clarified, with respect to the subgroup analysis comparing healthy with a high-risk population, that a study had to have at least 50% of participants defined as unhealthy for the study to be categorised as such, and vice versa.

INDEX TERMS

Medical Subject Headings (MeSH)

*Cardiovascular Diseases [epidemiology] [prevention & control]; Cholesterol; Dietary Sugars [adverse effects]; Primary Prevention; Sugars

MeSH check words

Adult; Aged; Humans; Middle Aged; Young Adult